

CLAUDE

Helix Constitution — Universal CLAUDE.md

Field	Value
Revision	16
Created	2026-05-14
Last modified	2026-06-03T00:00:00Z
Status	active
Status summary	Added §11.4.114–§11.4.121 mirrors (eight new universal anchors from the 1.1.8-dev live-defect remediation, 2026-06-03): §11.4.114 last-known-good-tag regression isolation (diff the broken state against the known-good tag FIRST — the tagged-good version is the regression oracle), §11.4.115 RED-baseline-on-the-broken-artifact + polarity-switch (one test reproduces the defect on the pre-fix artifact then flips RED_MODE=0 to GREEN-guard — refines §11.4.43), §11.4.116 real-time conductor↔autonomous-test-framework sync channel (append-only JSONL event stream + atomically-rewritten status snapshot the orchestrator tails live; every verdict event carries its evidence path), §11.4.117 CV/OCR pixel-oracle fallback for non-introspectable UIs (drive + assert via pixels when the accessibility tree is blank — refines §11.4.48/.52/.107), §11.4.118 discovery-pressure to confirm known-issue-set completeness (fixing the reported set is necessary-not-sufficient; provable enumerated discovery coverage), §11.4.119 single-resource-owner partitioning for parallel hardware testing (exactly ONE stream owns each device's exclusive resource — refines §11.4.58/.103), §11.4.120 fix-breaks-its-own-gate reconciliation (rewrite the gate to assert the NEW mechanism; never fake-pass, never revert the fix), §11.4.121

Field	Value
	no-commit-while-build-writes-tracked-artifacts (defer commit to build completion so partial/stale artifacts never land — build-output analogue of §11.4.84). §11.4.108-§11.4.113 + §11.4.78-§11.4.107 mirrors continue from earlier Revisions.
Issues	none
Issues summary	—
Fixed	§11.4.108 mirror
Fixed summary	§11.4.107 lands in lockstep with the Constitution.md §11.4.107 addition.
Continuation	—

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- MANDATORY HOST-SESSION SAFETY (Constitution §12)
 - Forbidden — directly OR indirectly
 - Required safeguards for heavy scripts
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- MANDATORY TESTING CONSTRAINTS
- Code conventions
- When in doubt

This is the **base CLAUDE.md** imported by every project that includes the Helix Constitution submodule. Project-level CLAUDE.md may extend or tighten any rule by adding an explicit `Override: <section> block` — but **MUST NOT** weaken them.

Last revision: 2026-05-14

How inheritance works

A consuming project's root CLAUDE.md **MUST** start with a clearly-marked inheritance pointer:

```
## INHERITED FROM constitution/CLAUDE.md
```

All rules in ``constitution/CLAUDE.md`` (and the ``constitution/Constitution.md`` it references) apply unconditionally. The project-specific rules below extend them.

Claude Code supports the `@path/to/file` import syntax natively, so a consuming project can also write `@constitution/CLAUDE.md` at the top of its own CLAUDE.md and Claude Code will resolve it recursively. For agents that do not support `@imports`, the pointer-block pattern above ensures the inheritance is at least readable.

MANDATORY DEVELOPMENT PRINCIPLES

NO BLUFF. Every change ships with positive-evidence validation on the real target environment. A test that passes without exercising the user-visible behaviour is a critical defect.

(Constitution §7.1 + §11.4 — these references point to `constitution/Constitution.md`.)

CRITICAL: All code changes MUST follow these principles WITHOUT EXCEPTION:

1. **Solutions MUST NOT be error-prone.** Every fix must be robust, not introduce new failure modes. If a fix solves problem A but creates problem B, it is NOT acceptable. Test the fix against ALL existing functionality before committing.
2. **No blocking operations inside synchronized / shared-lock regions.** Long computations, network calls, or sleeps inside synchronized / Lock-held regions cause deadlocks or timeouts in other threads.
3. **Always consider concurrent callers.** Any method called from multiple threads must be safe for rapid consecutive calls.
4. **Test the fix, not just the symptom.** Verify the fix works AND doesn't break anything else.
5. **Anti-bluff is mandatory** (Constitution §7.1) — every runtime test MUST source the project anti-bluff helper, call at least one explicit user-visible action before any PASS, capture state delta, and assert positive evidence.
6. **Real captured evidence for audio/video** (Constitution §7.1 + §11.4.5) — features producing audio output validated via captured audio; features producing video output validated via captured frames + OCR or pixel-diff.

MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE

Forensic anchor — verbatim user mandate (2026-04-28):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

This is the historical origin of the project's anti-bluff covenant. Every test, every Challenge, every gate, every mutation pair exists to make the failure mode (PASS on broken-for-end-user feature) mechanically impossible.

Operative rule: the bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS MUST carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and grep-based PASS without runtime evidence are all critical defects regardless of how green the summary line looks.

Tests AND Challenges (HelixQA, integration suites, smoke tests, acceptance suites) are bound equally — a Challenge that scores PASS on a non-functional feature is the same class of defect as a unit test that does.

Canonical authority: constitution/Constitution.md §11.4 and its sub-sections §11.4.1 through §11.4.16.

Non-compliance is a release blocker regardless of context.

§11.4.1 — FAIL-bluffs are equally forbidden

A test that crashes for a script-internal reason (undefined variable under set -u, regex error, malformed assertion, missing argument) and produces a FAIL exit code is just as misleading as a PASS-bluff. Both let real defects ship undetected. Every test MUST fail ONLY for genuine product defects — script-bug failures must be fixed at the source layer (helper library, shared lib, test source), not patched in individual call sites.

§11.4.2 — Recorded-evidence requirement

A test that emits PASS without captured visual or audio evidence of the user-visible feature actually working on the screen the user would see is a §11.4 PASS-bluff. Every PASS for a user-visible feature MUST be cross-checked against captured recording + action timeline.

§11.4.3 — Per-environment-topology test dispatch

Tests that depend on environment topology MUST detect topology at test entry and dispatch the topology-appropriate variant. SKIP-with-reason is the correct fallback when the required topology is absent; PASS-by-default is forbidden.

§11.4.4 — Test-interrupt-on-discovery + retest-from-clean-baseline

The moment any defect is re-discovered, re-produced, or newly identified during a test cycle, the cycle MUST stop. Then: systematic debugging → fix at root cause → four-layer test coverage (pre-build / post-build / runtime / meta-test paired mutation) → full rebuild → re-deploy on every target → full retest from beginning.

§11.4.5 — Captured-evidence quality analysis

Audio: presence (RMS amplitude), channel count, sample rate + bit depth, glitch census, coexistence-artifact census. Video: presence (non-zero frame count), routing target, frame health (drops / jitter / freeze), obstruction census (OCR for hostile overlays), resolution + codec. Every check is required for every PASS.

§11.4.6 — No-guessing mandate

Forensic anchor — verbatim user mandate (2026-05-08):

"'LIKELY' is guessing, we MUST NOT have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We MUST always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Forbidden vocabulary in tests / gates / status reports / closure narratives / commit messages when describing causes: likely, probably, maybe, might, possibly, presumably, seems, appears to, guess, seemingly, apparently, perhaps, supposedly, conjectured, and synonyms.

Either prove the cause with captured forensic evidence and state it as fact, OR explicitly mark UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: with a tracked-task ID for follow-up.

§11.4.7 — Demotion-evidence rule

Demotion from a FAIL classification to a lower-severity classification requires positive evidence captured under the SAME CONDITIONS (same target, same build, same cycle position, same load profile) that originally exposed the defect. "I cannot reproduce in isolation" is a hypothesis, not a finding.

§11.4.8 — Deep-web-research-before-implementation

Before designing a non-trivial fix, perform deep web research: official docs, vendor technical guides, open-source codebases, coding tutorials, issue trackers. Every non-trivial fix's commit message (or accompanying entry) MUST cite at least one external source OR the literal "NO external solution found — original work".

§11.4.9 — Batch-source-fixes-before-rebuild

All source-side fixes that DO NOT require runtime validation to design MUST be landed BEFORE the next artifact rebuild. The anti-pattern of "fix A → rebuild → flash → cycle → fix B → rebuild → ..." serializes operator time onto rebuild latency.

§11.4.10 — Credentials-handling mandate

Forensic anchor — verbatim user mandate (2026-05-12):

"Credentials or any secret and sensitive data MUST NOT leak!"

Credentials MUST NEVER be tracked in git. .env / .env.* / *.env patterns + scripts/testing/secrets/* (with .example + README.md exception) git-ignored project-wide. Test scripts MUST NEVER print or log credentials. Per-service file separation limits blast radius. chmod 600 on credential files, chmod 700 on parent directory. Rotation on suspected leak.

§11.4.10.A — Pre-store credential leak audit (User mandate, 2026-05-17)

Forensic anchor — verbatim user mandate (2026-05-17):

"Us these for all future testing (full automation testing) and make sure they are not leaking anywhere or get git versioned!"

When an operator provides credentials, API tokens, signing keys, or any other secret material to be stored in the project's gitignored configuration, the storing agent MUST FIRST execute a repo-wide audit for prior leaks of THOSE specific values BEFORE storing: (1) git ls-files | xargs grep -l <value> for tree-leaks, (2) git log -S<value> --all --source --remotes for history-leaks, (3) surface findings to operator BEFORE storing (operator may rotate, accept-as-compromise, or abort), (4) on finding open a §6/§7 sixth-law-incidents record + redact tracked files in-place to <redacted-per-§11.4.10> + record OPERATOR ACTION REQUIRED for rotation per §11.4.10 sub-clause 7, (5) extend pre-push hook credential-pattern grep to catch the escaped class in the same commit. See Constitution §11.4.10.A for the full mandate.

§11.4.11 — File-layout discipline

Project files organised by purpose, not historical accident. Source under canonical project roots. Tests under canonical test directories. Logs and forensic artifacts under operator-controlled directories — never scattered at repo root, never tracked unless they are reference assets.

§11.4.12 — Auto-generated docs sync

Every auto-generated document MUST be regenerated in the same commit as any edit to its source. All output formats (.md + .html + .pdf) MUST stay in sync at all times.

§11.4.13 — Out-of-band sink-side captured-evidence

Whenever a downstream consumer (HDMI sink, cloud monitor, downstream service) provides a network-accessible introspection API that reports what was actually received, the test suite MUST consume that report as captured-evidence. The on-source-side view alone is insufficient.

§11.4.14 — Test playback cleanup

Every test MUST leave the target in a quiescent state. Cleanup mandatory on every exit path (trap '<cleanup>' EXIT). The orchestrator MUST run a post-test sanity check and FAIL the just-completed test if it left orphan state.

§11.4.15 — Item-status tracking

Every active item in the project Issues file carries a **Status:** line within five lines of its heading. Six-state vocabulary: Queued, In progress, Ready for testing, In testing, Reopened, Fixed (→ Fixed.md). Status updated as the item progresses. All three Issues / Issues_Summary / Fixed file types kept in sync (Markdown + HTML + PDF).

§11.4.16 — Item-type tracking

Every active item in the project Issues file carries a **Type:** line within eight non-blank lines of its heading. Three-value CLOSED vocabulary: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream — refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The Issues_Summary file carries the Type column for every active

item. All three Issues / Issues_Summary / Fixed file types kept in sync (Markdown + HTML + PDF). Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION enforce the mandate.

§11.4.17 — Universal-vs-project classification of new rules (User mandate, 2026-05-14)

Before adding ANY new rule, mandatory constraint, covenant clause, gate, or "MUST"-bearing statement to a project's Constitution / CLAUDE.md / AGENTS.md (or to a submodule's equivalents), the author MUST classify it as **universal** (reusable across any project → goes into this constitution submodule) or **project-specific** (references particular hardware / vendor / package / region → stays in the project / submodule layer). The commit message MUST carry a Classification: line stating the choice + one-sentence rationale. Universal rules that leak project-specific assumptions (hardware part numbers, vendor names, geographic regions, internal asset names) MUST be genericised first or downgraded to project-specific. When uncertain, default to project-specific (the narrower scope — lifting to universal later is cheap; the reverse is expensive). Pre-build gate CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION audits new rule commits for the classification statement; paired mutation strips it and asserts gate FAILs.

§11.4.20 — Subagent-driven-by-default mandate (User mandate, 2026-05-14)

When the runtime supports subagent delegation (Claude Code Agent tool, Cursor task-runners, Aider sub-sessions, etc.), the primary agent MUST default to subagent delegation for any task that has multi-step scope (≥ 3 phases), parallelisable independent subtasks, long-running diagnostic loops, OR specialised domain workflows (code review, security audit, doc propagation). Foreground-only is reserved for single-file edits, mid-execution operator clarification, critical-state sequencing (commits / pushes / tags), or tasks so quick that subagent overhead exceeds the work. Sub-discipline: **tight scope** (4-6 tasks, not "do everything"), **checkpoint commits** after each major task, **anti-stall protection** explicit in prompts, **anti-bluff verification** of subagent claims via repo state. Parallel subagents MUST partition non-overlapping files; `commit_all.sh --auto-cascade` bundles both via `git add -A`. Gate CM-SUBAGENT-DELEGATION-AUDIT (when implemented in consuming project) flags foreground multi-step work as a §11.4.20 violation.

§11.4.18 — Script documentation mandate (User mandate, 2026-05-14)

Every Bash / shell / POSIX-sh script anywhere in a project (scripts/, bin/, tests/, library directories, deployment hooks, CI helpers — depth-N recursive) MUST carry: (1) an in-source documentation block (Purpose / Usage / Inputs / Outputs / Side-effects / Dependencies / Cross-references) at the top of the file; (2) an external user guide under docs/scripts/<script-name>.md covering Overview / Prerequisites / Usage examples / Edge cases / Internal behaviour / Related scripts / Last verified date. When a script is modified, BOTH the in-source block AND the external user guide MUST be updated in the SAME commit. **No documentation ever can be out of sync with its codebase.** Pre-build gate CM-SCRIPT-DOCS-SYNC walks every *.sh / *.bash under script directories, verifies a companion docs/scripts/<name>.md exists, AND verifies doc was modified in the same commit (or doc-mtime ≥ script-mtime as a softer floor). Paired mutation strips the doc-sync invariant and asserts gate FAILs.

§11.4.19 — Fixed-document column-alignment mandate (User mandate, 2026-05-14)

Every project that maintains an open-work tracker AND a closed-archive tracker MUST keep the two structurally aligned along the same lifecycle axes (Status + Type). For the Fixed archive that means: (1) every ### / #### heading in Fixed.md (or equivalent) carries a ****Status:**** line and a ****Type:**** line within 8 non-blank lines of its heading; (2) a Fixed_Summary.md companion exists with the same column structure as Issues_Summary.md (# | Level | Status | Type | One-line description); (3) all three file formats (.md + .html + .pdf) for BOTH Fixed.md and Fixed_Summary.md stay in sync via the same single-shot wrapper that handles Issues + Issues_Summary; (4) closure migration is atomic — when an Issues entry resolves it moves to Fixed.md in the same commit, disappears from Issues_Summary (open-only), and appears in Fixed_Summary (closed-only). Status values for closed items are drawn from {Fixed (→ Fixed.md) | Fixed – pending device verification | Fixed – RECLASSIFIED}. Type values follow §11.4.16: {Bug | Feature | Task}. Pre-build gate CM-FIXED-COLUMN-ALIGNMENT (5+ invariants) — Fixed_Summary.md exists, table header carries Status+Type columns, mtime (Fixed_Summary ≥ Fixed), generator script present, sync wrapper invokes it, HTML+PDF exports for both Fixed and Fixed_Summary present. Paired mutation strips Status column from Fixed_Summary table header → gate FAILs. Classification: universal (per §11.4.17). No escape hatch.

§11.4.21 — Operator-blocked status + self-resolution exhaustion (User mandate, 2026-05-14)

Operator-blocked is the §11.4.15 Status closed-set's 7th value: {Queued | In progress | Ready for testing | In testing | Reopened | Operator-blocked | Fixed (→ Fixed.md)}. It is a **last-resort classification**, earned only after the agent documents exhaustion of every applicable self-resolution path: (a) CLI / ADB / SSH / API access already available, (b) subagent delegation per §11.4.20, (c) existing repo tooling (scripts / helpers / libraries), (d) captured fallback (synthetic event, asset substitution, mock, §11.4.3 topology SKIP), (e) external research per §11.4.8. Every Operator-blocked item **MUST** carry an ****Operator-Block-Details:**** line within 8 non-blank lines of its heading stating: **WHAT** (concrete action), **WHY** (each exhausted alternative enumerated), **UNBLOCK CONDITION** (observable signal), **WHO** (handle / contact / doc pointer). Issues_Summary.md lists Operator-blocked as a sortable Status value. Items **MUST** be re-evaluated every Nth tag cycle (project- defined, recommended ≥3rd cycle) — operator dependencies change. Fake Operator-blocked (no exhaustion audit) is a §11.4 covenant violation at the planning layer, severity-equivalent to a PASS-bluff. Gates: CM-ITEM-OPERATOR-BLOCKED-DETAILS (every Operator-blocked heading has the details line), CM-OPERATOR-BLOCKED-SELF-RESOLUTION- AUDIT (NEW Operator-blocked commits contain "Attempted: a — ...; b — ...; c — ..." trail). Classification: universal (per §11.4.17). No escape hatch.

§11.4.22 — Document-sync commit discipline (User mandate, 2026-05-14)

Every project tracking work items through an Issues / Fixed lifecycle **MUST** provide a **lightweight commit path** distinct from the full-repo commit wrapper. The lightweight path stages, commits, and pushes **ONLY** the status-tracking doc set — Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + their HTML + PDF exports + any auto-generated audit artifact — so doc-status never drifts behind working-tree reality when the full-repo wrapper is unavailable (in-flight rebase, large submodule churn, partial network). The wrapper **MUST**: (a) auto-invoke the project's export-regeneration pipeline first so Markdown + HTML + PDF stay in sync; (b) stage **ONLY** the explicit doc-set list — **NEVER** git add -A; (c) use a separate flock disjoint from the full-tree wrapper's lock; (d) push to every parent-repo remote; (e) exit 3 on nothing-to-commit (informational, not error). The wrapper **MUST** be invocable standalone **OR** as a delegation flag on the full-tree wrapper (e.g. commit_all.sh --docs-only) so operators have a single mental model. Inherits the project's §9 preflight discipline (refuses to run mid-meta-test). Gate CM-COMMIT-DOCS-EXISTS verifies wrapper + guide + flag + doc-set enumeration. Paired mutation strips the doc-set array → gate FAILs. Classification: universal (per §11.4.17). Composes with §11.4.12 (export-sync), §11.4.15 (status

tracking), §11.4.18 (script documentation), §12.10 (CONTINUATION maintenance). No escape hatch — doc-status drift is a §11.4 PASS-bluff at the documentation layer.

§11.4.24 — Build-resource stats tracking mandate (User mandate, 2026-05-14)

Every project under this Constitution with a build exceeding 1 minute wall-clock MUST run a host-side resource sampler for every build that captures memory used, CPU%, load average, disk read/write at a fixed interval (recommended 5 s) and computes per-metric **min / max / mean / p95** at stop. Per-build summaries appended to a TSV registry; the registry is the single source of truth — the human-readable Markdown report (and its HTML + PDF exports per §11.4.12) is derived. Top of the report MUST surface **ever-values** (min / max / mean across all tracked builds). Per-build entries sorted most-recent-first; each row carries SUCCESS / FAIL / UNKNOWN + reason for FAIL. Sampler MUST itself stay under 50 MB RSS and 5% CPU (Heisenberg-class observer constraint). Stop hook MUST be called from both success AND failure paths of the build wrapper. The Stats. {md,html,pdf} triple is committed via the project's §11.4.22 lightweight doc-sync wrapper. Gate CM-BUILD-RESOURCE-STATS-TRACKER + paired mutation hiding the monitor aside → gate FAILs. Classification: mixed (per §11.4.17) — the discipline universal, the implementation paths project-specific. Composes with §11.4.12 / §11.4.18 / §11.4.22 / §12.6 / §12.7 / §12.9 (the host-safety forensic anchors are the empirical motivation for this telemetry). No escape hatch — build-resource debugging without time-series data is the bluff this anchor forbids.

§11.4.25 — Full-Automation-Coverage Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered! Nothing less than this does not give us a chance to deliver stable product!"

For every consuming project, no feature / functionality / flow / use case / edge case / service / application on any supported platform may be considered **deliverable** until it is covered by automation tests proving six invariants: (1) anti-bluff posture (captured runtime evidence per §7.1 + §11.4); (2) proof of working

capability end-to-end on the target topology (per §11.4.3, not in a mock); (3) working implementation matching the documented promise; (4) no open issues / bugs surfaced by the suite (cross-checked against §11.4.15 / §11.4.16 trackers); (5) full documentation (user manual entry + §11.4.18 for scripts) kept in sync per §11.4.12; (6) four-layer test floor per §1 (pre-build + post-build

- runtime + paired mutation). Consuming projects **MUST** publish a coverage ledger (feature × platform × invariant-1..6 × status), regenerated as part of release-gate sweeps, with gaps tracked per §11.4.15. Classification: universal (§11.4.17). No escape hatch. A project that ships a feature without all six invariants is **not delivering a stable product** — severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. See Constitution §11.4.25 for the full mandate (cross-cutting reach, composition, audit requirements).

§11.4.26 — Constitution-Submodule Update Workflow Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every time we add something into our root (constitution Submodule) Constitution, CLAUDE.MD and AGENTS.MD we **MUST FIRST** fetch and pull all new changes / work from constitution Submodule first! All changes we apply **MUST BE** committed and pushed to all constitution Submodule upstreams! In case of conflict, IT **MUST BE** carefully resolved! Nothing can be broken, made faulty, corrupted or unusable! After merging full validation and verification **MUST BE** done!"

Before ANY modification to constitution/Constitution.md, constitution/CLAUDE.md, or constitution/AGENTS.md, the agent or operator **MUST** execute the following pipeline in order:

1. **Fetch + pull first** — inside the constitution submodule worktree run `git fetch` against every remote, then `git pull --ff-only` (or `--rebase` if non-FF-mergeable; never `--strategy=ours` / `--allow-unrelated-histories` without explicit authorization). The submodule **MUST** be at upstream tip **BEFORE** any local edit.
2. **Apply the change** — classify per §11.4.17 (only universal additions belong here; project-specific clauses stay in the consuming project's governance). Cite the verbatim user mandate if one originated the change.
3. **Validate before commit** — run `meta_test_inheritance.sh` (or equivalent); verify no merge-conflict markers (`<<<<<<<`, `=====`, `>>>>>>>`); verify Constitution + CLAUDE + AGENTS cross-reference the new clause consistently.

4. **Commit + push to ALL upstreams** — stage only the governance files (NEVER `git add -A` inside the submodule); commit message cites the user mandate + §11.4.17 classification; push to every configured remote. A commit landing on one upstream but not others is a §2.1 violation AND a §11.4.26 violation.
5. **Conflict resolution** — if `pull --ff-only` reports non-fast-forward, merge carefully (preserve union of governance content, no clause silently dropped, re-classify, re-validate). Force-push to "make conflicts go away" is FORBIDDEN (§9.2). Nothing about the constitution may be broken, made faulty, corrupted, or rendered unusable.
6. **Post-merge validation + verification** — after the push lands, `git submodule update --remote --init` and re-run the consumer project's cascade verifier (per CONST-047) to confirm the new clause reaches every owned submodule. Any cascade gap closed in the same change-window.
7. **Bump consuming project pointer** — `.gitmodules`-tracked submodule pointer MUST be advanced to the new constitution HEAD in the SAME commit as any cascade work. Out-of-sync pointers are §11.4.26 violations.

Classification: universal (§11.4.17). No escape hatch. A constitution-submodule change violating §11.4.26 is a release blocker for every consuming project, severity-equivalent to a force-push without §9.2 authorization. See Constitution §11.4.26 for the full mandate (operational scope, cross-cutting reach).

§11.4.31 — Submodule-Dependency-Manifest Mandate (User mandate, 2026-05-15)

Every owned-by-us submodule MUST ship a machine-readable dependency manifest at canonical path `helix-deps.yaml` (or `.json/.toml`) listing its own-org Git SSH dependencies. Schema (per Constitution §11.4.31): `schema_version`, `deps: [{name, ssh_url, ref, why, layout: flat|grouped}]`, `transitive_handling.recursive: true`, `transitive_handling.conflict_resolution: operator-required`, `language_specific_subtree: bool`.

Tooling: `incorporate-submodule <ssh-url>` adds the submodule at its declared canonical path (CONST-051(C) `flat/grouped`), reads its `helix-deps.yaml`, recurses for each declared dep, aborts on conflicting refs, emits `<root>/helix-manifest.yaml` audit record.

Anti-bluff guarantee: every manifest paired with a Challenge that bootstraps a throwaway consuming project, runs `incorporate-submodule`, asserts produced layout matches manifest, runs the submodule's own tests against the bootstrapped layout, captures wire evidence per §11.4.2. A manifest without this proof is a §11.4.31 violation.

§11.4.31 is the operational complement of §11.4.28 / CONST-051(C): nested own-org submodule chains are FORBIDDEN, manifests are the bridge that lets consumers reconstruct the dependency graph at the parent root.

Classification: universal (§11.4.17). Composes with §1, §3, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, §11.4.29, §11.4.30, CONST-047. See Constitution §11.4.31 for the full mandate.

§11.4.32 — Post-Constitution-Pull Validation Mandate (User mandate, 2026-05-15)

Whenever a project's constitution submodule is fetched + pulled with any content change, the project **MUST** run a full-project + recursive-submodule validation sweep **BEFORE** the new constitution HEAD is treated as canonical for any other work.

Sweep contract (canonical script: `scripts/verify-all-constitution-rules.sh`): re-runs the governance- cascade verifier; for every rule with a programmatic gate (CONST-053 `.gitignore` audit, CONST-051(C) nested-own-org-chain audit, CONST-052 case audit, CONST-050(A) mock-from-production audit, CONST-035 anti-bluff smoke), runs the gate against post-pull tree. Failures produce directed FAIL entries → tracker per §11.4.15 with Status: Reopened, Type: Bug. Closure requires positive-evidence per §11.4 anti-bluff covenant.

Pull-time invocation: `git submodule update --remote constitution` triggers the sweep automatically (post-update hook or commit wrapper); operator-explicit manual invocation also available.

Anti-bluff: sweep's own meta-test (paired mutation §1.1) plants a known violation of each enforced gate and asserts sweep reports FAIL for the planted gate. A sweep that exits PASS without running every implementable gate is a §11.4.32 violation.

§11.4.32 is the **enforcement engine** for every other §11.4.x and CONST-NNN rule — without it, new rules cascade as anchors but never get enforced in the codebase.

Classification: universal (§11.4.17). Composes with every rule that has a programmatic gate. See Constitution §11.4.32 for the full mandate.

§11.4.30 — .gitignore + No-Versioned-Build-Artifacts Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"every project module, every Submodule, every service and application MUST HAVE proper .gitignore file! We MUST NOT git version build artifacts, cache files, tmp files, main .env file(s) or any files containing sensitive data, API keys or token! Any build derivative which we can recreate by executing proper mechanism for generating MUST NOT be versioned! We MUST pay attention what is going to be committed every time we are preparing to execute commit! If any violation is detected it MUST be fixed before commit is executed!"

Every project module / owned-by-us submodule / service / application MUST ship a proper .gitignore covering the forbidden-from-version-control classes:

1. **Build artefacts:** /bin/, /build/, /dist/, /out/, target/, *.exe, *.dll, *.so, *.dylib, *.a, *.o, *.class, *.pyc, generator-produced files (when the generator is committed).
2. **Cache files:** __pycache__/, .pytest_cache/, .mypy_cache/, .ruff_cache/, node_modules/, .next/, .nuxt/, .cache/, .gradle/, .terraform/, language-server caches.
3. **Temp files:** *.tmp, *.swp, *~, .DS_Store, Thumbs.db, *.orig, *.rej.
4. **Sensitive-data files:** .env, .env.* (allow .env.example placeholder), *.pem, *.key, *.crt, id_rsa*, id_ed25519*, .netrc, secrets/, api_keys.sh.
5. **Generated reports/logs:** *.log, coverage.out, htmlcov/, runtime captures unless reference assets.
6. **OS/IDE personal state:** .idea/, .vscode/ (except shared settings), .history/.

Anti-bluff invariant: .gitignore line alone is not sufficient — no file matching the forbidden patterns may be currently tracked. A tracked *.log despite the ignore-line is a violation of equal severity to no ignore-line at all.

Pre-commit attention: every commit author (human OR agent) MUST inspect `git diff --staged + git status` BEFORE the commit. Forbidden-class hits abort the commit until fixed (un-stage, add to .gitignore, scrub if already-tracked). Gate CM-GITIGNORE-PRECOMMIT-AUDIT + paired mutation.

Secret-leak intersection: §11.4.30 composes tightly with §11.4.10

- §12.1 (CONST-042) — a .env leak is BOTH a §11.4.30 and a §11.4.10 violation, requiring rotation + post-mortem.

Recreatable-content test: if a documented mechanism regenerates the file from sources, it's a build derivative and MUST be ignored. Generators MUST be committed so consumers regenerate on demand.

Classification: universal (§11.4.17). No escape hatch beyond enumerated exceptions. Severity-equivalent to §11.4 PASS-bluff at the repository-hygiene layer. Composes with §1, §2, §9.1, §11.4.10, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, §11.4.29, CONST-047. See Constitution §11.4.30 for the full mandate.

§11.4.29 — Lowercase-Snake_Case-Naming Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules MSUT HAVE lowercase names with space separator between the words of ' ' character (snake-case)! All existing Submodules and directories which are not following this rule MUST BE renamed! ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!"

Every directory, submodule, and file MUST use lowercase snake_case names (ASCII letters / digits / underscores, words separated by _). Existing non-compliant names MUST be renamed as part of the migration window opened by this clause. Every reference (configs, docs, links, source-code imports, governance files) MUST be updated atomically with the rename — reference drift after a rename is a §11.4.29 violation of equal severity to the rename itself.

Exceptions (common-sense, must-not-break-technology).

Language- mandated case (Java/Kotlin package paths, Android resource directories, Apple framework dirs, C#/Swift project layouts) is preserved inside the language-root. Submodule root directory follows our convention; language-specific subtree follows its own. Vendor/upstream third-party submodules keep their upstream names. Build artefacts (node_modules/, __pycache__/, .git/, target/, build/, bin/) keep tool-mandated names. "Does renaming break the technology?" trumps the rule.

Upstreams/ → upstreams/ transition. Constitution submodule's install_upstreams.sh (exported via .bashrc/ .zshrc) MUST support BOTH Upstreams/ and upstreams/ directory layouts during migration. Lowercase wins when both present. Uppercase fallback retires only by deliberate amendment.

Project-Toolkit Upstreamable synchronisation. Upstreamable / Project-Toolkit machinery MUST be fetched+pullled before any rename batch + MUST itself comply with this rule. Lacking BOTH- directory support is a release blocker.

Test coverage of renames. Each rename batch ships with: (i) regression test verifying every reference now resolves; (ii) full CONST-050(B) test-type matrix run on the post-rename tree; (iii) anti-bluff wire-evidence captured. All three or it's a §11.4.29 violation.

Phased execution. Comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks → every change covered by every applicable test type. Phases run in parallel with mainstream work (§11.4.20 subagent delegation).

Classification: universal (§11.4.17). No escape hatch beyond the common-sense exceptions enumerated. Severity-equivalent to §11.4 PASS-bluff at the reference-integrity layer. Composes with §1, §1.1, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, CONST-047. See Constitution §11.4.29 for the full mandate.

§11.4.28 — Submodules-As-Equal-Codebase + Decoupling + Dependency-Layout Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"All existing Submodules in the project that we are controlling and belong to some our organizations (vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — we can ALWAYS check dynamically using GitHub and GitLab CLIs) are equal parts of the project's codebase! We MUST work on that code as much as we do with main project's codebase! All on equal basis! Equally important! ... We MUST NEVER modify Submodules to bring into them any project specific context since they all MUST BE ALWAYS fully decoupled, project not-aware, fully reusable and modular (by any other project(s)), completely testable! All Submodule dependencies that are used by Submodule MUST BE accessed from the root of the project! We MUST NOT have nested Submodule dependencies but accessing each from proper location from the root of the project — directly from project's root project_name/ submodule_name or some more proper structure project_name/submodules/ submodule_name!"

Three cooperating invariants:

(A) Equal-codebase. Every owned-by-us submodule (orgs: vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — dynamically discoverable via gh / glab CLIs) is an **equal part** of the consuming project's codebase. Same engineering attention as main: analysis,

extension, test creation, gap-filling, bug-fix, documentation (user manuals, guides, diagrams, SQL, websites, all materials). A round that improves main while leaving an owned-submodule deficiency unaddressed is a §11.4.28 violation, severity-equivalent to a §11.4 PASS-bluff at the project-scope layer. Coverage ledgers (§11.4.25) list every owned submodule as in-scope.

(B) Decoupling / reusability. Owned submodules **MUST** stay fully decoupled, project-not-aware, reusable, modular, completely testable. **NEVER** inject project-specific context (hardcoded paths, hostnames, asset names) **INTO** a submodule. When a submodule needs parent-project info, use configuration injection (env var, config file, constructor parameter) — never a hardcoded reach.

(C) Dependency-layout. Every dependency consumed by an owned submodule **MUST** be accessible from the parent project's root at:

```
<project_root>/<submodule_name>/  
<project_root>/submodules/<submodule_name>/
```

Nested own-org submodule chains are FORBIDDEN. A submodule **MUST NOT** have its own .gitmodules entries pulling in further owned- by-us repos. Add the dependency at the parent's root path; the submodule reaches it via documented import / SDK / runtime resolver. Third-party submodules exempt.

Gates: CM-OWNED-SUBMODULE-EQUAL-ENGINEERING (release-gate sweep audits parity), CM-OWNED-SUBMODULE-DECOUPLING (pre-commit greps for parent-project context inside the submodule diff), CM-OWNED-SUBMODULE-LAYOUT (pre-merge verifies canonical location + no nested own-org submodules + dependency lookup from root). Paired mutations (§1.1) for all three. Classification: universal (§11.4.17). No escape hatch. Composes with §1, §3, §11.4.17, §11.4.20, §11.4.25, §11.4.26, §11.4.27, CONST-047. See Constitution §11.4.28 for the full mandate.

§11.4.27 — No-Fakes-Beyond-Unit-Tests + 100%-Test-Type-Coverage Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist **ONLY** in Unit tests! All other test types **MUST** interreact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project **MUST BE** 100% covered with every supported test type: unit tests, integration tests, e2e tests, full automation tests, security tests, ddos tests, scaling tests, chaos tests, stress tests, performance tests, benchmarking tests, ui tests, ux tests, Challenges (fully incorporating our

Challenges Submodule). EVERYTHING MUST BE tested using HelixQA (fully incorporating HelixQA Submodule). HelixQA MUST BE used with all possible written tests suites (test banks) for every applications, service, platform, etc and execution of the full HelixQA QA autonomous sessions! All required dependency Submodules MUST BE added into the project as well (fully recursive!!!)."

Two cooperating invariants:

(A) No-fakes-beyond-unit-tests. Mocks, stubs, fakes, placeholders, TODO, FIXME, "for now", "in production this would", or empty-implementation patterns are PERMITTED only in unit-test sources. Every other test type — integration, E2E, full automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA — MUST exercise the real, fully implemented system against real infrastructure. Production code MUST NOT import mock paths. Gate CM-N0-FAKES-BEYOND-UNIT-TESTS + paired mutation.

(B) 100% test-type coverage. Codebase MUST be covered by every supported test type the domain warrants: unit, integration, E2E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges (vasic-digital/Challenges submodule fully incorporated), HelixQA (HelixDevelopment/ HelixQA submodule fully incorporated). HelixQA autonomous sessions drive end-to-end execution of every registered test bank with captured wire evidence per check.

Required dependency submodules (recursive per CONST-047):

- Challenges — `git@github.com:vasic-digital/Challenges.git`
- HelixQA — `git@github.com:HelixDevelopment/HelixQA.git`
- Any other functionality submodules under `vasic-digital/* / HelixDevelopment/*` orgs the project depends on.

Pointers bumped to upstream HEAD in same commit as cascade work (§11.4.26 step 7); pointer drift = §11.4.27 violation.

Classification: universal (§11.4.17). Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. Composes with §1, §7.1, §11.4.1–§11.4.26 (esp. §11.4.25 — this is its strict expansion into per-type-of-test territory). See Constitution §11.4.27 for full mandate.

§11.4.33 — Type-aware closure-status vocabulary (User mandate, 2026-05-15)

Every project that tracks work items by Type per §11.4.16 MUST close them with the Type-appropriate closure-status word, drawn from this 3-element closed map:

Item	**Type:**	Closure	**Status:**	value
Bug		Fixed	(→ Fixed.md)	
Feature		Implemented	(→ Fixed.md)	
Task		Completed	(→ Fixed.md)	

The (→ Fixed.md) suffix is preserved across all three so the existing migration-discipline tooling (atomic Issues.md → Fixed.md move per §11.4.19) keeps working without per-Type branching. Generators (generate_issues_summary.sh, generate_fixed_summary.sh, status-counter helpers, the §11.4.23 colorizer) MUST treat the three terminal values as semantically equivalent (all map to "closed, positive evidence captured") while preserving the literal in the emitted document.

Closing a Feature with Fixed (→ Fixed.md) or a Task with Implemented (→ Fixed.md) is a §11.4.33 violation. Pre-build gate (recommended) CM-CLOSURE-VOCAB-TYPE-AWARE walks every Fixed.md heading + every Issues.md heading whose **Status:** is one of the three terminal values and asserts the Status-Type match. Composes with §11.4.15 (status tracking), §11.4.16 (type tracking), §11.4.19 (Fixed-document column alignment), §11.4.23 (colourisation). Classification: universal (per §11.4.17). No escape hatch.

§11.4.34 — Reopened-source attribution mandate (User mandate, 2026-05-15)

Every Issues.md (or equivalent project tracker) heading whose **Status:** is Reopened MUST carry, within 8 non-blank lines of the heading, a **Reopened-Details:** line capturing four sub-facts:

- **By:** AI or User (source-of-truth observer who flipped the status). AI covers in-loop reopens (test failure, gate regression, captured-evidence retrospect). User covers operator-side observations (manual testing, end-user report, design reconsideration).
- **On:** ISO date (YYYY-MM-DD).
- **Reason:** one-line cause classification — chosen from the closed vocabulary { test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered }. Other values are permitted with explicit Reason: <free text> annotation but the closed list MUST be tried first.
- **Evidence:** path to or short description of the captured artefact justifying the reopen — log file, recording, gate failure ID, operator quote, etc. Reopens without evidence are §11.4.6 / §11.4.7 violations: the reopen IS a demotion-from-Fixed classification change, and demotion requires positive evidence captured under the conditions that re-exposed the defect.

The Issues_Summary.md (or equivalent) Status column MUST distinguish the four Reopened sub-states by source so a sweep query for "reopens by AI in the last 30 days" is mechanically possible. Suggested column rendering: Reopened (AI: test-failed) vs Reopened (User: manual-testing).

A Reopened entry without ****Reopened-Details:**** is a §11.4.34 violation. Pre-build gate (recommended) CM-ITEM-REOPENED-DETAILS mirrors CM-ITEM-OPERATOR-BLOCKED-DETAILS (§11.4.21 walk pattern). Composes with §11.4.6 (no-guessing — Reason from closed vocabulary), §11.4.7 (demotion-evidence — reopen IS a demotion from Fixed), §11.4.15 (item-status tracking), §11.4.21 (Operator-blocked discipline — same audit-line pattern). Classification: universal (per §11.4.17). No escape hatch.

§11.4.35 — Canonical-root inheritance clarity (User mandate, 2026-05-15)

The constitution submodule's three files (constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md) ARE the canonical root — also called the parent files. They contain only universal rules per §11.4.17.

The consuming project's repository-root files (<project-root>/CLAUDE.md, <project-root>/AGENTS.md, optionally <project-root>/Constitution.md or equivalent) are consumer extensions. They open with the inheritance pointer (either the Claude-Code native @constitution/CLAUDE.md import or the portable ## INHERITED FROM constitution/CLAUDE.md heading defined in this file's "How inheritance works" section). They contain only project-specific rules per §11.4.17 — rules that reference particular hardware, vendor names, regulatory regions, internal asset names, or project-private conventions.

When in doubt about which file to edit: universal rule → edit constitution submodule's file; project-specific rule → edit consumer's file. Default consumer-side when uncertain (per §11.4.17, narrower scope is cheap to widen).

Terminology: when prose references "the parent CLAUDE.md" or "the root Constitution," the referent is the constitution-submodule file at constitution/<filename>, never the consumer's file. When it references "the project CLAUDE.md" or "this project's AGENTS.md," the referent is the consumer-side file at <project-root>/<filename>. AI agents resolve ambiguous references via this rule.

No silent demotion or silent promotion. Moving a rule between layers MUST be a visible commit — `git mv` of a section if it's a clean clone, or an explicit "Lifted from to constitution per §11.4.35" / "Demoted from constitution to per §11.4.35" line in the commit message.

Pre-build gate (recommended) CM-CANONICAL-ROOT-CLARITY verifies (a) consumer's CLAUDE.md opens with the inheritance pointer (either `@import` or `## INHERITED FROM constitution/CLAUDE.md` heading), (b) the constitution submodule's three files are present at the expected path, (c) no `## INHERITED FROM` block in the constitution submodule's own files (those ARE the source-of-truth, not consumers). Composes with §11.4.17 (universal-vs-project classification — §11.4.35 defines the file-layer split that §11.4.17 classifies INTO). Reading order: this anchor first, then §11.4.17. Classification: universal (per §11.4.17). No escape hatch.

§11.4.36 — Mandatory `install_upstreams` on clone/add Mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in `.bashrc` or `.zshrc`) using `install_upstreams` command executed from the root of the cloned (added) repository - only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository's upstreams!"

Every clone / add of a Git repository under any consuming project MUST be followed by `install_upstreams` invocation from that repository's root IF its tree contains an `upstreams/` directory (or legacy Upstreams/ per §11.4.29 transition) populated with `*.sh` recipe files declaring upstream Git SSH URLs.

`install_upstreams` is a host-system utility on operator's PATH (exported via `.bashrc/.zshrc`), implemented in this constitution submodule (`install_upstreams.sh`). The utility reads recipe files, configures every declared upstream as a named git remote, and fans out origin push URLs across all declared upstreams.

Skipping the invocation when `upstreams/` IS present silently breaks §2.1 (Multi-upstream push is the norm) — the next push lands on only one upstream. Gate CM-INSTALL-UPSTREAMS-ON-CLONE

- paired mutation (§1.1).

Automation: incorporate-submodule (§11.4.31) and scripts/init-submodules.sh patterns auto-invoke install_upstreams when applicable. Operator-explicit manual invocation remains supported.

Pre-commit attention: before the first commit in the newly-cloned working tree, verify `git remote -v | grep -c push` reports the expected upstream count.

Classification: universal (§11.4.17). Composes with §2, §2.1, §3, §9.2, §11.4.17, §11.4.20, §11.4.28, §11.4.29, §11.4.30, §11.4.31. See Constitution §11.4.36 for the full mandate.

§11.4.37 — Fetch-before-edit mandate (User mandate, 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that `feedback_fetch_before_edit` memory rule is part of our constitution Submodule - the root Consitution, AGENTS.MD and CLAUDE.MD. Validate and verify that Proejct-Toolkit and all Submodules do inherit all of them!"

The FIRST git-touching action of any session, on any consuming project, MUST be:

```
git fetch --all --prune
git log --oneline HEAD..@{u}          # parent
git submodule foreach --recursive 'git fetch --all --prune --
quiet'
```

If `HEAD..@{u}` is non-empty, integrate (ff-merge / rebase / surface to operator per §11.4.4) BEFORE any local edit, scanner run, or test cycle. Multi-agent / multi-upstream codebases (Claude Code + Cursor + Aider + operator sessions in parallel) routinely lap each other; a 30-second fetch prevents the agent from redoing work a parallel session already finished, from filing a false-confidence "completion" of already-done work, and from doubling the multi-upstream conflict surface (§2.1) with sibling commits of the same change.

The check is non-negotiable even when the operator says "do X immediately" — skipping it on the basis of "nothing could have changed in the last N minutes" is a §11.4.6 (no-guessing) violation: remote state is not knowable without a fetch. The fetch+log output (even if empty) is the captured evidence.

Scope: consuming project root + every owned submodule recursively (§11.4.28) + the constitution submodule itself (§11.4.26 step 1 made this explicit for constitution-side edits; §11.4.37 generalises to ALL edits) + any dependency cloned via `incorporate-submodule` (§11.4.31) or `git submodule add` (§11.4.36).

Pre-build gate `CM-FETCH-BEFORE-EDIT-AUDIT` (when implemented in the consuming project) audits the most-recent commit range against the upstream HEAD at the commit's parent — non-aligned parent = FAIL. Paired mutation (§1.1): synthetic commit whose parent was N commits behind the then-current upstream HEAD → gate FAILs.

Classification: universal (§11.4.17). No escape hatch. Composes with §2.1, §11.4.4, §11.4.6, §11.4.20, §11.4.26, §11.4.32. See Constitution §11.4.37 for the full mandate.

§11.4.38 — Installable-Asset Evidence Mandate (User mandate, 2026-05-17)

For any user-distributable build artifact (package, bundle, installer, or container image produced by the build pipeline and distributed to end users), tests and challenges **MUST** open the artifact and verify each user-visible asset is **present** and **non-degenerate**.

A PASS without opening the artifact and verifying the asset chain end-to-end is a §11.4 PASS-bluff. The specific failure mode: source file exists → build packages it → post-build checks pass at the source layer → artifact produced with the asset stripped or misconfigured, and no gate ever opens the artifact to verify.

Each consuming project ships one challenge script per artifact type that opens the produced artifact and verifies every declared user-visible asset. The challenge **MUST** run as part of the project's standard QA gate.

Classification: universal (§11.4.17). No escape hatch. See Constitution §11.4.38 for the full mandate.

§11.4.40 — Full-suite retest before release tag mandate (User mandate, 2026-05-17)

A release tag **MUST NOT** be created until a **COMPLETE** retest with **ALL** existing tests has been executed on a clean baseline **AFTER** every workable item in the batch is done, fixed, polished, and individually verified. Spot-check retests that run only the tests touched by the batch are **FORBIDDEN** — they miss interaction defects between batch fixes and previously-stable code.

The complete retest comprises: (1) pre-build full sweep, (2) post-build full sweep, (3) on-device 4-phase cycle on EVERY owned device, (4) meta-test full mutation sweep, (5) Challenge bank full sweep, (6) Issues.md/Fixed.md state audit, (7) CONTINUATION.md sync check.

Time is essential — typically 12-48h elapsed effort. NOT optional, NOT abbreviated. Skipping is the exact "tests pass but feature broken" failure mode §11.4 prohibits.

Composes with §11.4.4 (per-fix retest still required at fix granularity) + §11.4.7 (full-suite retest is authoritative baseline for closures) + §11.4.39 (per-feature on-device validation runs as step 3 of the full-suite retest).

Classification: universal (§11.4.17). No escape hatch. See Constitution §11.4.40 for the full mandate.

§11.4.41 — Pre-Force-Push Merge-First Mandate (User mandate, 2026-05-17)

Forensic anchor — verbatim user mandate (2026-05-17):

"make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides!"

Any force-push (--force, --force-with-lease, +<ref>, equivalent history-rewrite) authorised under CONST-043 MUST be preceded by a 4-step merge-first pipeline:

1. **Fetch every remote** — git fetch --all --prune --tags against origin + every upstream; capture output.
2. **Integrate every divergent commit locally** — rebase / merge / operator-confirmed cherry-pick per the appropriate strategy for every non-empty HEAD..<remote>/<branch> range.
3. **Audit the integrated tree** — no conflict markers anywhere (grep -rn '^<<<<<< \|^===== \$\|^>>>>>>' returns empty in governance + source + test files); no file silently dropped; previously-passing tests still pass; captured- evidence artefacts still validate.
4. **Force-push** — only after steps 1-3 produce clean integration evidence: git push --force-with-lease (NEVER --force alone unless authorised per §9.2 sub-clause 6).

Two-gate composition with CONST-043. §11.4.41 does NOT relax CONST-043's operator-approval requirement — it adds a SECOND mechanical gate. Both required: CONST-043 alone authorises a push that loses remote work; §11.4.41 alone risks pushing without operator awareness.

Three failure modes prevented: (a) remote-side content loss when parallel sessions land work between fetches; (b) stale-state acts when `--force-with-lease` reads stale local refs; (c) conflict-driven corruption when markers get committed verbatim (observed 2026-05-17 in `helix_qa` + containers governance files).

Verification artefact — `docs/changelogs/<tag>.md` "Force-push merge-first audit" section captures fetch output, per-remote divergence log, integration strategy, conflict- marker scan, test delta, push output with lease SHA, CONST-043 authorisation quote. Gate CM-FORCE-PUSH-MERGE-FIRST + paired mutation.

Classification: universal (§11.4.17). No escape hatch. Composes with §9.2, §11.4.4, §11.4.6, §11.4.26, §11.4.32, §11.4.37, §11.4.40, CONST-043, CONST-047. See Constitution §11.4.41 for the full mandate.

§11.4.42 — Iteration-discipline mandate (User mandate, 2026-05-18)

Project work proceeds in priority-ordered iteration cycles. Each cycle has five mandatory steps: (1) select TOP + MIDDLE critical items only (defer LOW until critical batch closed), (2) batch implementation with §11.4.4 four-layer coverage + §11.4.9 batch-source-fixes-before-rebuild, (3) smoke gate (<30 min — batch-touched tests + critical-path regression probe + anti-bluff baseline), (4) ONLY if smoke GREEN and no new operator/user report arrived, run §11.4.40 full retest (12-48 h), (5) release-ready OR loop back to step 1 with new evidence added to queue.

Composes with §11.4.4 (per-fix retest inside step 2), §11.4.7 (closures require same-conditions evidence; step 4 is authoritative baseline), §11.4.9 (source-side batching inside step 2), §11.4.34 (Reopened items attribute source), §11.4.40 (the multi-hour retest IS step 4 — §11.4.42 is the meta-loop conductor).

Anti-bluff coupling: every smoke and full-system PASS MUST carry positive captured evidence per §11.4.2 + §11.4.5. Tests AND HelixQA Challenges bound equally.

No escape hatch — no `--skip-priority-batch`, no `--skip-smoke`, no `--full-suite-only`. Subagents default to the §11.4.42 path.

Classification: universal (§11.4.17). See Constitution §11.4.42 for the full mandate.

§11.4.43 — TDD-Fix-Discipline mandate (User mandate, 2026-05-18)

Every fix MUST follow the 5-step TDD-fix workflow: **RED** (failing test FIRST, real product defect per §11.4.1, captured evidence per §11.4.2) → **LIVE-ADB-PROBE** (try fix on running device via adb shell for mutable surfaces — setprop persist.*, settings put, pm clear, am ..., boot-script push; INFEASIBLE for kernel / framework / HAL / vendor / init.rc / sepolicy / ro.* / Android.bp — must rebuild; cite LIVE_PROBE_INFEASIBLE: <reason> in commit message) → **GREEN** (source patch achieves same effect, batched per §11.4.9, four-layer coverage per §11.4.4) → **VERIFY** (re-run RED test, must PASS under SAME conditions per §11.4.7, captured positive evidence per §11.4.5, 10-iteration reliability per §11.4.42) → **DOCUMENT** (Issues.md → Fixed.md with type-aware closure vocabulary per §11.4.33, CLAUDE.md Applied Fixes row, changelog, guides, HelixQA bank, CONTINUATION.md per §12.10 — all in the SAME commit).

No escape hatch — no --skip-red-test, --no-live-probe, --skip-verify flag. "Test added after the fix" is a §11.4 PASS-bluff: it demonstrates the test agrees with the fix, not that the test catches the bug.

Classification: universal (§11.4.17). Composes with §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.7 / §11.4.9 / §11.4.40 / §11.4.42. See Constitution §11.4.43 for the full mandate.

§11.4.44 — Document revision header mandate (User mandate, 2026-05-18)

Every tracked document in scope (Issues.md, Issues_Summary.md, Fixed.md, Fixed_Summary.md, CONTINUATION.md, docs/guides/, **docs/research/**, docs/scripts/, **docs/changelogs/**, docs/superpowers/plans/, **docs/hardware/**, all other docs/* tracked Markdown) MUST carry a header block directly below the H1 title containing two MANDATORY fields: ****Revision:**** N (monotonic positive integer, never reset, never skipped) and ****Last modified:**** YYYY-MM-DDTHH:MM:SSZ (ISO 8601 UTC). Optional fields (Description, Authority, Maintainer, Scope) encouraged but not gated. CLAUDE.md / AGENTS.md / README / LICENSE / VERSION / rendered HTML / rendered PDF artifacts are explicitly OUT of scope (revision tracked via VERSION file or auto-derived from source Markdown).

Auto-bump: scripts/doc_revision_bump.sh <file> (idempotent), pre-commit hook for automatic staged-doc bumps, scripts/testing/sync_issues_docs.sh auto-bumps Issues_Summary / Fixed_Summary after regeneration, scripts/commit_docs.sh calls the bump before stage. CONTINUATION.md's existing Last updated: line per §12.10 IS the §11.4.44 Last modified: line —

composed, not duplicated. HTML/PDF exports inherit revision from source Markdown via pandoc pipeline. No escape hatch — no `--skip-revision-bump` flag exists anywhere.

Pre-build gates `CM-DOC-REVISION-HEADER-PRESENT` + `CM-COVENANT-114-44-PROPAGATION` + paired mutations per §1.1. Composes with §12.10 (`CONTINUATION.md` header reuse), §11.4.12 (`Issues_Summary` regen), §11.4.22 (`commit_docs.sh` hook entry), §11.4.23 (HTML colorizer preserves revision), §11.4.18 (companion doc for `doc_revision_bump.sh`).

Classification: universal (§11.4.17). See Constitution §11.4.44 for the full mandate.

§11.4.45 — Integration-status-doc maintenance mandate (User mandate, 2026-05-18)

Every non-trivial domain integration MUST have a `docs/<domain>/<integration>/Status.md` document that (1) exists when more than one `fix/test/gate` has landed for the integration, (2) carries the §11.4.44 revision header, (3) is auto-synced (HTML

- PDF) on every related test cycle and every fix touching the integration, (4) is auto-colored per §11.4.23, (5) has a sync wrapper invocable as `bash scripts/testing/sync_integration_status.sh` or a per-integration thin shell, (6) lives under `docs/<domain>/<integration>/`, (7) includes a captured-evidence- driven status table per §11.4.5 (every claim cites the test log / recording / sink-probe report that backs it), (8) uses the closed status vocabulary `PASS / FAIL / SKIP / PENDING FORENSICS / OPERATOR-BLOCKED`, (9) lists operator-blocked items at the top so operators find action items in $O(1)$, (10) is referenced from `docs/CONTINUATION.md` §3 (Active work) when any item is non- terminal.

§11.4.45 is the generic form of §12.10 (`CONTINUATION.md`) applied to every integration domain. Without this generalisation each new integration re-invents the same sync wrapper, revision-header discipline, captured-evidence requirement, and operator-blocked surface.

Pre-build gates `CM-COVENANT-114-45-PROPAGATION` + `CM-AF-INTEGRATION-STATUS-DOCS` + paired mutations (propagation strip / Revision-line delete / sync-staleness / vocabulary violation). Composes with §11.4.5 (captured evidence), §11.4.12 (sync wrapper pattern reused), §11.4.13 (sink-side evidence is a specific instance), §11.4.15 (status vocabulary), §11.4.22 (`commit_docs.sh` wrapper), §11.4.23 (colorizer), §11.4.44 (revision header), §12.10 (`CONTINUATION.md` references `Status.md` paths).

No escape hatch — no `--skip-status-sync`, no `--no-revision-bump-on-status`, no `--allow-stale-html` flag.

Classification: universal (§11.4.17). See Constitution §11.4.45 for the full mandate.

§11.4.46 — Validate-recent-work-before-post-flash-tests mandate (User mandate, 2026-05-18)

After every device flash, the orchestrator MUST first run a recent-work validation pass (targeted on-device tests for items currently in Issues.md In progress / Ready for testing / Reopened, Fixed.md items closed within last 7 days, CONTINUATION.md §3 active work). Only if that pass is 100% green does the orchestrator proceed to the full post-flash suite.

Helper: `scripts/testing/recent_work_validate.sh --device <serial>` writes `/data/local/tmp/.recent_work_validated` IFF green; the full suite (`test_all_fixes.sh`) refuses to start without it. Marker is invalidated on reboot (stores device boot epoch).

Composes with §11.4.4 (STOP-on-discovery) + §11.4.6 (no-guessing) + §11.4.7 (demotion-evidence) + §11.4.40 (full-suite gate) + §11.4.42

- §11.4.43 + §11.4.44 + §12.10. Each recent-item fix MUST have a paired §11.4.43 RED-then-GREEN — a GREEN with no prior RED is a bluff.

Pre-build gates CM-COVENANT-114-46-PROPAGATION + CM-AF-RECENT-WORK-VALIDATION-GATE + CM-AF-VALIDATION-ARTIFACT-FILE + paired mutations.

Classification: universal (§11.4.17). No escape hatch — no `--skip-validation`, `--full-suite-always`, `--ignore-recent-work` flag. See Constitution §11.4.46 for the full mandate.

§11.4.47 — Firebase data review mandate (User mandate, 2026-05-18)

Before every "bigger working round" (pre-build, pre-flash, pre-tag, daily, post-deployment burn-in) the operator/loop MUST execute `scripts/firebase/review_round.sh`. The pass queries Crashlytics (fatals + non-fatals + ANRs) + Analytics + Performance, classifies findings by severity, dedup-maps to existing Issues.md entries via a three-tier algorithm (exact Firebase Issue-ID match → stacktrace- similarity cluster hash → operator merge review), and drafts new Issue entries for unrecognised findings with full

Firestore Console URL refs + §11.4.4(a) systematic-debugging output. Skipping the pass is a §11.4 PASS-bluff — Firestore IS the captured evidence from real end-user devices.

Five mandatory elements: (1) 5-trigger cadence (pre-build / pre-flash / pre-tag blocking, daily / burn-in non-blocking), (2) 3-source query (all three sources), (3) Issues.md output with Firestore metadata (Issue IDs + URL + Cluster Hash / KPI / Funnel), (4) 3-tier dedup, (5) comprehensive systematic-debugging output per Issue.

Pre-build gates CM-COVENANT-114-47-PROPAGATION + CM-AF-FIREBASE-REVIEW-CADENCE + CM-AF-FIREBASE-ISSUE-XREF + 3 paired mutations. Composes with §11.4.4 / §11.4.4(a) / §11.4.6 / §11.4.7 / §11.4.10 / §11.4.12 / §11.4.14 / §11.4.15 / §11.4.16 / §11.4.34 / §11.4.42 / §11.4.43 / §11.4.44 / §11.4.45 / §11.4.46.

No escape hatch — no --skip-firebase-review, --firebase-review-not-applicable, --no-issue-from-firebase flag. Operator MAY filter with --severity-min but MUST execute the pass.

Classification: universal (§11.4.17). See Constitution §11.4.47 for the full mandate.

§11.4.48 — UI-driven video testing mandate (User mandate, 2026-05-18)

Every test that asserts video playback on a secondary display MUST traverse the user-equivalent UI path (launcher icon → app home → content list → tile tap → playback → in-app pause/resume → back button stop). NOT Intent/Broadcast shortcuts (am start -a VIEW, cmd media_session play). Real uiautomator dump element resolution

- input tap X Y. Coverage: every video-capable app in PRODUCT_PACKAGES + every stream type (progressive HTTP / HLS / DASH / RTMP / file-local / DRM) + every codec (H.264/265/VP9/AV1/MPEG-2/MP4V video; AC-3/E-AC-3/TrueHD/DTS/DTS-HD/MLP/Opus/AAC audio). Secondary display verified via ffprobe-on-captured-mp4 + VOM activeDecoder state. Arvus codec-state cross-check per §11.4.13 + §CG screenshot.

Per §11.4.4 four-layer: pre-build gate CM-AF-UI-DRIVEN-VIDEO-COVERAGE + propagation gate CM-COVENANT-114-48-PROPAGATION + on-device test framework at device/rockchip/rk3588/tests/ui_driven/ (Layer 1 helper + Layer 2 per-app drivers + Layer 3 scenarios) + Layer 4 orchestrator scripts/testing/run_ui_driven_video_suite.sh + paired meta-test mutations.

Carve-out (User mandate 2026-05-20). The 5 canonical tracker documents — docs/Issues.md, docs/Issues_Summary.md, docs/Fixed.md, docs/Fixed_Summary.md, docs/CONTINUATION.md — sit at docs/

root by design. They are architectural constants of the project layout, analogous to AOSP's Makefile, Android.bp, OWNERS files at repo root. Their location is encoded as literal path strings in §11.4.12 + §11.4.15 + §11.4.16 + §11.4.19 + §11.4.44 + §11.4.53 propagation gates plus the helper-script constellation that regenerates them. Moving them would require coordinated amendment of those 6 sister anchors plus 5 pre-build gates plus ~20 helper scripts plus 42 consumer files in a single PWU. Per §11.4.66, that scope is operator-blocked until explicitly authorised. Audit-snapshot files (docs/audit/anti_bluff_audit.md, docs/audit/PRE_SONOS_TAG_READINESS.md, docs/audit/D1_WIFI_FAIL_CLASSIFICATION.md, plus any future audit snapshots) DO move under docs/audit/ per the §11.4.11 general principle.

Canonical authority: constitution submodule Constitution.md §11.4.48.

Non-compliance is a release blocker regardless of context.

§11.4.49 — Dual-approach testing mandate (User mandate, 2026-05-18)

Every feature test exercising a user-visible behaviour MUST ship in TWO variants: a UI-driven variant (uiautomator-based, §11.4.48 surfaces A-E) AND an Intent/Broadcast-driven variant (am start --es / am broadcast-based). Either alone is a §11.4 PASS-bluff for the OPPOSITE half of the stack — UI catches app-side bugs; Intent catches framework/system-server bugs.

Shared assertion base at tests/lib/dual_approach_test_base.sh:
dat_init / dat_start_capture / dat_assert_codec_state /
dat_assert_video_frames / dat_assert_audio_channels /
dat_arvus_dashboard_capture / dat_cleanup / dat_report_finding.
Both variants gather identical evidence into mirror directories qa-results/dual_approach/<F>/<run-ts>/{ui,intent}/ so the orchestrator diffs results and pinpoints which half of the stack contains a bug.

Kinopoisk 5.1 EAC3 is the canonical first implementation. Both variants are RED per §11.4.43 until the §CN decoder pipeline fix lands.

Pre-build gates: CM-COVENANT-114-49-PROPAGATION (anchor across parent + 42 consumer files) + CM-AF-DUAL-APPROACH-COVERAGE (shared base contract) + CM-AF-KINOPOISK-5-1-DUAL-COVERAGE (both variants exist + share the base). Three paired meta-test mutations.

No escape hatch — no --ui-only / --intent-only / --skip- dual flag.

Canonical authority: constitution submodule Constitution.md §11.4.49.

Non-compliance is a release blocker regardless of context.

§11.4.50 — Deterministic consistency mandate (User mandate, 2026-05-18)

Every test that PASSes MUST have been executed N times (default N=3 normal tests, N=10 cycle-validation suites) against the same firmware MD5 + same device + same topology and produced IDENTICAL PASS in every iteration. A divergent N-iter run is auto-FAIL — there is no "first PASSEd therefore X was a flake" path. §11.4.7's intermittent / transient / flake / flaky vocabulary is enforced MECHANICALLY by this mandate, not merely textually.

Coverage: every public API path (Activity / Service / Broadcast receiver / ContentProvider URI / IPC interface / JNI entry / sysprop write / sysfs node / init.rc trigger) MUST have ≥ 1 dedicated test that drives it. Untested paths surface in a feature-coverage-matrix audit; the threshold ratchets 70 $\rightarrow$ 85 $\rightarrow$ 95 $\rightarrow$ 99 over phases.

Reliability mechanism: `ab_run_n_times <test_name> <N> <fn> [args...]` helper in the project anti-bluff library loops, captures evidence-hash per iter, asserts all N hashes + exit codes identical. NO operator-facing escape converts divergence to PASS.

Pre-build gates: CM-COVENANT-114-50-PROPAGATION + CM-AF-RELIABILITY-CHECK-WIRED + CM-AF-FEATURE-COVERAGE-MATRIX. Three paired meta-test mutations. No escape hatch — no `--allow-flake`, `--first-pass-suffices`, `--skip-n-iter`, `--skip-coverage-audit` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.50.

Non-compliance is a release blocker regardless of context.

§11.4.51 — Live-ADB-First Maximization Mandate (User mandate, 2026-05-18)

Every fix MUST be classified by rebuild-requirement before commit using the project's per-file-class decision matrix. If `LIVE_ADB_TESTABLE` (on-device test scripts, host scripts, atmosphere-*.sh boot scripts, persist.* properties, markdown docs, test fixture assets), the operator MUST first apply the fix to the running device via `adb push / setprop / pm install -r / mount -o remount, rw`, run the §11.4.43 RED test live, capture PASS, THEN commit + rebuild + reflash as belt-and-suspenders re-validation. Commit footer: `LIVE_ADB_VALIDATED: yes`. If `REQUIRES_REBUILD` (kernel, framework Java/AIDL, native C++ in APEX, sepolicy, init.rc, ro.* properties, XML overlays, codec XML in APEX, Android.bp/.mk), the operator proceeds directly to source-side + rebuild. Commit footer: `REQUIRES_REBUILD: <reason>`. Mixed batches use partial.

§11.4.51 REFINES §11.4.43 step 2 with mechanical enforcement. Helper: `scripts/testing/classify_fix_rebuild_requirement.sh` walks `git diff --name-only`, looks up each file against the matrix, emits per-file classification + recommended commit-message footer. Unmatched paths classify as `REQUIRES_REBUILD: unmatched-path` (safe default per §11.4.6). Pre-build gates: `CM-COVENANT-114-51-PROPAGATION` + `CM-AF-CLASSIFY-FIX-HELPER-EXISTS`

- `CM-AF-LIVE-ADB-FIRST-COMMIT-MARKER`. Three paired meta-test mutations. No escape hatch — no `--skip-classify` / `--assume-rebuild` / `--no-footer-required` flag.

Canonical authority: constitution submodule [`Constitution.md`](#) §11.4.51.

Non-compliance is a release blocker regardless of context.

§11.4.52 — Autonomous-Validation Mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18):

"Make sure we have full automation tests which will do all this work in full automation! IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Every user-facing feature MUST have at least one autonomous validation path: end-to-end via `adb shell` + scripted automation, captured runtime evidence per §11.4.5, PASS/FAIL verdict WITHOUT a human present to drive UI, observe screen, or make decisions. Operator-attended tests are SUPPLEMENTARY, never PRIMARY. A feature whose ONLY validation path is operator-attended is a §11.4.52 violation — the path does not scale to CI, does not run on every commit, does not survive operator unavailability, and produces the exact "tests pass but feature doesn't work for users" failure mode §11.4 forbids.

Acceptable autonomous paths: programmatic instrumentation APK (SDK-API exercises like `MediaCodec.createDecoderByName` + JSON result file), headless intent dispatch + state poll (`am start --es`

- `dumpsys / /proc/<pid>/maps / media.metrics` polling), ADB-driven `uiautomator` (ONLY if `uiautomator dump | grep -c clickable=true ≥ 1` — near-empty hierarchy proves UI-driven INFEASIBLE and demands fallback to APK/intent), network-side sink probe

(Arvus dashboard, Sonos REST, etc. per §11.4.13), HelixQA autonomous QA session (§11.4.27).

Per-feature coverage ledger (§11.4.25) MUST classify each row as AUTONOMOUS_VERIFIED / AUTONOMOUS_DESIGNED / OPERATOR_ATTENDED_ONLY / NOT_APPLICABLE. OPERATOR_ATTENDED_ONLY is a release blocker until promoted, citing a tracked migration work item per §11.4.15 + §11.4.16. Autonomous paths themselves MUST be anti-bluff: positive captured evidence per §11.4.5, paired meta-test mutation per §1.1.

Composes with §11.4.25 (full-automation-coverage — §11.4.52 strictens invariants 1+2), §11.4.27 (no-fakes-beyond-unit + 100% type coverage — operational layer closing the gap), §11.4.39 (per-feature on-device end-user validation — refined to mandate autonomous drivability), §11.4.43 (TDD-fix — autonomous path RED before fix, GREEN after), §11.4.48 (UI-driven — fallback to APK/intent when uiautomator hierarchy empty), §11.4.49 (dual-approach — Intent variant often IS the autonomous path), §11.4.50 (deterministic consistency — autonomous paths scale to N iterations), §11.4.51 (live-ADB-first — instrumentation APK is LIVE_ADB_TESTABLE).

Pre-build gates CM-COVENANT-114-52-PROPAGATION (anchor literal across canonical files) + CM-AF-AUTONOMOUS-PATH-PER-FEATURE (coverage-ledger classification column non-empty + valid value). Paired mutations strip the anchor literal AND inject an OPERATOR_ATTENDED_ONLY row without a tracked migration item. No escape hatch — no --allow-operator-attended-only, --skip-autonomous-path, --manual-validation-suffices flag.

Canonical authority: constitution submodule Constitution.md §11.4.52.

Non-compliance is a release blocker regardless of context.

§11.4.53 — Fixed_Summary parity mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18T17:55Z):

"Note: Just like for Issues we have Issues_Summary, for Fixed we MUST HAVE Fixed_Summary - like all other docs: ALWAYS in sync and up to date and ALWAYS exported into the PDF and HTML! Add this mandatory rule / constraint into the root (constitution Submodule) Constitution, AGENTS.MD and CLAUDE.MD."

docs/Fixed_Summary.md is the symmetric short-form summary of docs/Fixed.md. MUST be regenerated whenever Fixed.md changes. HTML + PDF exports MUST travel with the markdown (identical mtimes within sync_issues_docs.sh granularity). Stale exports are

§11.4.53 violations regardless of whether the underlying .md is correct — an operator (or future agent) reading the HTML or PDF gets a divergent view of which items are closed, and the §12.10 CONTINUATION resumption guarantee silently breaks. Same discipline as §11.4.12 Issues_Summary applied to Fixed.md.

Generator: scripts/testing/generate_fixed_summary.sh (canonical, MUST be executable, MUST emit a markdown table whose header columns include Status and Type per §11.4.19 column-alignment). Auto- sync wrapper: scripts/testing/sync_issues_docs.sh regenerates BOTH Issues_Summary AND Fixed_Summary in one shot (stages 1a + 1b), then exports HTML + PDF (stage 2), then colorizes per §11.4.23 (stage 3), then re-renders the PDFs from the colorized HTML (stage 3b). MUST be invoked after any edit to Fixed.md. No --issues-only flag exists, and §11.4.53 prohibits adding one.

Sort order: closure date DESC (most-recent-Fixed first), §-letter / Fix-# secondary. Documented at the top of the generated file.

Composes with §11.4.12 (Issues_Summary parity is the sibling rule; §11.4.12 + §11.4.53 are the canonical pair), §11.4.19 (atomic Issues→Fixed migration triggers Fixed_Summary regen — column-aligned structure), §11.4.23 (visual-cue + grouping colorizer post-processes both summaries), §11.4.33 (type-aware closure vocabulary — Fixed_Summary respects Fixed (→ Fixed.md) / Implemented (→ Fixed.md) / Completed (→ Fixed.md) terminal values literally), §11.4.44 (revision header applies to Fixed_Summary.md), §12.10 (CONTINUATION.md resumption guarantee depends on divergent-summary problem NOT existing).

Pre-build gates CM-FIXED-SUMMARY-SYNC (6 invariants: Fixed_Summary exists; HTML + PDF mtime ≥ md mtime; Fixed_Summary mtime ≥ Fixed mtime; generator script exists + executable; sync wrapper invokes generator) + CM-COVENANT-114-53-PROPAGATION (anchor literal across canonical files + per-consumer propagation). Paired mutations strip the anchor literal AND move the generator aside AND backdate Fixed_Summary mtime. No escape hatch — no --skip-fixed-summary-sync, --issues-only, --summary-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.53.

Non-compliance is a release blocker regardless of context.

§11.4.54 — ATM-NNN ticket identifier mandate (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every workable item in Issues.md / Issues_Summary.md / Fixed.md / Fixed_Summary.md MUST carry a stable, unique, auto-incremental ATM-NNN ticket identifier. ATM- prefix, monotonic, never renumbered, append-only."

Every workable item in docs/Issues.md AND docs/Fixed.md MUST carry a [ATM-NNN] ticket identifier in its heading (form ## \$X.Y. [ATM-NNN] <title>, zero-padded ≥ 3 digits). Identifiers are allocated by scripts/testing/assign_atm_ticket_ids.sh and persisted to an append-only state file scripts/testing/.atm_ticket_state.json (jsonl: atm_id, heading_hash, type, current_location, current_status, reopens_count, created_at, last_modified). Once assigned, an ATM-NNN MUST NEVER be renumbered, reused, decremented, or skipped. heading_hash is the binding key — wording reflows preserve the binding via the state-file lookup.

Issues_Summary.md and Fixed_Summary.md MUST carry an ATM ID column as the leftmost data column. Generators (generate_issues_summary.sh, generate_fixed_summary.sh) emit the column so operators / agents can sort + filter on it.

Composes with §11.4.15 (Status), §11.4.16 (Type), §11.4.19 (column-alignment), §11.4.33 (closure vocabulary), §11.4.12 + §11.4.53 (Issues_Summary + Fixed_Summary regen — helper invoked from sync_issues_docs.sh), §11.4.55 (per-item Reopens.md path key), §11.4.57 (README.md doc-link cross-reference key).

Pre-build gates CM-ATM-TICKET-IDS-COMPLETE (every heading carries [ATM-NNN]) + CM-ATM-TICKET-IDS-UNIQUE (no duplicates) + CM-ATM-TICKET-IDS-MONOTONIC (no gaps) + CM-COVENANT-114-54-PROPAGATION (anchor literal across canonical files). Paired mutations strip an [ATM-NNN] heading token, dup an ID in the state file, gap the sequence at NNN=2, strip the anchor literal. No escape hatch — no --skip-atm-assignment, --renumber, --no-atm-id-required flag.

Canonical authority: constitution submodule Constitution.md §11.4.54.

Non-compliance is a release blocker regardless of context.

§11.4.55 — Reopens-history tracking + per-item Reopens.md doc (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a Reopens-count column. For any item whose reopens-count > 0, create docs/issues/ATM-NNN/Reopens.md (+ HTML + PDF) with comprehensive reopen history + Fixed cycles. Each reopen MUST include By (AI / User), On, Reason, Evidence, and each Fixed-marking with reasoning chain."

Every workable item with `reopens_count > 0` MUST have a companion document at `docs/issues/<ATM-NNN>/Reopens.md` (+ HTML + PDF). The document MUST contain: (1) §11.4.44 revision header, (2) item identification (ATM ID, Title, Type, Current Status, Current Location, link back to live heading), (3) cycle counters (Total reopens, Total fixed cycles), (4) chronological timeline — one entry per state-change event, each with By (AI/ User per §11.4.34), On (ISO date), Event (Opened/Reopened/ Fixed/Implemented/ Completed), Reason (closed-vocabulary value from §11.4.34 for reopens; captured-evidence summary for closures), Evidence (path or short description), Outcome, (5) reasoning chain for each closure (root-cause analysis, captured-evidence under same conditions per §11.4.7, gate / mutation pair), (6) most-recent state-change pointer.

`Issues_Summary.md` and `Fixed_Summary.md` MUST carry a `Reopens` column; cells with count > 0 hyperlink to the per-item `Reopens.md`. The §11.4.23 colorizer MAY apply a visual cue when `reopens > 2`.

Composes with §11.4.34 (per-event capture in current heading — §11.4.55 is the per-item history aggregation), §11.4.54 (ATM-NNN provides the stable path), §11.4.44 (revision header), §11.4.45 (`Status.md` per-integration analogue), §11.4.53 (`Fixed_Summary` parity — `Reopens` column symmetric on both summaries).

Pre-build gates `CM-REOPENS-DOC-EXISTS-WHEN-COUNT-GT-ZERO` + `CM-REOPENS-DOC-REVISION-HEADER` + `CM-REOPENS-COL-IN-SUMMARIES` + `CM-COVENANT-114-55-PROPAGATION`. Paired mutations delete a `Reopens.md` for a `reopens_count=2` item, strip the revision header, remove the column from `Issues_Summary`, strip the anchor literal. No escape hatch — no `--skip-reopens-doc`, `--collapse-history`, `--reopens-not-applicable` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.55.

Non-compliance is a release blocker regardless of context.

§11.4.56 — `Status_Summary` parity + two-audience format (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every `Status.md` doc gets a `Status_Summary` parity companion ALWAYS in sync, exported to HTML + PDF. Two-page format: page 1 = non-developer audience (team-specific), page 2 = software engineer summary. Auto-generated after every main Status update."

For every `docs/<domain>/<integration>/Status.md` (§11.4.45) a companion `Status_Summary.md` MUST exist with: (1) §11.4.44 revision header, (2) **Page 1 — For the** — audience- specific

heading (audio team for docs/dolby/*, video team for docs/video/*, etc.) — plain-language summary, What works (1-3 bullets), What's broken or pending (1-3 bullets), Operator / team actions if any. NO code references, NO §-letter jargon, NO captured-evidence file paths, NO gate / mutation names. (3) **Page 2 — For software engineers** — §-letter references, gate names, commit hashes, captured-evidence paths, ATM-NNN cross-references. HTML + PDF exports travel with the markdown.

Generator: scripts/testing/generate_status_summary.sh <Status.md path> produces both pages. Invoked from scripts/testing/sync_integration_status.sh (§11.4.45 sync wrapper) on every Status.md update.

Composes with §11.4.45 (Status.md per integration — Status_Summary.md COMPLEMENTS, never replaces), §11.4.12 + §11.4.53 (parity discipline), §11.4.44 (revision header), §11.4.23 (colorizer for tracked-item references on page 2), §12.10 (CONTINUATION.md — non-developer stakeholders read Status_Summary; engineers read Status + CONTINUATION + Issues).

Pre-build gates CM-STATUS-SUMMARY-EXISTS-FOR-EVERY-STATUS + CM-STATUS-SUMMARY-TWO-AUDIENCE + CM-STATUS-SUMMARY-REVISION-HEADER + CM-COVENANT-114-56-PROPAGATION. Paired mutations delete a Status_Summary.md, remove the Page 1 heading, strip the anchor literal. No escape hatch — no --skip-status-summary, --engineer-only, --no-audience-split flag.

Canonical authority: constitution submodule Constitution.md §11.4.56.

Non-compliance is a release blocker regardless of context.

§11.4.57 — README.md doc-link section + revision metadata (User mandate, 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a doc-link section to README.md — links to Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + ALL Status docs + their exports. Each link shows revision + last-modified."

Every project's top-level README.md MUST contain a section titled Tracked-Items + Status Documents (heading MUST contain literal Tracked-Items). The section is a markdown table with columns: Document, Last modified (ISO 8601 UTC from §11.4.44 header), Revision (integer from same header), Markdown link, HTML link, PDF link. The section MUST link to: Issues.md + Issues_Summary.md (§11.4.12, §11.4.15, §11.4.16), Fixed.md + Fixed_Summary.md (§11.4.19, §11.4.53), CONTINUATION.md (§12.10), every docs/**/

Status.md + its Status_Summary.md pair (§11.4.45, §11.4.56). Status docs are auto-discovered by the generator via `find docs -name 'Status.md'`.

Generator: `scripts/testing/update_readme_doc_links.sh` extracts each doc's Revision + Last modified, renders the markdown table, replaces the section between markers (`<!-- doc-link-section:begin --> / <!-- doc-link-section:end -->`) in README.md. Invoked from `sync_issues_docs.sh` (Issues / Fixed edits) AND `sync_integration_status.sh` (Status edits).

Composes with §11.4.12 + §11.4.19 + §11.4.53 (Issues / Fixed / summaries — README links all four), §11.4.44 (revision header data source), §11.4.45 + §11.4.56 (Status pairs enumeration), §12.10 (CONTINUATION.md explicit link).

Pre-build gates CM-README-DOC-LINK-SECTION-PRESENT (literal Tracked-Items + section markers) + CM-README-DOC-LINK-ROWS-COMPLETE (every canonical doc appears as a row) + CM-README-DOC-LINK-FRESHNESS (Last modified matches source within sync-wrapper granularity) + CM-COVENANT-114-57-PROPAGATION. Paired mutations strip the markers, remove the CONTINUATION row, backdate a Last modified cell, strip the anchor literal. No escape hatch — no `--skip-readme-doc-links`, `--collapse-status-rows`, `--no-freshness-check` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.57.

Non-compliance is a release blocker regardless of context.

§11.4.58 — Parallel-development methodology (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19T~05:00Z MSK):

"We MUST DO one comprehensive research and planning in the background in parallel with current mainstream work: our current methodology of the development is very slow. ... We MUST CREATE adjusted improved version of working methodology where multiple workable items could be done in parallel, all come into the central (main) branch as they are done, then we at particular moment rebuild and reflash the System and in background full testing with validation and verification is done. Each parallel work on one workable (or more) item(s) must use parallel agents as much as possible! ... Writing in depth tests (all supported types of the tests) with Challenges and full HelixQA use is MANDATORY! Every test we execute besides executed with success MUST

RESULT in proof that actual functionality being tested
REALLY DOES WORK with NO BLUFF of any kind! Heavt
enforcement of no-bluff / anti-bluff policy IS MANDATORY!"

Project work proceeds through the **Parallel Work Unit (PWU) pipeline** rather than sequential phase-chain. Each PWU is a self-contained workable item with mandatory components: ATM-NNN identifier per §11.4.54, Issues.md entry per §11.4.15+§11.4.16, file-scope manifest, §11.4.43 RED test, source patch, pre-build gate per §11.4.4(b) layer 1, post-flash test per §11.4.4(b) layer 3, paired §1.1 meta-test mutation, §11.4.4(b) layer 4 HelixQA Challenge bank entry, captured-evidence directory per §11.4.5+§11.4.52.

5-stage pipeline: (Stage 1 DEVELOP — parallel PWU agents in isolated worktrees) → (Stage 2 MERGE — serial conductor via `commit_all.sh` flock

- §11.4.41 4-step merge-first) → (Stage 3 REBUILD+FLASH — parallel where hardware allows) → (Stage 4 VALIDATE — parallel on D3+D4+meta-test+ coverage) → (Stage 5 SWEEP — parallel HelixQA Challenges + Fixed.md migration + README refresh). Stage 1 of round N+1 overlaps with Stages 4-5 of round N — the throughput multiplier.

Synchronization mechanism: 4-layer lock hierarchy (L1 parent flock / L2 per-submodule git ops / L3 contention-path advisory locks for the 10 forbidden cross-PWU paths / L4 per-PWU git worktree). Disjoint-scope PWUs run fully parallel. Conflict detection at Stage 2: `git fetch --all --prune --tags + scope-overlap check` → overlap detected rejects PWU back to Stage 1.

Anti-bluff enforcement at merge time (all four required): C1 §11.4.43 RED-test captured-evidence (proves test catches regression), C2 §1.1 paired meta-test mutation (proves gate is not bluff), C3 §11.4.50 deterministic-consistency (3 or 10 iterations identical), C4 §11.4.5 captured-evidence (audio WAV / video screen recording / UI uiautomator dump / HelixQA `result.json`). Metadata-only / configuration-only / absence-of-error / `grep-without-runtime` PASS all REJECTED.

HelixQA mandatory. Every user-visible PWU MUST add a Challenge entry in `tools/helixqa/banks/atmosphere.yaml` referencing the PWU's ATM-NNN + dispatching to its on-device test + scoring PASS only on positive captured evidence per §11.4.5.

Pre-build gates CM-PWU-LOCK-HIERARCHY + CM-PWU-ANTI-BLUFF-COVERAGE + CM-PWU-MERGE-QUEUE-DISCIPLINE + CM-PWU-PARALLEL-AGENT-LIMIT + CM-COVENANT-114-58-PROPAGATION (anchor literal across 42 consumer files). Paired mutations strip the lock helpers / forge a READY

marker without evidence / bypass the merge queue / spawn a 7th concurrent agent / strip the anchor literal — every mutation FAILs its gate.

No escape hatch — no --skip-merge-queue, --allow-bypass, --no-anti-bluff-check, --unlimited-agents, --sequential-phase-chain-mode flag. Composes with §11.4.4, §11.4.5, §11.4.6, §11.4.9, §11.4.15, §11.4.16, §11.4.19, §11.4.33, §11.4.41, §11.4.42, §11.4.43, §11.4.45, §11.4.49, §11.4.50, §11.4.52, §11.4.54, §11.4.57, §12.6, §12.7, §12.8, §12.10, §9.2.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.58. Project-specific implementation reference in consumer-side docs/guides/PARALLEL_DEVELOPMENT_METHODODOLOGY.md.

Non-compliance is a release blocker regardless of context.

§11.4.59 — README always-sync mandate (User mandate, 2026-05-19)

README.md is a §11.4.12-class always-sync document: HTML + PDF exports refresh on every update via scripts/testing/sync_readme_export.sh (pandoc + weasyprint); auto-invoked by sync_issues_docs.sh so a single doc-sync run refreshes Issues / Issues_Summary / Fixed / Fixed_Summary / CONTINUATION / README (md + html + pdf). README carries a §11.4.44 revision header and a Documentation Map section linking to every Status.md + Status_Summary.md + spec + plan + guide + script-companion doc + changelog + the constitution submodule, plus per-audience navigation. Pre-build gate CM-README-EXPORT-SYNC enforces mtime parity (README.html + README.pdf ≥ README.md). Paired meta-test mutation backdates HTML+PDF → gate FAILs. No escape hatch — no --skip-readme-sync, --no-readme-export, --readme-stale-OK flag. Composes with §11.4.12 + §11.4.18 + §11.4.44 + §11.4.45 + §11.4.53 + §11.4.56 + §11.4.57 + §12.10.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.59.

Non-compliance is a release blocker regardless of context.

§11.4.60 — Documentation always-sync composite covenant (User mandate, 2026-05-19)

Eight documentation classes (Issues, Issues_Summary, Fixed, Fixed_Summary, CONTINUATION, README, every Status.md, every Status_Summary.md) MUST be in sync at all times across .md + .html + .pdf artefacts. Per-class anchors §11.4.12 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §12.10 govern individually; §11.4.60 binds them via single composite gate CM-DOCS-COMPOSITE-SYNC that FAILs if ANY instance's .html or .pdf mtime is older than .md mtime. Walks docs/**

recursively for Status.md fleet. Paired mutation backdates docs/Issues.html → gate FAILs. No escape hatch — no --skip-composite-doc-sync, --allow-stale-html, --summary-not-applicable flag exists. Composes with §11.4.12 + §11.4.15 + §11.4.16 + §11.4.19 + §11.4.23 + §11.4.33 + §11.4.44 + §11.4.45 + §11.4.53 + §11.4.56 + §11.4.57 + §11.4.59 + §12.10.

Canonical authority: constitution submodule Constitution.md §11.4.60.

Non-compliance is a release blocker regardless of context.

§11.4.63 — Workable-items procedure docs as single source of truth (User mandate, 2026-05-19)

Every workable-item action (open / update / close / reopen / migrate) MUST follow the canonical procedure document at docs/procedures/issues/<Action>.md. Closed-set of 5 procedure docs: Creation.md, Updating.md, Resolution.md, Reopening.md, Migration.md. Each carries §11.4.44 revision header + HTML + PDF exports (refreshed by scripts/testing/sync_procedure_docs_export.sh, invoked as the final stage of sync_issues_docs.sh). ALL work — new features, behavioural changes, tasks (refactor / doc / infra / gate / audit / cleanup), bug fixes, investigation, documentation — MUST flow through these procedures. No ad-hoc procedure permitted.

Pre-build gate CM-PROCEDURES-DOCS-PRESENT checks all 5 procedure docs exist + carry revision header + have current HTML/PDF exports + sync helper is wired into sync_issues_docs.sh. Paired mutation: rename one procedure doc → gate FAILs. Propagation gate CM-COVENANT-114-63-PROPAGATION enforces this anchor literal in every CLAUDE.md / AGENTS.md.

Canonical authority: constitution submodule Constitution.md §11.4.63.

Non-compliance is a release blocker regardless of context.

MANDATORY HOST-SESSION SAFETY (Constitution §12)

Every script, test, helper, and AI agent MUST respect host-session safety. Non-compliance is a release blocker.

Forbidden — directly OR indirectly

1. Suspending the host (systemctl suspend, etc.).
2. Hibernating / hybrid-sleeping the host.
3. Logging out the user (loginctl terminate-session, pkill -u <user>, anything targeting user@<uid>.service).

4. Unbounded-memory operations inside `user@<uid>.service` cgroup — any command expected to exceed ~4 GiB RSS **MUST** be wrapped in a bounded execution scope.
5. Programmatic `rkill` toggles, lid-switch handlers, power-button handlers — these cascade into idle-actions.
6. Disabling session managers "to make things faster".

Required safeguards for heavy scripts

1. Source the project's host-safety helper library.
2. Call its pre-flight check and abort if it fails.
3. Wrap any subprocess expected to exceed ~4 GiB RSS in a bounded execution scope.
4. Cap parallelism to fit available RAM.

§12.6 Memory-Budget Ceiling — 60% MAXIMUM

Forensic anchor — verbatim user mandate:

"First make sure that whatever we do through our procedures related to this project **MUST NOT** use more than 60% of total system memory! All processes **MUST** be able to function normally!"

Project procedures **MUST NOT** use more than **60%** of total system RAM. The remaining 40% is reserved for the operator's other workloads. No escape hatch — bypassing this is the bluff §11.4 forbids.

§12.10 Continuation document maintenance

`docs/CONTINUATION.md` **MUST** exist at the project root and reflect the live state of the work. Every non-trivial state change updates it in the same commit. Any agent must be able to resume work exactly where the previous session left off by reading this single file.

MANDATORY ABSOLUTE DATA SAFETY — ZERO RISK (Constitution §9)

Every destructive repository operation (history rewrite, force-push, branch deletion, bulk file removal, submodule de-init, object pruning) **MUST** follow the full §9 safety protocol **WITHOUT** EXCEPTION:

1. **Backup first, always** — hardlinked mirror of `.git` to a sibling backup directory (`cp -al .git <backup>/repo.git.mirror`) is near-instant and uses zero additional disk.

2. **Record metadata** — refs, tags, submodule pointers, HEAD commit, HEAD tree hash, tree content sha256.
3. **Define expected post-op state.**
4. **Run the destructive operation** — never with `--no-verify`, never with `--force` that bypasses hooks, never auto-force on failure.
5. **Post-op gate** — HEAD tree identical, all tags preserved, all submodule pointers intact, per-entry archive integrity 100%, pre-build gates green. If any check fails → restore immediately.
6. **Force-push authorization** — force-push is NEVER automatic. Each force-push event requires explicit human authorization AND requires the post-op gate to have passed.
7. **Audit trail** — every history rewrite gets a docs/changelogs/"Force-push audit" section.

Hardlinked backup is so cheap (zero disk, <2 s) that there is NEVER an excuse to skip it.

MANDATORY COMMIT & PUSH CONSTRAINTS

1. **Use the project's official commit wrapper** for the main repo (e.g. `scripts/commit_all.sh`).
2. **NEVER use `git add`, `git commit`, or `git push` directly** on the main repo unless the project Constitution explicitly carves out a use case.
3. **Multi-upstream push is the norm** — every commit pushed to ALL configured upstream remotes. The constitution submodule ships an `install_upstreams.sh` (and an `Upstreams/` directory) that configures all remotes locally.
4. **NEVER skip hooks** (`--no-verify`, `--no-gpg-sign`) unless the user explicitly authorizes it for the session.

MANDATORY TESTING CONSTRAINTS

Tests MUST be run at every stage WITHOUT EXCEPTION:

1. **Pre-build / pre-merge verification** — BEFORE every build.
2. **Post-build / packaging verification** — AFTER every build.
3. **Post-deploy verification** — AFTER every deploy / flash.

NEVER skip tests. NEVER mark a test as "broken — disable for now" without fixing the underlying issue. NEVER ship a release with unresolved WARNs.

Code conventions

Universal conventions applicable to most projects:

- Use the project's preferred language for new code. Don't mix languages in one module without explicit Constitution permission.
- Static analyzers / linters / type-checkers MUST run clean (zero warnings) before commit.
- Style is set by the project's formatter; do not hand-edit style in PRs.
- Public APIs are documented at the source.

When in doubt

- Read constitution/Constitution.md for the canonical text.
- Cross-reference the project's CLAUDE.md / AGENTS.md for project-specific extensions.
- If still unclear, ask the operator. Do NOT guess. Do NOT bluff.

§11.4.65 — Universal Markdown export mandate (User mandate, 2026-05-19)

Every Markdown document inside the project that is NOT part of an application or service's source-code tree MUST have synchronized .html and .pdf siblings. Includes: project-root *.md, docs/**/*.*md, scripts/**/*.*md (doc-format companion docs), owned-submodule top-level README.md / CLAUDE.md / AGENTS.md / CHANGELOG.md and their docs/**/*.*md, constitution/**/*.*md, owned HelixQA submodules' equivalents. Excludes: external/**, prebuilts/**, packages/modules/**, kernel-5.10/**, out/**, build/**, application/service source-code trees, and third-party submodules NOT in the owned set. Every edit triggers regeneration via scripts/testing/sync_all_markdown_exports.sh (pandoc HTML + weasyprint PDF, timeout 60 per file, capped at 500 candidates). HTML + PDF mtime MUST be $\geq$ source .md mtime at all times.

Pre-build gates CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC + CM-COVENANT-114-65-PROPAGATION. Paired meta-test mutations. Composes with §11.4.12 / §11.4.18 / §11.4.23 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.59 / §11.4.60 / §11.4.63 / §11.4.64. No escape hatch — no --skip-md-exports, --no-pdf-only, --md-export-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.65.

Non-compliance is a release blocker regardless of context.

§11.4.66 — Blocker-resolution interactive-clarification mandate (User mandate, 2026-05-19)

When any task is blocked (operator decision, hardware access, external authorization, ambiguous scope), the agent MUST: (1) research what's doable from the agent side without operator input; (2) calculate minimum-viable operator input; (3) construct 2-4 mutually-exclusive options with one marked "Recommended" and each stating what the agent does after that answer; (4) present via the platform's interactive question mechanism (AskUserQuestion on Claude Code) — NEVER free-text "what would you like?" for closed- set decisions; (5) after the answer, resume work without follow-up round-trips. Composes with §11.4.6 / §11.4.7 / §11.4.40 / §11.4.41 / §11.4.42 / §11.4.52. No silent waiting; no bulk-text questions when interactive options would do.

Pre-build gate CM-COVENANT-114-66-PROPAGATION enforces the anchor literal across the 42-file consumer fleet. Paired meta- test mutation strips the literal → gate FAILs. No escape hatch — no --skip-ask, --silent-wait, --free-form-only flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.66.

Non-compliance is a release blocker regardless of context.

§11.4.67 — Shell-script target-shell-parseability mandate (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19): "any issue we spot must be fixed, bash scripts as well if they are broken!"

- "Make sure that this is mandatory rule!"

Every shell script that may be invoked under a target shell other than the one in its shebang MUST parse cleanly under that target shell. Forensic incident: device/rockchip/rk3588/tests/test_all_fixes.sh:114 used bash-only `exec > >(tee -a "$f") 2>&1` on a sh script.sh callsite — Android mksh parses the whole script BEFORE executing, so the runtime `[ -n "${BASH_VERSION:-}" ]` guard could not save it. Fixed by wrapping in `eval 'exec > >(tee ...) 2>&1'` so the parser sees only a string.

Closed-set scope: every tracked .sh under device/rockchip/rk3588/tests/, scripts/, scripts/testing/ (and equivalent paths in owned submodules). OUT of scope: external/, prebuilts/, packages/modules/, kernel-5.10/, out/, build/, scripts/legacy/. Mandatory invariants: (1) every in-scope script parses under `sh -n`; (2) bash-only constructs (`>(...)`, `<(...)`, `[[ ]]`, `<<<`, arrays, `${var^^}`, etc.) MUST be wrapped in `eval` OR guarded by bash-only loading; (3)

shebangs honest — `#!/bin/bash` only if bash actually expected; (4) fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51.

Pre-build gate CM-SCRIPT-TARGET-SHELL-PARSEABLE runs `sh -n` on every in-scope script. Propagation gate CM-COVENANT-114-67-PROPAGATION enforces the anchor literal across the 44-file consumer fleet. Paired mutations: inject bash-only outside eval → parse gate FAILs; strip 11.4.67 literal → propagation gate FAILs. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.67.

Non-compliance is a release blocker regardless of context.

§11.4.68 — Positive sink-side / downstream evidence mandate (User mandate, 2026-05-20)

A test asserting audio or video routing PASS MUST capture and verify positive sink-side or downstream evidence — never config-only, never metadata-only, never PCM-open-state-only. Closed enumeration of acceptable evidence: (1) sink-side codec-state showing non-empty codec matching the expected regex during playback (Arvus / Sonos / Yamaha REST API); (2) PCM `hw_ptr` delta strictly positive across the playback window; (3) ALSA ELD demonstrating negotiated channel count + format; (4) `ffprobe-on-captured-mp4` with non-zero frames matching expected codec; (5) recording-analyzer matched event per §11.4.2 / §11.4.5 timeline; (6) `tinycap` RMS amplitude above floor.

Failure modes specifically forbidden: `arvus_probe_present` returning 1 silently dropping the test to SKIP; empty codec-state treated as evidence; `dumpsys media.audio_flinger` policy-side device field used as proof PCM actually opened; config-XML-only PASS without runtime evidence. All four are the §11.4 PASS-bluff pattern materialised on D3 2026-05-20 when the test suite reported audio-routing PASS while the user heard nothing and Arvus Codec-In-Use was empty.

Mandatory protections: (1) sink-side library helpers expose `*_require_reachable` returning exit code 2 (OPERATOR-BLOCKED) when sink unreachable; (2) test harness propagates 2 to summary as OPERATOR-BLOCKED, release tags blocked while OPERATOR-BLOCKED exists; (3) audio-routing tests rewritten to capture ≥1 positive sink/downstream evidence per the enumeration above; (4) anti-stickiness post-stop — test re-probes sink and asserts codec-state transitioned to N.E. / 0-frames-delta. No escape hatch — no `--skip-sink-evidence`, `--allow-empty-codec`, `--sink-unreachable-is-pass`, `--metadata-only-suffices` flag.

Composes with §11.4.2 / §11.4.5 / §11.4.13 / §11.4.14 / §11.4.46 / §11.4.49 / §11.4.50 / §11.4.52. Pre-build gates CM-COVENANT-114-68-PROPAGATION + CM-POSITIVE-SINK-EVIDENCE-PER-AUDIO-TEST with paired meta-test mutations per §1.1.

Canonical authority: constitution submodule Constitution.md §11.4.68.

Non-compliance is a release blocker regardless of context.

§11.4.69 — Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!"

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Closes the PASS-bluff pattern where tests reported green while end users hit broken features (2026-05-19→20 D3 audio "82/84 PASS" + empty Arvus Codec-In-Use).

The mandate. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediacodec_decode, mediacodec_encode, miracast, cast, boot_service, package_install, permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape.

Helper contracts (additive during grace; mandatory after 2026-06-19):

- ab_pass_with_evidence <description> <evidence_path> — the new canonical PASS helper. Verifies path exists AND non-empty; emits PASS: <description> [evidence: <path>].
- ab_skip_with_reason <description> <closed-set-reason> — reasons: geo_restricted, operator_attended, hardware_not_present,

topology_unsupported, network_unreachable_external,
feature_disabled_by_config. Forbids
network_unreachable_external for any taxonomy feature with a
sink-side probe.

- Bare ab_pass deprecated — WARN pre-grace, FAIL post-grace (2026-06-19).

Mechanical enforcement. Three pre-build gates + three paired §1.1 meta-test mutations:

- CM-SINK-EVIDENCE-PER-FEATURE — walks tests for # §11.4.69 FEATURE: <class> annotation + verifies taxonomy probe + ab_pass_with_evidence use.
- CM-NO-FAIL-OPEN-SKIP — audits sink-side probe helpers; FAILs if any code path converts empty/unreachable response to PASS-counting SKIP for a feature class with a sink-side probe.
- CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE — pre-grace WARN, post-grace FAIL on bare ab_pass calls.

Composes with §11.4.1 (FAIL-bluffs forbidden), §11.4.2 (recorded-evidence), §11.4.5 (audio + video 5-layer quality), §11.4.6 (no-guessing), §11.4.13 (sink-side captured-evidence), §11.4.27 (no-fakes-beyond-unit), §11.4.50 (deterministic consistency), §11.4.52 (autonomous-validation), §11.4.68 (audio-specific sink-side — §11.4.69 is the universal generalisation).

No escape hatch — no --skip-evidence, --config-only-pass, --allow-fail-open-skip, --legacy-ab-pass-permitted flag. The discipline exists because the 2026-05-20 forensic incident demonstrated the failure: tests reported audio-routing PASS while the user heard nothing and the Arvus Codec-In-Use field was empty.

Propagation gate CM-COVENANT-114-69-PROPAGATION enforces this anchor literal across the ~44-file consumer fleet. Paired mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.69.

Non-compliance is a release blocker regardless of context.

§11.4.70 — Subagent-driven execution is the default (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"Always do if possible Subagent-driven! Add this into our root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md. This should be the default choice ALWAYS!"

When executing implementation plans authored via `superpowers:writing-plans` (or any equivalent task-decomposed execution flow), the **default execution model is subagent-driven** per `superpowers:subagent-driven-development`. Inline execution via `superpowers:executing-plans` is permitted ONLY when (a) the task is trivial AND fits in a single sub-300-line edit, OR (b) the operator explicitly requests inline execution at brainstorm-handoff time.

Subagents bring an isolated context window per task (no conductor context bloat), a structurally separated review seam (conductor reviews subagent output, eliminating self-review blind spots), parallel-PWU compatibility (§11.4.58 — subagents ARE the parallel work units), and resumability across operator absence (subagents resume from on-disk plan + spec inputs).

Composes with §11.4.4 (four-layer coverage), §11.4.6 (no-guessing — subagent's captured output IS the evidence), §11.4.42 (iteration discipline), §11.4.43 (TDD-fix), §11.4.50 (deterministic consistency), §11.4.51 (LIVE_ADB_FIRST), §11.4.58 (parallel-development PWU).

No escape hatch — `--inline-execution-required`, `--no-subagents`, `--monolithic-execution` are NOT permitted flags. Skipping subagent-driven for non-trivial work without recorded operator authorisation is itself a §11.4 PASS-bluff. Pre-build gate `CM-COVENANT-114-70-SUBAGENT-DEFAULT-PROPAGATION` enforces this anchor literal across the ~44-file consumer fleet. Paired meta-test mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.70.

Non-compliance is a release blocker regardless of context.

§11.4.71 — Pre-Push Fetch + Investigate + Integrate Mandate (User mandate, 2026-05-20)

Everyday-push variant of §11.4.41 (force-push merge-first). Before pushing to any upstream for any repository (main repo or submodule), the agent MUST follow the 5-step pre-push cycle: (1) `git fetch --all --prune --tags`, (2) `git pull --no-rebase <remote> <branch>` for each remote whose tip differs from local, (3) investigate the diff vs OUR previous HEAD by reading every foreign commit's body (what changed, why, how it affects OUR project), (4) integrate mandatory changes with full §11.4.4(b) four-layer anti-bluff test coverage producing REAL captured-evidence proofs, (5) THEN push to every configured remote in cascade order.

Composes with §11.4.41 (force-push case) + §11.4.26 / §11.4.32 / §11.4.37 / §11.4.40 / §11.4.42 / §11.4.43 / §11.4.4(b) / §11.4.5 / §11.4.6.

Applies to parent repo + constitution submodule + every owned submodule + every nested submodule + every HelixQA dependency. Audit-trail per push reconstructable from docs/changelogs/<tag>.md + per-repo git log evidence.

No escape hatch — no --skip-fetch, --no-investigate, --fast-push, --trust-upstream flag. Pre-build gate CM-COVENANT-114-71-PROPAGATION + paired mutation.

Canonical authority: constitution submodule Constitution.md §11.4.71.

Non-compliance is a release blocker regardless of context.

§11.4.72 — Audio Top-Priority Mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure all fixes for audio are always top priority in main working stream!" The conductor (main working stream — Claude Code session, AI agent, or human operator) MUST treat audio fixes as the highest-priority class on the serial dispatch queue. Audio scope includes §EU HDMI rejection, §EM multichannel HDMI, §ET Arvus integration, §EV/§EW/§EX D3 audio defects, HiFi (Fix #74, #76), AC3 (Fix #106), ES8388 (Fix #103/§104/#105), multichannel LPCM (Fix #112), and every future audio-stack improvement.

Composes with §11.4.42 (iteration-discipline — audio sits at apex of priority order) + §11.4.58 (parallel-development PWU — background research subagents run concurrently but do NOT preempt audio on the main-stream serial dispatch queue).

Operative rule: any time the conductor faces a choice between dispatching an audio task vs a non-audio task on the SAME serial resource, the audio task wins. No escape hatch — no "but this non-audio task is faster" or "but this research is more interesting" override. Audio-stack regressions are user-perceptible and high-impact (D3 silent post-flash is a release blocker), while research and refactors can wait.

Canonical authority: constitution submodule Constitution.md §11.4.72.

Non-compliance is a process violation regardless of context.

§11.4.73 — Main-specification document versioning + revision discipline (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure everything we add now in previous and upcoming requests IS ALWAYS applied to the main specification — if we have one. Since all these are not major changes we could increase Specification version per change for secondary version instead of the primary. Primary version MUST BE increased for much bigger levels of changes! Document MUST BE updated ALWAYS to follow the versioning rules we are applying here + revision and other properties we have!"

Applies only when a project recognises a main specification document (e.g. docs/specs/**/specification.V<n>.md). Two-axis versioning: **primary** (V1/V2/V3/...) bumps for major rewrites (old primary archived to <spec-dir>/archive/); **secondary** (Revision) bumps for additive operator requirements, refinements, polish (matches the §11.4.61 metadata-table Revision integer). Every operator-mandated requirement MUST land in the spec as part of the work that implements it. Cross-doc propagation copies MUST reference the active spec file, not a stale archived version. Composes with §11.4.44, §11.4.61, §11.4.59, §11.4.65.

Canonical authority: constitution submodule Constitution.md §11.4.73.

Non-compliance is a release blocker.

§11.4.74 — Submodule-catalogue-first discovery + extend-don't-reimplement (User mandate, 2026-05-20)

Direct user mandate (verbatim): "We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in vasic-digital and HelixDevelopment organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times. For any missing features we MUST IMPLEMENT them properly and extend those Submodules further!"

Before scaffolding any new module / package / helper / utility, the contributor (human or AI agent) MUST: (1) survey the vasic-digital + HelixDevelopment orgs on GitHub + GitLab — the canonical inventory is submodules-catalogue.md (142 repos categorised); (2) reuse an existing Submodule when it covers $\geq$ 80% of the functionality; (3) extend in-place via upstream PR when 80%+ matches but features are missing — never duplicate the 80% in-project; (4) document the survey result in the relevant tracker row with a Catalogue-Check: reuse|extend|no-match <org/repo>@<sha> line.

Every Submodule in the catalogue is subject to the same development-cycle rules (§11.4 anti-bluff, §1.1 paired mutations, §11.4.10 credentials, §11.4.61 metadata + ToC, §11.4.65 universal export, §11.4.73 spec versioning, §2.1 multi-mirror push, §3 propagation order).

Canonical authority: constitution submodule Constitution.md §11.4.74.

Non-compliance is a process violation; severe cases (duplicate implementation landed without catalogue check) are release blockers.

§11.4.75 — Mechanical Enforcement Without Exception (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Why do these violations still happen!? This is a serious problem! We cannot rely on stability nor consistency if we cannot respect our Constitution, mandatory rules and constraints! Is there a way to make this always respected, followed and applied without exception fully and unconditionally!? WE MUST HAVE THIS WORKING FLAWLESSLY!!! Do investigate the root causes of such problems! Once all problems are identified WE MUST apply proper mechanisms for this not to happen NEVER EVER AGAIN!"

The §11.4 covenant historically relied on agent + operator vigilance. Three forensic incidents in 2026-05-19→20 (two stalled remediation subagents + post-§11.4.66 propagation drift requiring catch-up commit f93b25a92eb) demonstrated that late-binding enforcement at `pre_build_verification.sh` time fires hours-to-days AFTER the violator commit has already reached every remote. §11.4.75 closes the gap with FIVE independent mechanical enforcement layers — bypassing any single layer does not bypass the discipline:

1. **Local pre-commit git hook** — refuses staged `.md` lacking sibling `.html` + `.pdf` (and other staged-only invariants).
2. **commit_all.sh integration** — the canonical project commit script invokes the same checks + auto-runs `sync_all_markdown_exports.sh` to self-repair before commit (`_constitution_sibling_check` function).
3. **Local pre-push git hook** — re-runs siblings + propagation-gate subset on every commit in the push.
4. **post-commit auto-repair hook** — detects orphan `.md` in just-committed manifest, auto-generates siblings via `pandoc` + `weasyprint`, creates a chore(§11.4.75): `auto-export ... follow-up commit`. Idempotent + recursion-guarded.
5. **Local-only equivalent** (Phase 39.GF, User mandate 2026-05-20). Remote CI surfaces (GitHub Actions, GitLab pipelines, Jenkins, CircleCI, etc.) are DISABLED — the workflow file is preserved at `.github/workflows/constitution-`

compliance.yml.disabled-local-only (NOT .yml; GitHub Actions ignores it). Layer 5 enforcement migrated to the LOCAL pre-build-verification + meta-test ritual the operator MUST execute before tagging per §11.4.40. Layers 1-4 remain authoritative; Layer 5 is the operator's local final gate. A future re-enable PWU may re-establish remote CI.

Helper contracts (mandatory): scripts/install_git_hooks.sh (idempotent installer wired into scripts/setup.sh), scripts/git_hooks/{pre-commit,pre-push,post-commit,commit-msg}, _constitution_sibling_check in scripts/commit_all.sh. The commit-msg hook enforces a Bypass-rationale: <reason> footer when --no-verify is detected (touch-file marker .git/ATMO_LAST_BYPASS_ATTEMPT); docs/audit/bypass_events.md accumulates the audit trail per §11.4 captured-evidence requirement.

Five pre-build gates with paired §1.1 meta-test mutations: CM-COVENANT-114-75-PROPAGATION (anchor literal across canonical files), CM-GIT-HOOKS-INSTALL-SCRIPT (installer present + executable), CM-GIT-HOOKS-SOURCE-DIR (4 hook bodies present + executable), CM-COMMIT-ALL-SIBLING-CHECK (_constitution_sibling_check in commit_all.sh), CM-CI-WORKFLOW-PRESENT (CI workflow + ≥ 3 required jobs).

Composes with §1.1 (paired meta-test mutations), §9 (data safety), §11.4 (end-user-quality covenant — Layer 5 CI makes the §11.4 promise mechanical), §11.4.41 (this anchor's renumber from §11.4.74 → §11.4.75 was itself driven by §11.4.41 merge-first discipline), §11.4.65 (universal Markdown export — Layer 1+3+4 are §11.4.65's mechanical seam), §11.4.66 (interactive clarification), §11.4.67 (target-shell-parseability — hooks parse under bash AND mksh), §11.4.71 (pre-push fetch + integrate — Layer 3 + 5 honour the merge-first pipeline), §11.4.72 (audio top-priority — hooks hold no lock themselves, so do not race against in-flight audio commits), §11.4.73 (main-spec versioning — when spec exists, hooks include it), §11.4.74 (submodule-catalogue-first — hooks can be added to the canonical catalogue for cross-project reuse).

No escape hatch — no --skip-hooks, --bypass-enforcement, --allow-orphan-md, --ci-not-applicable, --mechanical-enforcement-not-needed flag exists. The --no-verify route IS the deliberate audit-trail bypass; §11.4.75 makes the audit trail mechanical via the commit-msg footer requirement.

Canonical authority: constitution submodule Constitution.md §11.4.75.

Non-compliance is a release blocker regardless of context.

§11.4.76 — Containers-submodule mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "For any work or requirements of running services or codebase inside the Containers (Docker / Podman / Qemu / Emulators, and so on) we MUST USE / INCORPORATE the Containers Submodule properly: <https://github.com/vasic-digital/containers> (git@github.com:vasic-digital/containers.git). Containers Submodule contains all means for us to Containerize our code and services! If any feature or Containing System is missing or not supported we MUST EXTEND IT properly like we do all of our projects! No bluff work is allowed of any kind!"

§-slot history note. Originally drafted as §11.4.75, renumbered to §11.4.76 per §11.4.71 fetch-before-push after the concurrent 0a70083 landing of §11.4.75 (Mechanical Enforcement). Same collision-resolution pattern as §11.4.75's own renumber from drafted §11.4.74.

For ANY containerized workload (Docker / Podman / Qemu / Kubernetes / container-backed emulators), every consuming project MUST: (1) install vasic-digital/containers (digital.vasic.containers) as a Git submodule; (2) consume via replace directive during development + pinned commit SHAs in production; (3) boot infra on-demand via pkg/boot + pkg/compose + pkg/health so operators are never required to start podman machine / docker compose up manually — the boot is part of the test entry point (the **on-demand-infra invariant**); (4) extend the Submodule (PR upstream) for missing runtimes / lifecycle primitives — never reimplement in-project (per §11.4.74); (5) anti-bluff: integration tests claiming to exercise containerized components MUST actually boot them via the Submodule — short-circuit fakes that bypass boot are a §11.4 violation.

Tracker rows touching containerization MUST record Catalogue-Check: extend vasic-digital/containers@<sha> (or reuse); no-match requires demonstrating the Submodule cannot model the workload.

Planned anti-bluff gate CM-CONTAINERS-USED scans container-touching PRs for digital.vasic.containers/... imports. Paired mutation strips the import + asserts FAIL.

Composes with §11.4.74 (catalogue-first), §11.4.75 (mechanical enforcement — this clause IS one of the things mechanical enforcement protects), §3 (propagation), §11.4.31 (Submodule-Dependency-Manifest), §11.4.36 (install_upstreams on clone), §11.4.28 (Submodules-As-Equal-Codebase), §1.1 (paired-mutation gates protect the Submodule's own primitives).

Canonical authority: constitution submodule Constitution.md §11.4.76.

Non-compliance: reinventing compose orchestration in-project is a release blocker.

§11.4.77 — Regeneration-mechanism-required mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "We must be sure that after excluding anything from Git versioning we still have the mechanism which will out of the box obtain or re-generate missing content!"

Forensic anchor. 2026-05-20T15:00Z: ATMOSphere parent's audio Tier 1 commit `all.sh` stalled 4 h on `git add -A` scanning 274 GiB `.git-backup-*` + 159 GiB `RKTools/linux/` + 167 GiB `qa-results/` — all untracked but un-gitignored. Bare `.gitignore` fix would orphan every fresh clone (missing `RKTools`, missing test infra, missing build outputs).

Every `.gitignore` entry excluding (a) $>\sim 100$ MiB OR (b) any artefact essential to building / running / testing the project **MUST** carry a documented + automated mechanism to either **re-obtain** (download from authoritative source: vendor tarball, SDK installer, npm / pip / cargo / go-mod / container registry, dedicated git submodule, S3/GCS) OR **re-generate** (run from tracked source code via build pipeline, code-gen, asset render, captured-evidence replay, container build, kernel build). Required artefacts per qualifying entry: (1) `.gitignore-meta/<entry-slug>.yaml` declaring pattern + mechanism-type + script-path + expected-disk-usage + vendor-url-or-source + integrity hash + requires-network + requires-credentials; (2) entry in `scripts/setup.sh` (post-clone bootstrap) that runs the mechanism non-interactively; (3) pre-build gate verifying regenerated content present OR stamp `.gitignore-meta/.regenerated/<slug>.ok` recent; (4) README + relevant docs/guides/*.md describing the mechanism + manual fallback + time/disk budget + per-§11.4.10 credentials.

No escape hatch: bare `.gitignore` additions without the mechanism are themselves a §11.4 PASS-bluff variant — codebase appears complete to casual eye but fresh clone cannot build / run. No `--skip-regen-mechanism`, `--gitignore-is-enough`, `--operator-already-has-content` flag.

Planned anti-bluff gate CM-GITIGNORE-REGEN-MECHANISM scans every `.gitignore` addition for matching `.gitignore-meta/` sibling + verifies script-path exists + parses YAML + checks bootstrap reference. Paired §1.1 mutation strips one required YAML key → gate FAILs.

Composes with §11.4.6 (no-guessing — mechanism **MUST** be verified working on sandbox clone, not assumed), §11.4.65 (universal Markdown export — generated HTML/PDF siblings are a re-generate instance), §11.4.66 (interactive clarification — ASK operator if mechanism unknown), §11.4.71 (pre-push fetch + integrate — re-validate vendor integrity before push), §11.4.74

(catalogue-first — extend a reusable downloader Submodule rather than reimplement), §11.4.75 (Mechanical Enforcement — pre-commit + CI replay layers refuse bare additions), §11.4.76 (Containers-submodule — container images regenerated via vasic-digital/containers), §9 / §9.2 (zero-risk data safety — pre-test mechanism in sandbox with hardlinked-backup), §3 (propagation — consuming submodules inherit §11.4.77).

Canonical authority: constitution submodule Constitution.md §11.4.77.

Non-compliance is a release blocker regardless of context.

§11.4.78 — CodeGraph code-intelligence mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make codegraph MANDATORY CHOICE for this purpose for all of our project ... All project which do not have configured and installed codegraph yet MUST DO IT and MUST USE IT!"

Every consuming project worked on by AI coding agents MUST install, initialize, and use **CodeGraph** (<https://github.com/colbymchenry/codegraph>, npm @colbymchenry/codegraph) — a local SQLite semantic code-knowledge-graph exposed to agents over MCP (100% local, no cloud, no external API). (1) Install globally via npm — no sudo (the npm prefix MUST be user-writable). (2) codegraph init + codegraph index: .codegraph/config.json is tracked, .codegraph/codegraph.db is gitignored with codegraph index as its §11.4.77 regeneration mechanism; the config.json exclude list MUST exclude other-owned submodules AND — non-negotiably — every credential/secret path per §11.4.10 (a secret reaching the index is a §11.4.10 violation). (3) Wire the codegraph serve --mcp server into every CLI agent the developers use — project-scoped + committed where supported (Claude Code .mcp.json, OpenCode opencode.json, Qwen Code .qwen/settings.json, Crush .crush.json), host-local otherwise (Kimi CLI ~/.kimi/mcp.json); every config references the bare codegraph command on PATH (no hardcoded host path). (4) Cover the integration with an anti-bluff verification suite whose per-agent end-to-end layer uses an **unforgeable challenge** — a fact obtainable only by calling a CodeGraph MCP tool (e.g. the index node count via codegraph_status) so an agent answering from its own file-reading tools cannot produce a false PASS; an agent that genuinely cannot be driven end-to-end (missing credentials / quota / environment incompatibility) is a documented SKIP gap per §11.4.3, never a faked PASS. (5) Document everything in docs/CODEGRAPH.md, kept in sync per §11.4.12 / §11.4.65. CodeGraph is consumed as the published npm package (§11.4.74) — not added as a git submodule, and it adds no Git remote.

Composes with §11.4.3, §11.4.10, §11.4.12, §11.4.65, §11.4.30, §11.4.74, §11.4.77, §11.4 (anti-bluff — CI green is necessary, never sufficient), §1.1. Planned gate CM-CODEGRAPH-WIRED + paired §1.1 mutation (strip a secret-exclusion from config.json → gate FAILs).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.78.

Non-compliance is a process violation; a project worked on by AI agents without CodeGraph installed, wired, and anti-bluff-verified is in breach of this mandate.

§11.4.79 — Own-org submodules MUST be included in the CodeGraph index (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "All Submodules we use in the project and that are part of organizations to which we have the full access via GitHub, GitLab and other CLIs MUST BE included into the codegraph database and initialized / scanned / synced!"

Refines §11.4.78 step 2's exclude-list with a per-submodule-ownership split: (a) **Own-org submodules** — full write access via the project's CLIs (canonical orgs: vasic-digital on GitHub/GitLab/GitFlic/GitVerse + HelixDevelopment on GitHub) — MUST be **INCLUDED** in the index. (b) **Third-party submodules** (the §11.4.74 no-match → vendor path, e.g. gopkg.in/telebot.v3) — MUST be **EXCLUDED**. Operational steps: (1) git submodule update --remote --merge to pull latest from every submodule before re-indexing — respect load-bearing pins on third-party submodules (e.g. roll back if --remote advances gopkg.in/telebot.v3 past the v3 import-path pin). (2) Adjust .codegraph/config.json exclude list to keep own-org paths in scope. (3) Re-index via scripts/codegraph_setup.sh. (4) Verify via scripts/codegraph_validate.sh with at least one probe that resolves a symbol living ONLY inside an own-org submodule. (5) Paired §1.1 mutation: temporarily add the own-org submodule to exclude → validate MUST FAIL on the cross-submodule probe → restore.

Composes with §11.4.74 (catalogue ownership classifier), §11.4.78 (CodeGraph parent mandate), §11.4.10 (credentials still excluded regardless), §1.1 (paired mutation), §107 (an index that lies about reachable symbols is a §107 PASS-bluff against AI agents).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.79.

Non-compliance is a process violation; severe cases (own-org submodules silently excluded WITHOUT an audit trail in .codegraph/config.json comments) are release blockers.

§11.4.80 — CodeGraph regular-update + sync automation mandate (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "We MUST regularly check for the updates and execute codegraph npm updates so the latest version of it is always installed on the host machine! [...] Make sure we have proper full automation bash scripts which will run regularly and that these are part of the constitution Submodule since they MUST BE available to all projects which do respect and follow our constitution rules and mandatory constraints! Make sure all updates, sync processes we do and important codegraph related events are all documented under docs/codegraph in Status and Status_Summary documents (another area / context to regularly update and sync) and regularly export them like all other Status docs into the PDF and HTML!"

Three deliverables (all in this constitution submodule): (1) scripts/codegraph_update.sh — npm-installs latest @colbymchenry/codegraph after npm-registry version check; appends old/new version to docs/codegraph/Status.md; §107 anti-bluff: verifies codegraph --version reflects the new version after install (npm exit 0 ≠ working binary). (2) scripts/codegraph_sync.sh — after a successful update, runs codegraph status → codegraph sync . → codegraph status → project's scripts/codegraph_validate.sh in the consuming project; appends every step's output to BOTH the project's and the constitution's docs/codegraph/Status.md. (3) docs/codegraph/Status.md + Status_Summary.md append-only ledgers tracking all CodeGraph-related events; exported to .html + .pdf siblings per §11.4.65.

Operational cadence: weekly floor (per §11.4.45 status-digest cadence). Scripts are inherited by reference (per §3 submodule inheritance) — consuming projects invoke them at \${CONST_DIR}/scripts/codegraph_*.sh, never copy.

Composes with §11.4.78 (CodeGraph parent), §11.4.79 (own-org submodule inclusion), §11.4.10 (credentials excluded), §11.4.45 (status-digest cadence), §11.4.53 (Fixed_Summary backfill), §11.4.65 (multi-format export), §107 (observed-version evidence per update; observed PASS/FAIL per sync), §1.1 (paired mutation: downgrade installed version → script detects drift → restore).

Canonical authority: constitution submodule Constitution.md §11.4.80.

Non-compliance is a process violation; severe cases (consuming project has not run codegraph_update.sh in >2 weeks AND has open AI-agent work) are release blockers.

§11.4.81 — Cross-platform-parity mandate (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!"

Every consuming project whose supported-platforms manifest lists more than one OS MUST, for every feature/test/gate/challenge/mutation that depends on platform-specific primitives, ship a per-OS-equivalent implementation chosen at runtime via `uname -s` (or equivalent platform detection). Three sub-mandates: **(A) Per-OS implementation REQUIRED.** Linux `cgroup/systemd//proc` primitives MUST have documented per-OS equivalents (POSIX `setrlimit/ulimit`, macOS `launchd`, BSD `rctl`, Windows Job Object) chosen via runtime dispatch — same operator-path invocation, OS-correct mechanism. **(B) Per-OS tests REQUIRED.** Every gate test that exercises platform-dependent behaviour MUST have case `"$(uname -s)"` in branches with positive captured evidence per §11.4.2 + §11.4.5 in each branch. SKIP-with-reason is acceptable ONLY when the platform genuinely cannot enforce the invariant. **(C) Honest kernel-gap citation + adjacent equivalent test REQUIRED.** Where a Linux primitive has NO macOS/other-OS equivalent due to a documented kernel limitation (canonical: XNU does NOT enforce `RLIMIT_AS` for unprivileged processes), the test MUST detect the gap at runtime, SKIP with exact kernel reason + reproducer + link to honest-gap doc, AND provide an ADJACENT test exercising the closest invariant the platform CAN enforce (e.g. `RLIMIT_CPU+SIGXCPU` as the macOS proxy for "process is bounded"). The adjacent test MUST itself be anti-bluff with a paired §1.1 mutation.

Per-OS equivalence catalogue (canonical, not exhaustive): `systemd-run --user --scope ↔ POSIX ulimit -t -u / launchd; cgroup MemoryMax ↔ XNU gap (use RLIMIT_CPU adjacent) / rctl / Job Object; cgroup TasksMax ↔ RLIMIT_NPROC; /proc/<pid>/oom_score_adj ↔ no equivalent on Darwin/BSD (no kernel OOM-killer).`

Composes with §11.4.1 (FAIL-bluffs forbidden — Linux-PASS / Darwin-SKIP without honest reason is a §11.4.81 violation), §11.4.2 (recorded evidence per branch), §11.4.3 (§11.4.81 strictens §11.4.3: topology SKIP only when kernel CANNOT, not when "we didn't implement yet"), §11.4.4 (test-interrupt triggers when discovering a Linux-only test without per-OS equivalent), §11.4.5 (captured evidence per platform), §11.4.6 (no guessing about platform capability — prove via reproducer), §11.4.20 / §11.4.70 (per-OS branches are parallel-subagent dispatchable), §11.4.27 (100% test-type coverage applies per-platform), §11.4.69 (sink-side evidence per platform), §107 (multi-platform projects must guarantee end-user usability on every platform). Pre-build gate CM-CROSS-PLATFORM-PARITY (when implemented): scans every gate test for case `"$(uname -s)"` blocks, asserts a non-SKIP branch

(or honest-gap citation per (C)) exists for each platform in the supported-platforms manifest. Paired §1.1 mutation: strip a Darwin branch → gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.81.

Non-compliance is a release blocker on multi-platform projects. No escape hatch.

§11.4.82 — Iteration-speedup discipline mandate (User mandate, 2026-05-22)

Forensic anchor — direct user mandate (verbatim, 2026-05-22): "How can we speed-up this whole development and fixing process? ... Do not forget to all speed optimizations critical rules and mandatory constraints MUST BE all added into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md and QWEN.md and all other relevant constitution Submodules files!"

Iteration cycle time is a first-order quality enabler. Slow cycles bound the project's defect-discovery rate; fast cycles multiply it. The 2026-05-22 forensic session witnessed ~3 hours of operator wait time on a job whose intrinsic compute was ~90 min — every wasted minute came from a missing speedup discipline.

Every consuming project's build / test / commit / debug pipeline MUST adopt the following speedup disciplines AS MANDATORY (each independently enforceable):

- **(A) Phase 1 forensic before any speculative source patch** — before any non-trivial source patch, complete superpowers:systematic-debugging Phase 1 to identify FACT-grade root cause. Speculative patches without Phase 1 evidence are §11.4.6 + §11.4.82 violations.
- **(B) Live-ADB-First (or live-equivalent) before any rebuild** — strengthens §11.4.51 to a release-blocker mandate. Skipping live-probe for a LIVE_ADB_TESTABLE change is a §11.4.82 violation.
- **(C) Pre-flight before launching rebuild orchestrators** — 30-second pre-flight verifies device reachability, sink reachability, host memory/disk, no stale locks, no orphan processes. A rebuild against a broken precondition wastes 45 min.
- **(D) Persistent build caches outside containers** — ccache + Soong / sccache / Gradle daemon state bind-mounted to host, NOT in ephemeral container layer. Cold rebuild ~40 min → warm rebuild 5-15 min.
- **(E) Module-only rebuild for loadable-module-only changes** — make -C kernel M=<driver-path> modules for CONFIG_*=m drivers (~2-5 min). Full rebuild required only for =y built-ins.

- **(F) Parallel multi-device testing** — every owned validation device runs the autonomous cycle concurrently with separate qa-results/<TS>/<device-tag>/ outputs.
- **(G) Subagent scope discipline + worktree isolation** — subagents ≤30 min budget, single-responsibility, intermediate output emitted as they progress, isolation: "worktree" by default.
- **(H) Lock-file + stale-process hygiene** — detect and clean .git/index.lock, .lock files, orphan git/build processes on session start. Disable auto git-gc (gc.auto 0) in concurrent multi-agent repos.
- **(I) Cycle telemetry per §11.4.24** — every iteration logs commit hash, per-phase wall-clock, speedup-discipline flag set, outcome. Aggregated weekly to surface which disciplines deliver the biggest empirical return.

Composes with §11.4.4, §11.4.6, §11.4.9, §11.4.20 / §11.4.70, §11.4.24, §11.4.42, §11.4.43, §11.4.50, §11.4.51, §11.4.52, §11.4.58, §12.7, §107. Pre-build gate CM-ITERATION-SPEEDUP-DISCIPLINE (when implemented): audits the most recent N cycles for telemetry citing which of (A)-(I) applied. Paired §1.1 mutation: strip the speedup-flag column → gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.82.

Non-compliance is a release blocker. No escape hatch — no --skip-phase1-forensic, --no-pre-flight, --rebuild-everything-always, --unlimited-subagent-scope, --ignore-locks, --no-telemetry flag exists.

§11.4.83 — docs/qa/ end-user evidence mandate (User mandate, 2026-05-22)

Forensic anchor — verbatim user mandate (2026-05-22):

"every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under docs/qa/<run-id>/ (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation MUST preserve full bidirectional communication threads as proof."

Every feature that ships MUST carry a recorded end-to-end communication transcript plus any attached materials (screenshots, request/response payloads, audio, file uploads) committed under docs/qa/<run-id>/ — one directory per feature run. A feature with no QA transcript is itself a §11.4 / §107 PASS-

bluff: it claims to work but has no auditable runtime evidence that an end user actually exercised the feature through the same interface they will use in production.

Operative rule. (1) Every consuming project MUST maintain a docs/qa/ tree. Each new feature lands under docs/qa/<run-id>/ where <run-id> is monotonic + greppable (timestamp, HRD-NNN, ATM-NNN, other workable-item ID per §11.4.54). (2) Transcripts MUST be full bidirectional — every prompt/command sent + every response received. One-sided is not a transcript. (3) Attached materials MUST be committed in-repo (no external-only links — that is §11.4.13 sink-side violation). (4) Bot-driven / agent-driven QA automation MUST preserve the full conversation thread as the proof artefact. (5) CI / release gates MUST refuse to tag a version that has any feature-shipping commit without its matching docs/qa/<run-id>/ directory.

Composes with §11.4.2, §11.4.5, §11.4.13, §11.4.65, §11.4.69, §107, §1.1.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.83.

Non-compliance is a release blocker. No --qa-evidence-optional escape hatch.

§11.4.84 — Working-tree quiescence rule for subagent commits (User mandate, 2026-05-22)

Short tag: working-tree quiescence.

Forensic anchor — verbatim user mandate (2026-05-22):

"no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before git add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT."

No subagent (or main-thread) commit may proceed while any concurrent mutation gate, paired-mutation experiment, or other in-flight mutation is live in the same checkout. Before git add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, // MUTATION / # MUTATION annotations, _mutated_* filename suffixes, etc.) and explicitly account for every modified file in the staging area. Any unexplained file → ABORT.

Lesson (forensic case study). A consuming project's logo-fix subagent (Herald commit 72e81ab, 2026-05-21) ran in a checkout where a paired §1.1 mutation gate had temporarily introduced an

// always pass shortcut into a JWT verify path. The subagent's git add swept the mutation residue into the same commit as the unrelated logo fix, and the resulting commit was pushed to all four mirrors before any other agent caught it. The fix (Herald d5bd360, "SECURITY FIX: restore commons_auth/middleware.go JWT verify") landed within the hour, but the window during which production-equivalent binaries shipped with a bypassed JWT verify is a real security-defect window. The lesson is now constitutional.

Operative rule. (1) Pre-git add MUST grep for mutation markers + cross-check git status --porcelain against the subagent's declared scope; unaccounted entries → ABORT. (2) Any active mutation gate MUST be serialised — mutate → assert FAIL → restore → assert PASS — and the working tree MUST be verifiably clean BEFORE any unrelated commit. (3) Concurrent subagents in the SAME checkout MUST coordinate through a lockfile (.git/MUTATION_IN_PROGRESS); the cleaner solution is git worktree add per subagent (composes with §11.4.20/§11.4.70). (4) Post-commit mutation-residue-scanner MUST run before push; any commit containing a mutation marker → push BLOCKED.

Composes with §1.1, §11.4.20, §11.4.70, §11.4.27, §11.4.10, §11.4.71, §107.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.84.

Non-compliance is a release blocker. A mutation marker that lands in a tagged commit is a critical defect regardless of how briefly it persisted.

§11.4.85 — Stress + Chaos Test Mandate (User mandate, 2026-05-24)

Short tag: stress-chaos-mandate.

Forensic anchor — verbatim user mandate (2026-05-24):

"Every fix or improvement you do MUST BE covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!"

Every fix or improvement landed in a consuming project MUST ship with full-automation **stress** AND **chaos** test suites that exercise edge cases, sustained load, concurrent contention, and failure-injection. Happy-path coverage alone is a §11.4 / §107 PASS-bluff at the resilience layer.

Stress (closed-set, mechanically auditable): sustained load ($N \geq 100$ iterations OR ≥ 30 s wall-clock, per-iteration latency p50/p95/p99 recorded) + concurrent contention ($N \geq 10$ parallel invocations, no deadlock, no resource leak) + boundary conditions (empty / max / off-by-one input — every boundary produces a categorised result).

Chaos (closed-set, mechanically auditable, applied per fix-class appropriateness): process-death injection (kill primary or upstream mid-call, categorised recovery) + network-fault injection (drop/delay/reorder, category=network|upstream per §11.4.69) + input-corruption injection (corrupt .env / config / input file mid-test, detected + reported) + resource-exhaustion injection (disk full, OOM, FD exhaustion — refuse cleanly OR degrade gracefully, NEVER crash) + state-corruption injection (mid-flight lock loss, partial-write fault — recovery restores consistent state).

Anti-bluff (mandatory). Every stress + chaos test PASS MUST cite a captured-evidence artefact path per §11.4.5 + §11.4.69 (per-iteration latency.json, categorised_errors.txt, state_delta_snapshot.json, recovery_trace.log). Helper library stress_chaos.sh provides ab_stress_run, ab_stress_concurrent, ab_chaos_kill_pid_during, ab_chaos_drop_network_during, ab_chaos_corrupt_file_during, ab_chaos_oom_pressure_during, ab_chaos_disk_full_during, each composing with ab_pass_with_evidence / ab_skip_with_reason per §11.4.69. Chaos-injection cleanup is non-negotiable — corrupt-restore, disk-fill-cleanup, process-restart MUST run in trap '...' EXIT; cleanup failure = §11.4.14 violation.

4-layer coverage per §11.4.4(b): pre-build gate (test files exist + executable + sh -n + bash -n parseable; library exists; fix's pre-build gate cites the stress+chaos test path) + paired meta-test mutation per §1.1 (strip chaos-injection or evidence-capture → gate FAILs) + on-device test (if LIVE_ADB_TESTABLE per §11.4.51, dispatched against a real device with captured evidence under qa-results/<run-id>/stress_chaos/) + HelixQA Challenge entry (if user-visible feature per §11.4.4(b) layer 4).

Composes with §11.4 / §107 (resilience IS end-user quality), §11.4.1 (FAIL-bluffs forbidden — set-u crashes from chaos step are bluffs), §11.4.5 (captured-evidence content quality applies to latency distribution + error categories), §11.4.6 (no guessing — categorised errors only), §11.4.43 (TDD RED-first under load/chaos), §11.4.50 (N iterations produce identical exit + identical evidence-hashes), §11.4.52 (autonomous validation), §11.4.69 (universal sink-side positive-evidence taxonomy), §11.4.83 (recovery transcripts ARE end-user-channel proofs).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.85.

Non-compliance is a release blocker regardless of context. No escape hatch — no --skip-stress, --no-chaos, --happy-path-suffices, --stress-test-later flag exists.

§11.4.86 — Roster/corpus-backed Status-doc auto-sync mandate (User mandate, 2026-05-25)

Forensic anchor — verbatim user mandate (2026-05-25):

"Make sure that assets and players Status docs are ALWAYS regularly updated and in sync like all others Status docs — any time we add or modify the assets content(s) or we change or add new / remove existing pre-installed video and audio player apps! This MUST WORK OUT OF THE BOX!"

Some Status docs (§11.4.45) are backed by a **tracked roster** (installed apps/components) or a **tracked asset corpus** (test/media asset directory) rather than narrative alone. Their freshness MUST NOT depend on operator vigilance — the moment a roster/corpus member changes (player app added/removed/renamed; asset added/modified/removed) the Status doc + Status_Summary + HTML + PDF MUST resync **out of the box**, mechanically.

Mechanism (all must hold): (1) **drift-proof fingerprint** — sha256 of the sorted member list (NOT mtime), persisted in a sidecar beside the Status doc; (2) **sync helper** that regenerates the fingerprint + re-exports HTML+PDF (via the §11.4.65 exporter), wired so sync is automatic; (3) **pre-build gate** that FAILs when the live fingerprint differs from the persisted one (mirrors §11.4.12 CM-ISSUES-SUMMARY-SYNC + §11.4.45 sync_integration_status); (4) **paired §1.1 mutation** that corrupts the fingerprint and asserts the gate FAILs.

Composes with §11.4.12 / §11.4.45 (roster/corpus specialisation) / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.6. Classification: universal (§11.4.17) — the consuming project supplies the specific docs, roster/corpus sources, helper, and gate name per §11.4.35.

Canonical authority: constitution submodule Constitution.md §11.4.86.

Non-compliance is a release blocker regardless of context. No escape hatch — no --skip-roster-sync, --allow-status-drift, --roster-sync-not-applicable flag exists.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing mandate (User mandate, 2026-05-26)

When the operator instructs an AI agent to "continue in endless loop fully autonomously" (or any semantically-equivalent phrasing), the agent MUST treat this as a HARD-CONTRACT covenant: (A) continue working until docs/Issues.md Status-column has zero non-terminal entries AND docs/CONTINUATION.md §3 Active work is empty AND no background subagent is mid-execution AND no external dependency is in-flight; (B) dispatch background subagents for parallelisable work — main + every subagent operate concurrently, "waiting for results" is the ONLY acceptable idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence (audio/video/network/UI/sysfs — "physical proofs"); (D) the §11.4 anti-bluff covenant family (§11.4.1/§11.4.2/§11.4.6/§11.4.7/§11.4.27/§11.4.50/§11.4.52/§11.4.68/§11.4.69/§11.4.83) is the operative truth-discipline — tests AND HelixQA Challenges bound equally per the historical forensic anchor "we had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used"; (E) loop terminates ONLY on all-conditions-met, explicit operator STOP, host-session-safety demand, or scheduled wake on a known-future-actionable signal.

Composes with §11.4 / §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.7 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.58 / §11.4.68 / §11.4.69 / §11.4.70 / §11.4.83 / §11.4.85 / §11.4.86 / §12.10. Pre-build gate CM-COVENANT-114-87-PROPAGATION + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule Constitution.md §11.4.87.

Non-compliance is a release blocker regardless of context. No escape hatch — no --idle-OK, --skip-endless-loop, --bluff-permitted-for-this-task, --metadata-only-test-suffices, --no-physical-proof-required flag exists.

§11.4.88 — Background-push mandate: commit-lock release immediately after commit, push runs detached (User mandate, 2026-05-26)

Forensic anchor: 2026-05-26 a single commit_all.sh held its flock ~5 hours because do_push ran synchronously after the commit landed — every subsequent commit was blocked on a slow mirror push that had nothing to do with the local commit's durability. Implementation seam for §11.4.87(B) zero-idle.

The mandate: (A) `.git/.commit_all.lock` MUST be released IMMEDIATELY after `git commit` returns 0 — the commit is durable on local disk regardless of remote push outcome; (B) push runs detached via `nohup ./push_all.sh ... > <log> 2>&1 & + disown` — orchestrator's exit code reports COMMIT success, NOT push success; (C) `push_all.sh` acquires per-remote flock `.git/.push.<remote>.lock` so concurrent invocations targeting same remote serialize but different-remote invocations run in parallel; (D) backgrounded push failures land in `qa-results/push_failures/<ts>_<remote>.log` — next autonomous-loop tick checks per §11.4.87(A) "no external dependency in-flight" gate; (E) synchronous-push escape: explicit `--sync-push` CLI flag preserves legacy behaviour for §11.4.41 force-push merge-first audit paths.

Composes with §2.1 / §9.2 / §11.4.41 / §11.4.42 / §11.4.71 / §11.4.87. Pre-build gates `CM-COVENANT-114-88-PROPAGATION` + `CM-BACKGROUND-PUSH-WIRED` + paired §1.1 meta-test mutations.

Canonical authority: constitution submodule [`Constitution.md`](#) §11.4.88.

Non-compliance is a release blocker. Synchronous push (without `--sync-push`) = §11.4 PASS-bluff at execution layer. No escape hatch beyond `--sync-push` for force-push events.

§11.4.89 — Background test execution mandate (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27):

"Any tests we are executing, especially long test cycles, MUST BE performed in background in parallel with main work stream! This MUST NOT block our capabilities to work on queued workable items. Main work stream can be blocked or sit iddle only if absolutely needed and if it depends hard on results of some background execution."

Forensic incident: 2026-05-27 conductor invoked `pre_build_verification.sh` synchronously with foreground timeout 360, blocking main stream for 6-7 minutes on `$JV/$JW/$JX/$JY` scaffolding work. Symmetric anchor to §11.4.88 (background push) at the test-execution layer.

Mandate: (A) Long-running tests (>30s expected: `pre_build`, `meta_test`, `test_all_fixes`, `recent_work_validate`, HelixQA banks, 4-phase cycles, full-suite retests, `audio_loop_supervisor`, `dual_display_record`) MUST run via `nohup ... > <log> 2>&1 & + disown` with log placed under known dir (`qa-results/<test_id>_<ts>.log`); (B) Main stream proceeds to §11.4.42 priority queue immediately; (C) Hard-dependency gating: poll exit-status file or `pgrep -af <test>` before steps that need exit code — surface as §11.4.66 interactive options if test still running; (D) Failures land in `<log>` files, next loop tick checks; (E) Foreground execution

permitted ONLY for <30s tests OR explicit operator authorisation;
(F) Per-script flock serialises same-script invocations, different-script invocations parallel.

Composes with §11.4.42 / §11.4.66 / §11.4.82 / §11.4.84 / §11.4.85 / §11.4.87 / §11.4.88. Pre-build gates CM-COVENANT-114-89-PROPAGATION + CM-BACKGROUND-TEST-EXECUTION-WIRED + paired §1.1 meta-test mutations.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.89.

Non-compliance is a release blocker. No escape hatch beyond explicit per-invocation operator authorisation.

§11.4.90 — Obsolete status + per-item obsolescence audit (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "Bug No 6 - Albums cover etc. seems obsolete after latest request for new behavior when audio and video content is being played ... mark obsolete tickets with some light gray background ... text - the description to be strikethrough styled ... review all existing open or resolved workable items if they are obsolete - not valid any more ... There MUST NOT be any mistake! No bluff is allowed of any kind!"

§11.4.15 Status closed-set extended with terminal Obsolete (→ Fixed.md) value (orthogonal to Type per §11.4.16). Obsolescence reasons (closed vocabulary): superseded-by-design-change | superseded-by-later-mandate | feature-removed | duplicate-of | unsupported-topology. Every Obsolete heading MUST carry ****Obsolete-Details:**** line (Since + Reason + Superseding-item + Triple-check evidence) within 8 non-blank lines. §11.4.23 colorizer adds cell-status-obsolete class — light-gray #E0E0E0 background + strikethrough description. Audit cadence: every release-gate sweep per §11.4.40 + §11.4.42. Triple-check non-negotiable per operator mandate. Composes §11.4.15 / §11.4.16 / §11.4.19 / §11.4.21 / §11.4.23 / §11.4.33 / §11.4.34 / §11.4.40 / §11.4.42 / §11.4.66 / §11.4.71. Pre-build gates CM-COVENANT-114-90-PROPAGATION + CM-ITEM-OBSOLETE-DETAILS + CM-OBSOLETE-COLORIZER-WIRED + paired §1.1 mutations.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.90. Non-compliance is a release blocker.

§11.4.91 — Summary-doc clarity mandate (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27):

"Summary docs - Issues_Summary some not clear one line descriptions - like 'Composes with' ... For each workable item we MUST HAVE clearly understandable meaning ... every team member can clearly understand what that particular workable item is exactly about! There cannot be misunderstanding or unclarity of any kind and no bluff allowed!"

Every summary entry (Issues_Summary, Fixed_Summary, README doc-link, Status_Summary page 1+2, all one-liners) MUST contain self-contained meaningful description ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden one-liner anti-patterns: section labels (Composes with, Closure criteria, Fix direction, etc.); bare metadata fragments (Critical, Bug, In progress, etc.); section-marker echoes; \$-letter alone. Generators (generate_issues_summary.sh / generate_fixed_summary.sh / update_readme_doc_links.sh / generate_status_summary.sh) MUST extract from H1/H2 heading line per §11.4.54 ATM-NNN convention, NEVER from arbitrary downstream text. Generators refuse anti-pattern rows — emit (MISSING DESCRIPTION – fix source heading) placeholder with visual highlight. Pre-build gate CM-SUMMARY-CLARITY-DESCRIPTIONS scans every summary; anti-pattern match = FAIL. Audit cadence: every §11.4.40 + §11.4.42 sweep. Composes §11.4.12 / §11.4.19 / §11.4.23 / §11.4.44 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.74.

Canonical authority: constitution submodule Constitution.md §11.4.91. Non-compliance is a release blocker.

§11.4.92 — Multi-pass change-evaluation discipline (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "Every change to the project or codebase we do MUST BE evaluated in several passes and in in-depth analysis for potential new issues or problems it can introduce! ... no bluff of any kind! After we do change or set of changes this mandatory steps MUST BE taken!"

Every non-trivial change MUST pass 5-pass evaluation BEFORE commit-ready: **(Pass 1)** Main-task verification — change achieves stated goal, captured-evidence per §11.4.5/§11.4.69; **(Pass 2)** Regression-blast-radius analysis — enumerate every direct dependency, demonstrate no contract break; **(Pass 3)** Cross-feature interaction analysis — parallel features sharing state/timing/hardware/shell environment audited; **(Pass 4)** Deep-research validation per §11.4.8 — external precedent OR "NO external solution found — original work" + CodeGraph queries

per §11.4.78/§11.4.79; **(Pass 5)** Anti-bluff confirmation per §11.4/§11.4.1/§11.4.6/§11.4.27/§11.4.50/§11.4.52/§11.4.69/§11.4.83 — no new bluff surface introduced. Documentation per pass (commit footers OR docs/ entries OR qa-results/ evidence). Only AFTER all 5 passes complete may commit/push/test/release proceed. Trivial exemption: typo / revision-bump / MD-export-regen IF zero source touched AND commit message cites exemption explicitly. Composes §11.4.4 / §11.4.5 / §11.4.6 / §11.4.8 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.78 / §11.4.79 / §11.4.82 / §11.4.83 / §11.4.85 / §11.4.87 / §11.4.89. Pre-build gates CM-COVENANT-114-92-PROPAGATION + CM-MULTI-PASS-EVALUATION-EVIDENCE + paired §1.1 mutations.

Canonical authority: constitution submodule Constitution.md §11.4.92. Non-compliance is a release blocker.

§11.4.93 — SQLite-backed single-source-of-truth for workable items (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "There MUST be single source of truth for all of our workable items - SQLite database ... proper scripts (we recommend Go programs) ... reduce a chance for sync to be broken ... generate always all docs from DB or to re-generate Db from all docs we have in opposite direction".

Text-based Issues/Fixed/Summary/CONTINUATION constellation converted to SQLite-DB-backed single source of truth. DB at docs/.workable_items.db (gitignored per §11.4.30 + §11.4.77 regen mechanism). Schema mandatory tables: items (atm_id PK + Type + Status incl. Obsolete + Severity + title + description ≥40chars + created/modified + composes_with JSON + current_location); item_history (append-only audit per §11.4.34 By/Reason/Evidence); obsolete_details (§11.4.90); operator_block_details (§11.4.21); firebase_metadata (§11.4.47); meta (schema version + last sync + integrity hash). Go binary at cmd/workable-items/: sync md-to-db / db-to-md / diff / validate / add / close. Bidirectional regen byte-identical round-trip (closed-set tolerance for whitespace/section-order). commit_all.sh refuses on diff non-empty. sync_issues_docs.sh invokes Go binary. pre_build runs workable-items validate. Anti-bluff: unit + integration + stress (1000-row insert + 10 concurrent writers) + chaos (mid-write SIGKILL + corrupt-DB recovery + disk-full) + paired §1.1 mutation + HelixQA Challenge CME-WORKABLE-ITEMS-001. 6-phase migration: §LA Issues entry → Go binary scaffold + DDL → md-to-db → db-to-md round-trip CI → generator shims → text-direct edits prohibited. Cross-project: Go binary lives in constitution submodule (constitution/scripts/workable-items/) per §11.4.74. Composes §11.4 / §11.4.12 / §11.4.15 / §11.4.16 / §11.4.17 / §11.4.19 / §11.4.21 / §11.4.27 / §11.4.30 / §11.4.33 / §11.4.34 / §11.4.42 / §11.4.43 / §11.4.44 / §11.4.45 / §11.4.50 /

§11.4.52 / §11.4.53 / §11.4.54 / §11.4.55 / §11.4.56 / §11.4.57 /
§11.4.58 / §11.4.60 / §11.4.65 / §11.4.74 / §11.4.83 / §11.4.85 /
§11.4.86 / §11.4.87 / §11.4.89 / §11.4.90 / §11.4.91 / §11.4.92. Pre-
build gates CM-COVENANT-114-93-PROPAGATION + CM-WORKABLE-ITEMS-DB-
PRESENT + CM-WORKABLE-ITEMS-MD-DB-IN-SYNC + paired §1.1 mutations.

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.93. Non-compliance is a release blocker — text-based-only
trackers are §11.4 PASS-bluff at data-architecture layer.

§11.4.94 — Zero-idle priority-first parallel-by- default operating mode (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "We
MUST NEVER sit iddle / wait or sleep if there is possibility for us
to work on something ... Always check if there is a possibility to
work on something while we are not working actively on
something! Pick always by priority - most critical workable items
and other tasks MUST BE done first! ... Stay still / iddle if nothing
is left to be done at all or waiting for something that is blocking
us / you!!!"

§11.4.94 binds
§11.4.20+§11.4.42+§11.4.58+§11.4.70+§11.4.72+§11.4.82+§11.4.
87+§11.4.88+§11.4.89 into single always-on enforcement: (A) Idle
ONLY when every queued item genuinely blocked on external dep
(hardware / network upstream / build/test completion conductor
cannot accelerate) OR operator STOP OR §12 host-safety; "don't
see what to do" is NEVER valid; (B) before ANY wake/sleep MUST
survey parallel-work feasibility per §11.4.42+§11.4.72+§11.4.87,
identify non-contending items, dispatch in parallel per §11.4.20/
§11.4.70 (subagent) + §11.4.58 (PWU disjoint scope) + §11.4.89
(background long tests); (C) priority order MANDATORY — pick
highest-severity+§11.4.72 audio-first the conductor can
autonomously progress; (D) subagent-driven default for non-
trivial; (E) background default for >30s wall-clock work via
nohup+disown; (F) stability-preserving — composes with §11.4.92
multi-pass + §11.4.84 quiescence + §12.6-§12.9 host safety; (G)
progress updates surfaced when operator asks OR at milestone
boundaries. Anti-pattern forbidden: scheduling wake without
queue survey; "nothing to do — sleeping" with non-trivial items
progressable; serialising parallelisable work; picking lower
priority over higher. Pre-build gates CM-COVENANT-114-94-PROPAGATION
+ CM-PARALLEL-WORK-AUDIT + paired §1.1 mutations.

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.94. Non-compliance is a release blocker.

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "We shall not Git ignore our workable items SQLite DB since it is our single source of truth ... workable items SQLite DB regularly committed and pushed to all upstreams!"

§11.4.93's earlier "gitignored per §11.4.30" clause is AMENDED — the DB at docs/workable_items.db is TRACKED in git, NEVER gitignored. It IS authoritative source data, NOT a build artefact. Every workable-items sync md-to-db that mutates state MUST stage+commit+push the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. WAL-checkpoint (PRAGMA wal_checkpoint(TRUNCATE)) required before commit-stage so transient .db-wal + .db-shm sidecars (gitignored per §11.4.30) are safely discardable. §11.4.77 regeneration mechanism does NOT apply — DB IS the source. Destructive DB ops require §9.2 hardlinked-backup + operator authorization; §11.4.41 force-push merge-first applies if DB history ever needs rewrite. Pre-build gates CM-COVENANT-114-95-PROPAGATION + CM-WORKABLE-ITEMS-DB-TRACKED + paired §1.1 mutation.

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.95. Non-compliance is a release blocker.

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "Are there except AOSP build process any other active jobs being done at the moment? Can we work on something in parallel while build is in progress so we slowly cleanup our slate? ... If yes, and that was not already clear by our root Constitution, we should add all mandatory details into it ... do as much as possible work in background in parallel with main work stream and oreferably using subagents-driven approach!"

Operational catalogue for the canonical long-running workload (5-7h AOSP containerised build per §12.9):

SAFE during build: (A) MD/docs work under docs/+README+CLAUDE/AGENTS/QWEN+Status+Issues/
Fixed+changelogs+research notes; (B) generator/helper script work under scripts/ scripts/testing/ scripts/llm/ scripts/firebase/; (C) pre-build + meta-test gate authoring + paired §1.1 mutations; (D) on-device test scripts under tests/test_*.sh; (E) constitution submodule edits + push to 6 remotes; (F) any submodule commit + push per §11.4.88; (G) D3/D4 read-only live-ADB probes (dumpsys/getprop/cat /proc/asound/screencap/logcat); (H) subagent dispatch per §11.4.20/§11.4.70 + §11.4.84 quiescence;

(I) web research + external API queries with §11.4.10 credentials;
(J) workable-items DB ops per §11.4.93+§11.4.95; (K) pre-build + meta-test execution backgrounded per §11.4.89.

UNSAFE during build: (α) git checkout/reset --hard/clean -df on source tree (use git worktree instead); (β) mass file deletes/renames under device/+frameworks/+hardware/+vendor/+kernel-5.10/+external/+packages/+bionic/+bootable/+build/+system/+cts/+tools/+prebuilts/; (γ) submodule pointer updates affecting built APKs (defer to between-build windows); (δ) out/directory mutations; (ε) make clean/m clobber/rm -rf out/; (ζ) container destruction; (η) disk-filling operations breaching §12.9 free-space minimum; (θ) §12 host-session-safety breaches.

Conductor responsibility: before EVERY pause point during long build, consult catalogue, identify (A)-(K) queue items per §11.4.42+§11.4.72, dispatch ≥ 1 per §11.4.20/§11.4.70 subagent default + §11.4.89 background. "Build running, nothing else to do" NEVER true per §11.4.94+§11.4.96. Pre-build gates CM-COVENANT-114-96-PROPAGATION + CM-PARALLEL-WORK-DURING-BUILD-AUDIT + paired §1.1 mutations.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.96. Non-compliance is a release blocker.

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "keep it working, we should do as much as possible, if not it all but as much as we can as long as there is iddle time! it MUST be used! ... keep us updated about all progress and all phisycal proofs and gathered data as you progress through all open workable items!"

Operating-mode capstone strengthening
§11.4.87+§11.4.94+§11.4.96: (A) every minute of conductor idle time during which work could autonomously progress AND not genuinely blocked = §11.4.97 violation; "as much as possible, if not it all but as much as we can" is operative — dispatch CONTINUOUSLY through entire idle window, not just at scheduled wakes; (B) progress-update cadence — emit operator-facing 1-line update at every: commit landed / subagent return / constitutional anchor / captured evidence / milestone closure; no operator prompt required; (C) continuous physical-proof gathering per §11.4.5+§11.4.6+§11.4.69 — every autonomous closure cites captured-evidence; evidence path goes into §11.4.93 item_history.evidence_path when DB lands; (D) composes with §11.4.5/6/13/20/27/42/50/52/69/70/72/83/85/87/88/89/94/96; (E) idle-only-when-blocked closed-set unchanged from §11.4.94(A). Pre-build gates CM-COVENANT-114-97-PROPAGATION + CM-IDLE-TIME-AUDIT + paired §1.1 mutations.

Canonical authority: constitution submodule Constitution.md
§11.4.97. Non-compliance is a release blocker.

§11.4.98 — Full-Automation Anti-Bluff Mandate — Live tests MUST be re-runnable end-to-end without manual intervention (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28): "Make sure we have full automation testing of all scenarios with real bot, main group and users without any manual intervention or contribution of real user! Everything MUST BE fully automatic and autonomous! These tests MUST BE able to rerun endless times when needed! ... Make sure there is no false positives in testing! Every test and its results MUST obtain real proofs of everything working! No bluff is allowed!"

Composes with §11.4 + §11.4.2 + §11.4.5 + §11.4.50 + §11.4.85 + §11.4.87 + §11.4.89 + §11.4.94 — closes the **manual-intervention gap** they did not explicitly forbid. A live/integration/e2e/Challenge test requiring a human action during execution (typing a chat message, clicking a UI, hand-triggering a webhook, anything beyond test start + PASS/FAIL report) is **by definition a §11.4 PASS-bluff at the automation layer**, regardless of how thorough the manual run is — cannot run continuously in CI, cannot validate regressions between manual runs, human dependency masks drift.

(A) Binding rule: every test this Constitution governs — unit/integration/e2e/Challenge/stress/chaos/live — MUST be fully self-driving end-to-end; reports PASS/FAIL/SKIP-with-reason without any further human action after startup. (B) Single permissible exception: one-time credential bootstrap OUTSIDE test execution (.env from vault, shell exports in ~/.bashrc, OAuth approval at first install, MTProto session activation at first run) — configuration, not test driving. (C) Concrete requirements for live messenger/channel/agent tests: (1) no "operator MUST type a message" prompts — drive programmatically (MTProto for Telegram, real-user-token API for Slack, IMAP-test-account for email, webhook fixture, in-process loopback — never human keystrokes); (2) no hard-coded session UUIDs that collide with active dev session (Herald 2026-05-28 lesson: `claude --resume <UUID>` on same UUID as dev session returns silent exit -1); (3) no 60s human-response windows (§11.4.50 determinism violation); (4) re-runnability proof — PASS at -count=3 consecutive automated invocations with self-cleaning state; (5) §11.4.98 obsolescence audit — every existing test classified COMPLIANT vs NON-COMPLIANT; (6) no false-positive PASS — silent-skip-reported-as-PASS forbidden, stale-evidence forbidden, SKIP-with-reason per §11.4.3 is correct. (D) Composes with §11.4.85 + §11.4.89 + §11.4.87 + §11.4.94 — together = continuously-validated, fully-automated, non-flake, anti-bluff regime. (E) Inheritance per §11.4.35 — every consuming

repo's CLAUDE.md/AGENTS.md/QWEN.md restates citing literal anchor 11.4.98; pre-build gate CM-COVENANT-114-98-PROPAGATION enforces literal presence; paired §1.1 mutations strip → gates FAIL. (F) Enforcement: commit adding manual-action test BLOCKED at release-gate; manual-dependency test not rewritten within 30 days graduates to §11.4.90 Obsolete citing §11.4.98 as obsolescence reason.

Canonical authority: constitution submodule Constitution.md §11.4.98. Non-compliance is a release blocker.

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate — instructions/guides/manuals MUST be verified against latest official online sources BEFORE publication (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28): "Make sure we ALWAYS check against latest versions of services we use web / online docs before creating instructions! This situation is illustration of how we can misguide ourselves or get banned! ... These are mandatory rules / constraints and the result is consistency and safety of created instructions, guides and manuals!"

Case study (Herald 2026-05-28): first-draft Herald MTProto setup guide recommended VoIP/Google-Voice/Twilio fallback AND omitted the recover@telegram.org pre-login email — both directly contradicted Telegram's official docs + gotd/td maintainer guidance, and following the guide could have caused a permanent Telegram account ban. Misguidance-by-stale-docs is the same severity class as a §11.4 PASS-bluff at the documentation layer.

Composes with §11.4.4 + §11.4.5 + §11.4.8 + §11.4.92 Pass 4 + §107 + §11.4.98. Closes the gap §11.4.92 Pass 4 alludes to but does not explicitly mandate. (A) Pre-commit cross-reference: every operator-facing instruction document MUST (1) fetch the LATEST official online docs of the service/library via WebFetch/MCP/direct browsing (NEVER training data, memory, or prior committed docs); (2) cross-reference each instruction against the source; (3) seek secondary authoritative sources when the official docs are sparse/silent (library maintainer SUPPORT.md, official changelogs, vetted community FAQs); (4) cite source URL + date in a "## Sources verified" footer on the doc; (5) cite the cross-reference in the commit-message footer (Sources verified <date>: <urls>). (B) Negative findings: gaps/silences/contradictions MUST be documented explicitly so the next reader doesn't assume absence-of-contradiction is authoritative agreement. (C) Re-verification cadence: documents older than 6 months are STALE — re-verify before citing as operator authority, at every vN.0.0 release boundary, on service breaking-change announcements, or

when an operator reports an error following the guide. (D) Risk-classified services (Telegram/WhatsApp messengers, cloud APIs, payment systems, AI/LLM providers, code-hosting services, package managers) — 90-day max staleness; explicit safety warnings cited against latest policies. (E) Composes with §11.4.92 Pass 4 but is INDEPENDENT — cannot substitute. (F) Inheritance per §11.4.35 — restate literal anchor 11.4.99 in every consumer's CLAUDE/AGENTS/QWEN; pre-build gate CM-COVENANT-114-99-PROPAGATION (when implemented) enforces; paired §1.1 mutations strip → gates FAIL. (G) Enforcement: commit without "Sources verified " doc-footer AND commit-message-footer BLOCKED at release-gate; stale-beyond-grace docs graduate to §11.4.90 Obsolete with Reason=stale-documentation.

Canonical authority: constitution submodule Constitution.md §11.4.99. Non-compliance is a release blocker.

§11.4.100 — RETIRED. Demoted to consumer project (ATMOSphere video-color/visual-quality fidelity) per §11.4.17/ §11.4.35 — project-specific (RK3588/MPV/Arvus), not universal. See the consuming project's Constitution/CLAUDE/AGENTS/QWEN.

§11.4.101 — Autonomous-decision-over-blocking mandate (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28):

"when working in endless working loop fully autonomously try to decide most properly about points which would block execution and wait for us. If we haven't answered now work would be blocked whole night! If possible and if that will not cause any issues make proper and most reliable and safe decision so we achieve maximal efficiency and work gets fully done!"

When operating in an autonomous / endless-loop mode (per §11.4.87), the agent MUST minimize operator-blocking and instead make the safe, reliable, reversible decision itself — so work is NOT stalled (e.g. overnight) waiting for input. §11.4.87 says keep working; §11.4.101 says HOW to clear the decision points that would otherwise force a stop-and-wait.

Decision rule (closed-set — proceed autonomously when ALL hold): (a) the action is reversible OR has a captured pre-op backup per §9.2; (b) the agent can determine the safe choice from captured evidence per §11.4.6 (no guessing — LIKELY is not a determination); (c) a wrong choice's blast radius is bounded AND recoverable; (d) it composes with anti-bluff §11.4, host-safety §12, data-safety §9.

Block-only-when rule (BLOCK via §11.4.66 ONLY when ALL hold): the action is irreversible AND high-blast-radius AND the safe choice cannot be determined from evidence — e.g. external-account state the agent cannot inspect, hardware it cannot access, destructive ops without backup, force-push (also §9.2 + §11.4.41), spending / sending to third parties. Operator-blocked per §11.4.21 is reached only after this rule fires AND the self-resolution-exhaustion audit completes.

Maximize-progress-while-blocked: an unavoidable block parks one work unit, it does not pause the loop — the agent MUST keep progressing every NON-blocked item in parallel per §11.4.87 + §11.4.94. Posing the question and going idle is a §11.4.94 + §11.4.97 violation.

Composes with §11.4.6 / §11.4.21 / §11.4.40 / §11.4.41 / §11.4.66 / §11.4.87 / §11.4.94 / §9.2 / §12. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-101-PROPAGATION enforces the literal anchor 11.4.101 across the consumer fleet; paired §1.1 meta-test mutation strips the literal → gate FAILs (gate-code = separate work item). No escape hatch — no --always-block-on-decision, --never-decide-autonomously, --skip-decision-rule, --block-without-self-resolution flag exists.

Canonical authority: constitution submodule Constitution.md §11.4.101.

Non-compliance is a release blocker regardless of context.

§11.4.102 — Mandatory systematic-debugging activation + always-loaded skill-discovery + plugin-dependency availability (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29):

"Make sure that we ALWAYS trigger / start the "/superpowers:systematic-debugging" skills when any issues happen! If this is possible to activate and use in this situations out of the box when we spot problems / issues / bugs / misalignments / inconsistencies we MUST activate the skill(s) and make strongest efforts in full in depth analysis / debugging and determine root causes of all problem or obtain relevant data and information we need! ... we MUST make sure that "/using-superpowers" skill is ALWAYS loaded, applied and used! All dependencies (plugins) that Claude Code or other market places are offering MUST BE installed if these are not already available for loading and use!"

Three cooperating invariants — the difference between guess-and-retry and investigate-to-root-cause-first.

(A) Mandatory systematic-debugging activation. On ANY spotted issue / bug / test failure / gate failure / regression / misalignment / inconsistency / unexpected behaviour, the agent **MUST** activate `superpowers:systematic-debugging` (or the platform-equivalent structured-debugging discipline) **BEFORE proposing, writing, or applying any fix** — the **Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST**. Full four-phase arc: root-cause (read the real failure, reproduce, gather facts) → pattern (classify the defect, search for the same pattern elsewhere) → hypothesis (form a falsifiable cause, prove/disprove with captured evidence) → implementation (design the fix only against the proven root cause). Guess-and-retry, symptom-patching, and re-running a failed test hoping it passes ("probably transient / flaky") **WITHOUT** a completed investigation are §11.4.102 violations. Real-world signal: calling a failure transient / flaky / intermittent / probably-timing without captured forensic evidence is simultaneously a §11.4.6 (no-guessing) and §11.4.7 (demotion-evidence) violation — §11.4.102 makes the corrective response mechanical (the classification is forbidden until the systematic-debugging arc proves it).

(B) Mandatory always-loaded using-superpowers. `superpowers:using-superpowers` (or the platform-equivalent skill-discovery / capability-index discipline) **MUST** be loaded and applied at session start and consulted before any task. Operative rule: before acting on ANY request, survey available skills; if ANY skill could apply — even at 1% relevance — it **MUST** be invoked rather than improvised from memory. Skipping skill-discovery at session start, or improvising a task a loaded skill exists for, is a §11.4.102 violation.

(C) Mandatory plugin / dependency availability. Every skill plugin / marketplace package / capability dependency the project relies on (Claude Code, its skill marketplaces, or any other runtime's plugin ecosystem) **MUST** be installed + loadable **BEFORE** the dependent work proceeds. A missing plugin that blocks a mandated skill (e.g. `superpowers` absent so systematic-debugging cannot launch) is a **release-blocker** until installed + confirmed loadable. Install mechanism: the runtime's own plugin/marketplace install path — for Claude Code the in-session /plugin marketplace flow (add marketplace → install plugin → confirm the skill appears in the available-skills list); for other runtimes the documented package/extension installer. Anti-bluff: confirm by observing the skill in the live capability list, never assume install succeeded (install exit 0 ≠ skill loadable, per the §11.4.80 lesson).

Composes with §11.4.4 (test-interrupt — first action after STOP is launch systematic-debugging), §11.4.6 (no-guessing — (A) is the procedural enforcement producing the facts §11.4.6 demands),

§11.4.7 (demotion-evidence — the arc captures same-conditions evidence), §11.4.8 (deep-web-research — inside the hypothesis phase), §11.4.43 (TDD-fix — RED test written against proven root cause), §11.4.70 (subagent-driven — the investigation is a natural subagent dispatch), §11.4.82(A) (generalises Phase-1-forensic-before-speculative-patch to ANY fix + adds skill-discovery + plugin-availability), §11.4.92 (feeds Pass 1 + Pass 4).

Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-102-PROPAGATION (literal 11.4.102 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --skip-systematic-debugging, --guess-and-retry-OK, --symptom-patch-permitted, --skip-skill-discovery, --plugin-optional, --missing-plugin-is-warning flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.102.

Non-compliance is a release blocker regardless of context.

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29):

"Do this working approach continuously and make it part of regular working routine and add it to the Constitution Submodule documented fully."

Promotes the proven multi-stream operating pattern into the project's **standing default working routine** (not a per-request opt-in). Binds §11.4.87/§11.4.88/§11.4.89/§11.4.94/§11.4.96 and adds the load-bearing invariant: **the main work stream MUST always stay FREE.**

(A) Main stream stays FREE. ALL commit AND push operations run detached (nohup ... & + disown, per §11.4.88 commit-lock-release-immediately + detached-push) — the main stream returns to the priority queue the moment the local commit is durable, never blocking on a push or a slow mirror. **(B) ≥3 parallel background streams at all times + auto-backfill** (User mandate 2026-05-29; raised from ≥2) — run **at least three** subagent-driven background streams (per §11.4.70/§11.4.20, isolated per §11.4.58 PWU worktree + §11.4.84 quiescence) alongside the main stream whenever three-plus non-contending actionable items exist; **the moment any one stream is FULLY done, a new stream MUST immediately start and take its place** (claim next-highest-priority non-contending item), so the active-stream count NEVER drops below 3 while actionable items remain. Idle below 3 is permitted ONLY when no remaining non-contending actionable items OR all remaining are externally blocked (§11.4.94/§11.4.97/§11.4.101). Standing band 3-6,

bounded above by §12.6 60% memory + §11.4.58 6-agent cap. **(C) Most-critical + most-visible first; audio always top** per §11.4.72 + §11.4.42 priority order. **(D) Safe-during-build scope only** — while the 42 GB containerised AOSP rebuild (§12.9) or any heavy gradle/m -j build runs, streams restrict to the §11.4.96 SAFE catalogue (investigation/forensic/docs/test-authoring/gate-authoring/read-only probes/submodule edits/research/DB ops/backgrounded pre-build+meta-test); NEVER a second concurrent heavy build (§12.8). **(E) Heavy anti-bluff on every closure** — root cause proven or UNCONFIRMED:/UNKNOWN:/PENDING_FORENSICS: per §11.4.6, captured evidence per §11.4.5+§11.4.69, deterministic consistency per §11.4.50, paired §1.1 mutations, §11.4.102 systematic-debugging on any spotted problem. **(F) Idle ONLY when genuinely externally blocked** (hardware/network upstream/in-flight build-test-push completion) OR operator STOP OR §12 host-safety, per §11.4.94(A)+§11.4.97; "nothing visible to do" with progressable items is NEVER valid; a block parks one work unit, not the loop (§11.4.101).

Composes with §11.4.58 / §11.4.70 / §11.4.72 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.94 / §11.4.96 / §11.4.97 / §11.4.101 / §11.4.102 / §11.4.42 / §11.4.84 / §12.6 / §12.7 / §12.8 / §12.9 / §9.2. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-103-PROPAGATION (literal 11.4.103 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --block-main-stream, --synchronous-commit, --synchronous-push, --single-stream-only, --skip-parallel-streams, --serialise-actionable-work, --idle-without-queue-survey flag.

Canonical authority: constitution submodule Constitution.md §11.4.103.

Non-compliance is a release blocker regardless of context.

§11.4.104 — Participant identity, attribution & notification-tagging (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31):

"Every supported messenger must relate messages to participants (Subscribers/Users); the same logical person may have a different username on every messenger. Workable items must carry who created them and who they are assigned to. Notifications must @-tag the right participant — but never the operator (who drives the system) and never the system agent."

MANDATORY for every consumer that ships a messenger/ notification surface (Herald + flavor binaries are the reference impl; others inherit per §11.4.35). Detailed spec: Herald docs/design/PARTICIPANT_ATTRIBUTION.md — restated, not redefined. **(A)**

Participant identity. Every messenger MUST relate messages to a **Participant** (logical Subscriber/User); the SAME person MAY have a DIFFERENT username per messenger — modelled as a logical subscriber (canonical messenger-neutral handle, kind ∈ {human, agent, service}) + per-channel aliases (channel, channel_user_id, the @username used for tagging). Canonical handle closed set: Claude (reserved system-agent sentinel; never tagged) OR a subscriber @username. **(B) Operator = env var, not a DB flag** — the one human who drives via the agent CLI, designated by HERALD_<CHANNEL>_OPERATOR_USERNAME (e.g. HERALD_TGRAM_OPERATOR_USERNAME); a normal Participant whose handle equals that value. **(C) Workable items MUST carry created_by + assigned_to** (canonical handles): CLI-prompt→Operator; system/agent-detected→Claude; received-message→sender's resolved @username. assigned_to defaults to Operator, overridable. Legacy items carry "" and MUST still parse + validate. **(D) Tagging matrix:** tag assigned_to if it is a human ≠ Operator; tag created_by if it is a human ≠ Operator ≠ Claude; NEVER tag Claude (system) or the Operator (no self-ping); de-dup; resolve each handle to the channel @username, skip if not on that channel. **(E) Anti-bluff (composes §11.4):** real SQLite round-trip with the new columns byte-identical (incl. legacy fixtures WITHOUT the fields); tagging matrix proven by a truth-table + a cell-flip mutation forcing FAIL; E2E real-event → real-message asserting the exact @username + a NEGATIVE case proving the Operator is NOT tagged; evidence under docs/qa/<run-id>/.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.5 / §11.4.69 / §11.4.50 / §11.4.91 / §11.4.93 / §11.4.95 / §1.1. Classification: universal (§11.4.17) — projects with no messenger surface inherit it latently (binds the moment they ship one, per the §11.4.96 pattern). Propagation gate CM-COVENANT-114-104-PROPAGATION (literal 11.4.104 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --skip-attribution, --no-participant-tagging, --tag-operator-anyway, --attribution-later flag.

Canonical authority: constitution submodule Constitution.md §11.4.104; detailed spec Herald docs/design/PARTICIPANT_ATTRIBUTION.md.

Non-compliance is a release blocker.

§11.4.105 — Natural-language intent recognition & clarification (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31):

"Users must NOT need to know command syntax (no COMMAND: ... prefix). They send a clear natural-language message; the System determines the intent. The System recognizes the commands it has; if none matches it infers the exact intent; if it is totally unable it replies, tags the user (@user ...), and asks to clarify precisely. We MUST always do our best to determine exact intent so we never annoy end users. This is a CORE part of the System."

MANDATORY for every consumer that ships a messenger/command surface (Herald + flavor binaries are the reference impl; others inherit per §11.4.35). Detailed spec: Herald docs/design/INTENT_RECOGNITION.md — restated, not redefined. **(A) No required command syntax.** Users MUST NOT be required to know any command syntax (no COMMAND: prefix); they send plain natural language and the System determines the intent. **(B) Three-tier resolution** (first that succeeds wins): TIER 1 — recognize the System's existing command set from natural language → action (confident deterministic match); TIER 2 — when no command matches, infer the exact intent (LLM dispatch maps language → action), NEVER guessing; TIER 3 — when neither a command nor a confident intent can be determined, REPLY to the message, TAG the sender (@username, resolved via the §11.4.104 IdentityResolver) and ask a PRECISE clarifying question NAMING the candidate intents — no guessing, no silent drop. **(C) Never guess, never drop:** a wrong action is worse than a clarifying question (composes §11.4.6 no-guessing); a message is NEVER silently dropped; only genuine ambiguity reaches Tier 3, which always replies-tags-and-asks. **(D) Anti-bluff (composes §11.4):** every tier ships unit + integration + E2E + full-automation tests with real captured evidence — Tier 1 truth-table (natural-language → action+fields, plus conservative negatives that MUST fall through to "no match"); Tier 3 E2E whose recorded reply body is EXACTLY @<sender> <specific question> + a NEGATIVE proving a clear command does NOT trigger clarify; a paired §1.1 mutation breaking the confidence guard (false-match) OR dropping the clarify tag MUST FAIL a test; evidence under docs/qa/<run-id>/.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.6 (no-guessing) / §11.4.104 (clarify reply tags the sender) / §11.4.5 / §11.4.69 / §11.4.98 / §1.1. Classification: universal (§11.4.17) — projects with no messenger surface inherit it latently (binds the moment they ship one, per the §11.4.96 pattern). Propagation gate CM-COVENANT-114-105-PROPAGATION (literal 11.4.105 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --require-command-syntax, --guess-intent-ok, --skip-clarify, --drop-on-ambiguous flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.105; detailed spec Herald docs/design/INTENT_RECOGNITION.md.

Non-compliance is a release blocker.

§11.4.106 — Docs Chain — mechanical documentation/DB sync engine (Operator mandate, 2026-05-31)

Forensic anchor — operator mandate (2026-05-31): Docs Chain is the canonical mechanical enforcer of the documentation-sync mandates; consumers **MUST** use the engine instead of ad-hoc per-project doc-sync scripts, register their chains via per-context YAML, and never accept a faked transform.

MANDATORY for every consumer. Docs Chain (the vasic-digital/docs_chain engine — a universal Go bidirectional document-and-database dependency-propagation engine) is the canonical mechanical enforcer of the documentation-sync mandates. Detailed spec: the engine docs ~/Projects/docs_chain → docs/CONSTITUTION_INTEGRATION.md (distribution + inheritance + anchor mapping table) + docs/USE_CASE_CATALOGUE.md (chain recipes) — restated, not redefined. **(A) Use the engine, never ad-hoc scripts** — consumed **by reference** (the flat-layout sibling ~/Projects/docs_chain / the constitution-exposed path), inherited like §11.4.80's codegraph_* scripts: referenced, NEVER copied; ad-hoc sync_*/generate_*_summary/update_readme_doc_links scripts are superseded and retired per registered context. **(B) Consumer-owned contexts** — the engine is project-agnostic; the consumer registers its chains as data via .docs_chain/contexts/*.yaml (§11.4.28 decoupling); state.json + *.docs_chain.tmp are gitignored. **(C) Anchors it mechanizes** (per the CONSTITUTION_INTEGRATION mapping table): §11.4.12 / §11.4.53 / §11.4.45 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.86 / §11.4.93 / §11.4.95 / §12.10 / §11.4.44 — with content-hash change detection (NOT mtime, §11.4.86), atomic-rename + SQLite-txn commit + rollback (§9.2), both-dirty sync → conflict-not-silent-merge (§11.4.6), verify as the deterministic CI/pre-build gate (§11.4.50), per-run captured evidence to qa-results/docs_chain/<run-id>/ (§11.4.69). **(D) NOT a replacement for authoring discipline** — the source author still writes the §11.4.44 revision header; the engine only keeps exports in sync. **(E) Anti-bluff (composes §11.4):** the engine NEVER fakes a transform — a missing pandoc/weasyprint surfaces a typed ToolAbsentError + honest §11.4.3 SKIP-with-reason, never a fake PASS / partial write; every sync/verify carries real captured evidence.

Composes with §11.4 + §11.4.1..§11.4.16 (anti-bluff covenant) / §11.4.6 (no-guessing — conflict not silent merge) / §11.4.28 (engine/context decoupling) / §11.4.80 (inherited-by-reference,

never copied) / §9.2 / §11.4.50 / §11.4.69 / §11.4.5 / §1.1, plus the sync anchors it mechanizes (§11.4.12/.53/.45/.56/.57/.59/.60/.65/.86/.93/.95/.44, §12.10). Classification: universal (§11.4.17) — projects with no derived-export/DB-sync surface inherit it latently (binds the moment they ship one, §11.4.96 pattern). Status (§11.4.6): engine Phases 1-3 IMPLEMENTED+tested; CLI/YAML loader (Phase 4) + submodule distribution (Phase 6) PLANNED + OPERATOR-GATED — wire as phases land, claim no unshipped behaviour. Propagation gate CM-COVENANT-114-106-PROPAGATION (literal 11.4.106 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --ad-hoc-sync-ok, --skip-docs-chain, --fake-transform, --sync-evidence-optional flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.106; detailed spec the Docs Chain engine docs docs/CONSTITUTION_INTEGRATION.md + docs/USE_CASE_CATALOGUE.md (vasic-digital/docs_chain).

Non-compliance is a release blocker.

§11.4.107 — Anti-bluff AV/test-validation techniques mandate (User-driven research, 2026-06-02)

Forensic anchor (genericised, 2026-06-02): a test PASSEd on a SINGLE captured frame showing "a picture" on the target output — but the picture was a FROZEN / STALE frame from the previously-played content (stale-producer / stuck-decoder), so the feature was broken for the user while the test was green; a sibling incident FLASHED media on the WRONG output for ~1 s before routing with no test sampling that window; a third class shipped a comparator that PASSEd its own deliberately-degraded fixture (the analyzer, not the feature, was the bluff). §11.4.5 mandates captured evidence + a presence pass; §11.4.107 raises the bar to **liveness + correct-routing + self-validated-analyzer**.

Every test asserting audio/video output is genuinely playing/advancing for the end user MUST satisfy ALL of: (1) **a single captured frame is NOT proof** — prove LIVE, ADVANCING frames over a steady-state window via a **freeze-detection oracle** (near-duplicate / freeze-detect-class filter OR perceptual-hash adjacent-frame distance, **NOT byte-identical compare** — byte-identity is only a zero-cost pre-filter); (2) an **independent frame-advance counter from the platform's compositor/decoder telemetry** must increase across the window (a different-domain oracle — flat counter ⇒ stuck decoder ⇒ FAIL even if pixels appear to move); (3) **loading/buffering is a distinct state** — wait for genuine playback-start before judging liveness, never false-FAIL a still-loading stream nor false-PASS a spinner;

timeout+unreachable ⇒ SKIP-with-reason per §11.4.3,
timeout+reachable ⇒ FAIL; (4) **not-stale-from-previous cross-check** — new content's first frame ≠ previous content's last frame; (5) **measured FPS / no-lost-frames within tolerance**; (6) **no-flash-on-the-wrong-output** — sample the non-target output at high frequency during a routing transition, any content frame there ⇒ FAIL (content-protection regime classified explicitly, never guessed); (7) **drive through the realistic feed/UI path, not deep-link shortcuts** (shortcuts bypass the transition paths where bugs live; UI-not-introspectable ⇒ operator-attended fallback per §11.4.52, never fake-PASS); (8) **metamorphic relations solve the oracle problem when there is no golden source** (same content on output-A vs output-B must match; paused ⇒ counter stops; 2× speed ⇒ ~2× advance rate); (9) **full-reference quality metrics (SSIM/VMAF/ΔE2000) vs a golden source for owned content**; (10) **mutation-test every analyzer with a golden-good + golden-bad fixture pair** so the analyzer itself provably cannot bluff (an analyzer that PASSES its golden-bad fixture is a bluff gate — §1.1 applied to the analyzers); (11) **per-channel audio RMS/loudness (EBU R128) + XRUN/underrun census** — a single aggregate RMS misses a dead channel; (12) **OCR overlay/subtitle detection needs a per-word confidence floor + ROI** to avoid BOTH false-positive (false-FAIL) and false-negative (false-PASS); (13) **thresholds calibrated on the project's own fixtures, not hardcoded from literature** (§11.4.6 no-guessing).

Honest gap (§11.4.6): true photon FPS at the sink has no clean software oracle — compositor/decoder counters measure the presentation pipeline; flag the gap, never claim it.

4-layer coverage per §11.4.4(b): pre-build gate (a CM-AV-LIVENESS-NO-FROZEN-FRAME-class gate asserting every output-is-playing test references the liveness battery not a single frame + an analyzer-self-validation gate wiring the golden-good/golden-bad fixtures into meta-test) + on-device/runtime test + paired §1.1 meta-test mutation (single-frame-only assertion → gate FAILs; analyzer that PASSES its golden-bad fixture → self-validation FAILs) + HelixQA Challenge. Every PASS via `ab_pass_with_evidence` citing the motion / not-stale / frame-advance / fps / per-channel-loudness / metamorphic / self-validation artefacts (`video_display / audio_output / subtitle_render` per §11.4.69) — never a single screenshot.

Classification: universal (§11.4.17) — platform-neutral AV/test-validation techniques reusable by ANY project validating media playback or any pixel/audio output; the project supplies its concrete capture mechanism + calibrated thresholds per §11.4.35. Composes with §11.4.5 (its strict expansion), §11.4.6, §11.4.50, §11.4.68, §11.4.69, §11.4.85, plus §11.4.2 / §11.4.3 / §11.4.13 / §11.4.48 / §11.4.52 / §1.1. Propagation gate CM-

COVENANT-114-107-PROPAGATION enforces the literal anchor 11.4.107 across the consumer fleet; paired §1.1 meta-test mutation strips the literal → gate FAILs (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.107.

Non-compliance is a release blocker regardless of context. No escape hatch — no --single-frame-proves-playback, --skip-liveness, --byte-identical-freeze-OK, --no-frame-advance-counter, --skip-not-stale-check, --allow-wrong-output-flash, --deep-link-shortcut-OK, --unvalidated-analyzer-OK, --aggregate-rms-suffices, --hardcoded-thresholds-OK flag exists.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate (systematic-debugging Phase 4.5, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): across one batch, multiple fixes were "green" at every gate yet NOT working for the end user — a change present in source and passed by both the source pre-build gate AND the post-build gate never reached the boot-time command-line embedded in the deployed boot artifact (output feature dead despite all-green); sibling fixes correctly built into the system image were masked by stale per-user overlay copies from a previous deployment (the running code was the stale shadow, not the fresh build); deployment did not wipe the mutable overlay so the shadow survived; validation ran against whatever was running — the stale shadow — and reported green on code that was never exercised. Per superpowers:systematic-debugging Phase 4.5: each fix revealing a fresh "fixed-but-not-working" in a DIFFERENT place is NOT independent bugs — it is ONE architectural VERIFICATION flaw; patching each symptom is thrashing.

A fix crosses FOUR distinct layers, and "fixed" at one does NOT imply fixed at the next: (1) **SOURCE** (committed in the source file — what a grep-the-source pre-build gate checks), (2) **ARTIFACT** (the change's BYTES actually landed in the produced build artifact — image / bundle / installer / embedded command-line), (3) **RUNTIME-ON-CLEAN-TARGET** (active on a CLEAN/fresh deployment — the layer the end user experiences — with no stale overlay shadowing the deployed code), (4) **USER-VISIBLE** (the feature works for the end user — the §11.4.5/§11.4.69 captured-evidence layer). Green at layer 1 is the cheapest, least conclusive signal and says nothing about layers 2-4. A gate verifying ONLY the source layer is itself a §11.4 bluff surface.

The mandate (ALL must hold): (1) **Runtime-signature-as-definition-of-done** — a fix is DONE only when its declared runtime signature verifies on a CLEAN/fresh deployment; source-

committed $\neq$ artifact-contains-it $\neq$ active-on-clean-target $\neq$ user-visible-working. (2) **Every fix declares ONE machine-checkable runtime signature** — a single observable on a clean target proving the fix is BOTH active AND working (a running-system property, a downstream/sink-side report per §11.4.13, a §11.4.5/§11.4.69 captured-evidence assertion, or a counter/state delta — NEVER a re-grep of the source); this **registry of per-fix runtime signatures is the SINGLE SOURCE OF TRUTH for "fixed"** — it REPLACES "a gate greps the source" as the definition of done. (3) **Gates span all four layers** — source (pre-build), artifact (post-build — assert the change's BYTES landed in the artifact, not merely that the source still contains the change), runtime-on-clean-target (post-deploy — assert the runtime signature on a freshly-deployed clean target), user-visible (§11.4.5/§11.4.69). (4) **Eliminate the stale-deployment / shadow layer by construction** — deployment MUST yield a CLEAN state (wipe the mutable overlay) OR a pre-validation assertion MUST prove running-artifact == built-artifact BEFORE any validation runs; validation against possibly-stale deployed state is INVALID and any PASS it produces is a §11.4 PASS-bluff (the test exercised code that was never deployed). (5) **Meta-rule** — ≥ 3 "fixed-but-not-working" discoveries in one cycle signal an architectural VERIFICATION flaw, NOT three independent bugs; on the 3rd, STOP patching symptoms (§11.4.4), fix the VERIFICATION pipeline, and re-certify EVERY item through it on a clean target. (6) **A batch is "validated" only after COMPREHENSIVE per-item runtime-signature verification on a clean baseline** — NOT after the touched items' own tests pass.

Classification: universal (§11.4.17) — platform-neutral verification-pipeline disciplines reusable by ANY project that builds an artifact, deploys it, and validates the deployed result; the project supplies its concrete artifact format, clean-deployment / equal-artifact mechanism, and per-fix runtime-signature observables per §11.4.35. Composes with §11.4.1 / §11.4.2 / §11.4.4 (clause 5's STOP-and-fix-pipeline is its §11.4.108 specialisation; "clean baseline" = clause 4's clean deployment) / §11.4.5 / §11.4.6 (every layer asserted by evidence, never assumed to propagate) / §11.4.27 / §11.4.40 (§11.4.108 adds the cross-layer per-item runtime-signature dimension) / §11.4.46 (clean-baseline / equal-artifact pre-flight) / §11.4.50 / §11.4.52 / §11.4.69 (a runtime signature is a taxonomy-class observable) / §11.4.102 (Phase 4.5 architectural-flaw recognition IS clause 5's trigger). Propagation gate CM-COVENANT-114-108-PROPAGATION (literal 11.4.108 across the consumer fleet) + recommended per-fix gate CM-RUNTIME-SIGNATURE-REGISTRY + paired §1.1 meta-test mutations (strip the literal $\rightarrow$ propagation gate FAILs; downgrade a fix to source-only verification $\rightarrow$ registry gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.108.

Non-compliance is a release blocker regardless of context. No escape hatch — no --source-green-is-done, --skip-artifact-byte-check, --validate-against-running-state, --no-clean-deployment, --skip-runtime-signature, --spot-validate-touched-only flag exists.

§11.4.109 — Mandatory Anti-Forgetting Enforcement: PreToolUse Guard Hook + Subagent Constitutional Preamble + Orchestrator Pre-Action Checklist (Operator mandate)

Short tag: anti-forgetting-enforcement. **UNCONFIRMED forensic anchor** (pending operator's verbatim mandate quote). Background: emulator subagents ran raw host-direct emulator/adb because the orchestrator forgot to inject the Containers-submodule rule. A rule forgotten at dispatch is not enforcement. Fix: (A) a PreToolUse guard hook (constitution/scripts/hooks/guard-forbidden-commands.sh) that blocks host-direct emulator, force-push/bypass, sudo, and host-power commands at the tool-call boundary regardless of agent memory; (B) a canonical docs/AGENT_GUARDRAILS.md preamble the orchestrator pastes verbatim into every subagent dispatch; (C) an **ORCHESTRATOR PRE-ACTION CHECKLIST** in the same document. Hook = the floor; preamble = the ceiling.

Consuming projects **MUST**: (1) wire constitution/scripts/hooks/guard-forbidden-commands.sh as a PreToolUse hook in .claude/settings.json (or equivalent runtime settings); (2) maintain docs/AGENT_GUARDRAILS.md containing the SUBAGENT CONSTITUTIONAL PREAMBLE and ORCHESTRATOR PRE-ACTION CHECKLIST headings, with the anchor literal 11.4.109; (3) provide a hermetic hook test suite (≥ 20 cases: every blocked class exits 2, every allowed command exits 0, escape hatch fires for non-power classes, host-power rejects even with escape marker). The hook is inherited by reference — NEVER copied locally (a copy diverges silently).

Gates: CM-ANTI-FORGETTING-ENFORCEMENT (hook present + wired + guardrails doc present + test present) + CM-COVENANT-114-109-PROPAGATION (anchor literal 11.4.109 across every consumer CLAUDE.md / AGENTS.md / QWEN.md). Paired §1.1 mutations: remove hook entry → gate (2) FAILs; delete guardrails doc → gate (3) FAILs; strip hook from constitution → gate (1) FAILs; strip 11.4.109 → propagation gate FAILs.

Classification: universal (§11.4.17). Composes with §11.4.6 / §11.4.10 / §11.4.75 / §11.4.76 / §11.4.78 / §11.4.79 / §11.4.80 / §11.4.81 / §11.4.84 / §11.4.98 / §11.4.102 / §12.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.109; reference implementation `constitution/scripts/hooks/guard-forbidden-commands.sh` + `constitution/docs/AGENT_GUARDRAILS.md`.

Non-compliance is a release blocker. No escape hatch — no `--skip-pretooluse-hook`, `--no-guardrails-doc`, `--anti-forgetting-optional`, `--single-layer-sufficient` flag exists.

§11.4.110 — Pre-build build-readiness verdict + change-impact clash detection mandate (operator mandate, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a fix shipped a new system-property *read* but introduced no matching security-policy grant + no property-context type entry — the read was silently denied at runtime + the feature was dead, while every pre-build gate stayed green because the gate only grepped the source file. The defect class generalises: a change introduces a new dependency on a *second* artifact (security policy, context file, service registry, interface freeze-snapshot, symbol table, build-graph node) that the pre-build never cross-checks. This is §11.4.108's SOURCE→ARTIFACT gap shifted LEFT to pre-build time — most such clashes are statically catchable from the change diff itself, before any build. The mandate (ALL hold): (1) a single deterministic **READY-FOR-BUILD verdict** gates the rebuild (orchestrator refuses to start the build unless READY); (2) a **diff-driven change-impact + clash detector** cross-checks every newly-introduced second-artifact dependency (new property read $\Rightarrow$ property-context type + read-grant; new service $\Rightarrow$ service-context entry; new init service $\Rightarrow$ security label; new/changed stable interface $\Rightarrow$ freeze-snapshot updated; new native-lib dep $\Rightarrow$ module/prebuilt resolves; new policy rule $\Rightarrow$ every type/attribute defined; two batch changes on the same seam $\Rightarrow$ collision acknowledged); (3) **coverage-completeness is a gate** — every changed file maps to ≥ 1 gate + ≥ 1 deployed-target test + ≥ 1 paired §1.1 mutation, baseline ratchets upward per §11.4.50; (4) **two-speed honesty** — grep-speed always-on gates vs REQUIRES_BUILD heavy gates (build-graph parse-only dry-run, full neverallow compilation, ABI diff) as diff-gated opt-in stages, bounded per §12.6/§12.7; (5) every gate + wired analyzer is **anti-bluff by paired §1.1 mutation**; (6) **honest boundary** — a READY verdict proves static internal-consistency + ready-to-build, NOT that the feature works on the deployed target; the regime empties the *preventable* defect class, not the *all-defects* class (runtime/USER-VISIBLE remains §11.4.108's job).

Classification: universal (§11.4.17) — platform-neutral pre-build-rigor disciplines reusable by ANY project that builds an artifact from source; the consuming project supplies its concrete property/service/policy registries, build-graph parse-only

command, interface-freeze mechanism, and changed-file→gate mapping per §11.4.35. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.9 / §11.4.27 / §11.4.50 / §11.4.67 / §11.4.75 / §11.4.92 / §11.4.108 (§11.4.110 is the SOURCE→ARTIFACT half shifted left to pre-build; §11.4.108 owns RUNTIME-ON-CLEAN-TARGET→USER-VISIBLE — together they span all four layers). Propagation gate CM-COVENANT-114-110-PROPAGATION (literal 11.4.110) + recommended per-family gates CM-READY-FOR-BUILD-VERDICT / CM-CHANGE-IMPACT-CLASH-DETECTOR / CM-COVERAGE-COMPLETENESS-GATE / CM-BUILDGRAPH-DRYRUN-WIRED / CM-SEPOLICY-NEVERALLOW-WIRED + paired §1.1 mutations (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.110. Non-compliance is a release blocker. No escape hatch — no --source-green-is-ready, --skip-clash-detector, --skip-coverage-gate, --no-ready-verdict, --grep-proves-neverallow, --skip-buildgraph-dryrun, --build-without-ready-verdict flag.

§11.4.111 — Resolve-by-stable-name-not-by-enumeration-index mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a platform bound an audio output to a kernel-enumerated device index (card=0); a second device of a different class enumerated FIRST at boot and took slot 0, shifting the intended device to slot 1 — the static index binding now pointed at the wrong device (policy mis-attached the sink, mis-assigned its TYPE, the output switcher labelled the AV receiver "Wired Headphone" + routing collapsed to stereo). The lower layer of the SAME stack already resolved the device correctly **by name** (scanning the controller-name registry) — proving the brittleness was the *index* binding, not the resolution capability. Generalises to every enumerated resource whose ordinal is assigned at discovery/boot/hotplug time (non-deterministic across reboots, device additions, topology changes). The mandate: any binding to a hardware device / resource handle / enumerated entity (audio cards, display connectors, network interfaces, storage devices, GPU render nodes, input/camera devices, container/process slots) MUST resolve by a **stable identifier** (name / UUID / serial / label / controller-name / content-hash / sink-reported identity) and MUST NOT bind by enumeration index / ordinal / slot, UNLESS the platform documents that ordinal as deterministically pinned AND the pin is itself captured + asserted as part of the binding. Where a stable identifier exists at one layer, every other layer binding the same resource MUST use the same identifier (mixed by-name-here / by-index-there is the structural weak link forbidden here). Honest boundary (§11.4.6): when only an ordinal exists, pin it deterministically via the platform's own mechanism, capture the

pin in the binding's §11.4.108 runtime signature, and document the residual fragility as UNCONFIRMED:-class risk — never silently trust an unpinned ordinal.

Classification: universal (§11.4.17) — platform-neutral binding-robustness discipline reusable by ANY project binding to enumerated hardware/resources/handles; the consuming project supplies its stable-identifier mechanism (ALSA card name, DRM connector name, NIC predictable name, block-device UUID, etc.) + the layers that must agree per §11.4.35. Composes with §11.4.6 (no-guessing — "the index is *usually* stable" is the exact guess forbidden) / §11.4.8 (mature stacks resolve by name/UUID — reproducing a known-brittle index binding when the by-name path is documented is a §11.4.8 omission) / §11.4.69 (sink-side evidence verifies the by-name binding's correctness) / §11.4.108 (the by-name binding's runtime signature asserted on a clean target across the topology that broke the index) / §11.4.110 (a new ordinal binding in a diff is a statically-catchable clash class). Propagation gate CM-COVENANT-114-111-PROPAGATION (literal 11.4.111) + recommended gate CM-RESOLVE-BY-NAME-NOT-INDEX + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.111. Non-compliance is a release blocker. No escape hatch — no --allow-index-binding, --ordinal-is-stable-enough, --skip-name-resolution, --trust-unpinned-index flag.

§11.4.112 — Structural-impossibility won't-fix classification mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): deep research (§11.4.8) into relocating protected (content-protection / secure-surface) video to a secondary display PROVED — from authoritative platform/HDCP docs + reproducible captured behaviour (a secure surface is blanked on any output lacking the secure flag; mirror/screenshot of a secure layer returns black) — that the goal is **structurally impossible by platform design**, not a missing feature or unsolved engineering problem. Without a durable classification, such a goal is re-investigated every cycle (re-read the same sources, re-run the same probes, re-derive the same impossibility — compounding wasted effort). The mandate: when deep research per §11.4.8 PROVES (cited authoritative sources AND, where applicable, reproducible captured evidence) a goal is structurally impossible on the target platform (forbidden by platform design / hardware-protocol constraint / documented kernel-or-API limitation — NOT merely unimplemented or hard), the goal MUST be: (1) classified won't-fix + closed per §11.4.90 with closure reason structurally-impossible; (2) documented with the impossibility evidence — cited authoritative source URLs (per §11.4.99 latest-source verification) + the reproducible probe/

captured evidence — in the tracker entry + relevant docs/ guide;
(3) NOT re-attempted in future cycles — a reopen MUST cite NEW evidence the platform constraint changed (per §11.4.34 + §11.4.7), never merely re-derive the same impossibility; (4) paired with the correct posture — the entry states what the project DOES instead, so "impossible" is never confused with "broken/unhandled". Honest boundary (§11.4.6): structurally-impossible is reserved for *proven* platform/hardware/protocol impossibility — "could not find a way" / "very hard" / "no time" are operator-blocked (§11.4.21) or open work, NOT won't-fix; mislabelling them to avoid the work is a §11.4 planning-layer bluff. A future platform change can make the impossible possible; the classification is durable but not eternal.

Classification: universal (§11.4.17) — platform-neutral effort-conservation + honesty discipline reusable by ANY project; the consuming project supplies the specific impossible goal, its platform constraint, and the cited evidence per §11.4.35. Composes with §11.4.6 (FACT with cited evidence, never "probably can't") / §11.4.7 (reopening requires NEW positive evidence) / §11.4.8 ("NO external solution found — structurally impossible" is the citation) / §11.4.34 (reopen attribution) / §11.4.90 (structurally-impossible is a closure reason in the closed vocabulary) / §11.4.99 (latest-source so the verdict is not stale). Propagation gate CM-COVENANT-114-112-PROPAGATION (literal 11.4.112) + recommended gate CM-WONT-FIX-STRUCTURAL-IMPOSSIBILITY + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.112. Non-compliance is a release blocker. No escape hatch — no --wont-fix-without-proof, --reattempt-closed-impossible, --skip-impossibility-evidence, --impossible-equals-broken flag.

§11.4.113 — Absolute no-force-push + merge-onto-latest-main mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03):

"Any force-push is strictly forbidden! We must for every Submodule take as a base latest commit on Submodule's main (or master) branch, then on top of it carefully to merge all changes that have to be pushed! Once all merging is carefully done we perform commit and push to all Submodule's upstreams!"

Force-push is STRICTLY FORBIDDEN with NO exception — git push --force, --force-with-lease, +<ref>, or any history-rewriting overwrite of a remote ref, against EVERY repository this Constitution governs (main repo, this constitution submodule, every owned + nested submodule, every upstream). No operator-approval path, no "after a merge-first audit" path. The mandated 6-step integration procedure for any repo/submodule whose local has commits to publish OR whose mirrors diverged: (1) git fetch

--all --prune --tags all remotes; (2) set the base to the LATEST commit on the canonical main/master branch (the most-advanced mirror tip); (3) carefully MERGE every change to be published on top of that base — union, preserve BOTH sides, NEVER -s ours / rebase / reset that drops commits (per §9 no-commit-loss); (4) resolve every conflict carefully — no conflict markers, no file dropped, gates/tests still pass; (5) commit the merge (stage only intended files, NEVER git add -A in a submodule per §11.4.30); (6) push to ALL upstreams — each push is a fast-forward because the merge commit descends from every mirror tip, so NO force is ever needed (if an upstream still rejects, return to step 1 for it, merge its new tip, re-validate, re-push). **TIGHTENS §11.4.41 / §11.4.71 / §9.2 / CONST-043 — the force-push escape hatch is REMOVED:** even WITH operator approval, even after a clean merge-first audit, force-push is forbidden, because the merge-onto-latest-main path is always available so force is never necessary. Those clauses' merge-first/fetch-first machinery stays in force as the integration discipline; only their terminal "...then force-push" step is struck.

Classification: universal (§11.4.17) — a platform-neutral integration discipline reusable across every repository. Composes with §2.1 (multi-upstream push — step 6 fans out) / §9 / §9.2 (absolute data safety — this is the no-loss push discipline that makes force unnecessary) / §11.4.4 / §11.4.6 (remote state unknowable without the step-1 fetch) / §11.4.26 (constitution-update conflict resolution IS this procedure) / §11.4.37 (fetch-before-edit → fetch-before-push) / §11.4.40 / §11.4.41 (merge-first stays, force-push step removed) / §11.4.71 (same tightening) / §11.4.88 (background-push still fast-forward-only) / CONST-043 (no force-push authorisable). Propagation gate CM-COVENANT-114-113-PROPAGATION (literal 11.4.113) + recommended gate CM-NO-FORCE-PUSH-ABSOLUTE (scan tracked scripts/hooks for push --force / --force-with-lease / push +<ref> + reject; a §11.4.109-class PreToolUse guard blocks the class at the tool-call boundary) + paired §1.1 mutation (inject a git push --force into a tracked script → gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.113. Non-compliance is a release blocker. No escape hatch — no --force, --force-with-lease, --allow-force-push, --force-push-authorised, --skip-merge-onto-main flag.

§11.4.114 — Last-known-good-tag regression isolation mandate (1.1.8-dev remediation, 2026-06-03)

When a previously-working feature/behaviour is observed broken, the FIRST diagnostic action MUST be to identify the last release tag (or commit/build/deploy) at which it was KNOWN-GOOD and diff/bisect the broken state against it — BEFORE any open-ended

root-cause hunt or speculative fix. The known-good revision is the regression oracle: it bounds the search to the commits between good and now (`git diff <good-tag>..HEAD --stat` of the feature's files = captured evidence), tells you it is a regression REPAIR not a from-scratch design problem, and gives each suspect file a behavioural oracle. When the operator volunteers a known-good tag, that lead is load-bearing and MUST be acted on first. Default to a SURGICAL forward-fix (keep the post-good-tag features, revert ONLY the broken sub-part) over a wholesale revert that loses the batch's other working features — unless the operator prefers wholesale revert. Honest boundary (§11.4.6): "it worked before" is a HYPOTHESIS until the known-good tag is identified AND the feature is confirmed working there; "probably regressed in the last batch" without the diff is a guess; a feature that NEVER worked is not a regression. Composes §11.4.4 / §11.4.6 / §11.4.7 / §11.4.40 / §11.4.43 / §11.4.102 / §11.4.108. Propagation gate CM-COVENANT-114-114-PROPAGATION (literal 11.4.114) + recommended gate CM-REGRESSION-ISOLATED-AGAINST-KNOWN-GOOD + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.114. Non-compliance is a release blocker. No escape hatch — no `--skip-known-good-diff`, `--root-cause-from-scratch`, `--assume-regression-source`, `--wholesale-revert-without-isolation` flag.

§11.4.115 — RED-baseline-on-the-broken-artifact + polarity-switch mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.43's RED step. Every RED test MUST be authored to REPRODUCE the defect on the CURRENT, pre-fix artifact (the actual broken build/deployment), capturing positive evidence per §11.4.5 / §11.4.69 / §11.4.107 that the defect is genuinely present — never a synthetic failure the fix is then written to agree with. The SAME test source MUST carry a single polarity switch (env flag / parameter, canonical `RED_MODE`, default 1 = reproduce-and-assert-defect-present) that flipped to 0 post-fix converts the test into the GREEN regression-guard asserting the defect is ABSENT. One source, two roles: the bug-catcher IS the regression-guard; no separate happy-path test is authored as the primary guard (that demonstrates only that the test agrees with the fix — the §11.4.43 PASS-bluff). RED-on-broken-artifact then GREEN-on-fixed-artifact (on a clean target per §11.4.108) MUST both be captured. Honest boundary (§11.4.6): if the RED run does NOT fail on the broken artifact, that is a finding (close per §11.4.7 with negative evidence, or fix the test) — a RED test that passes on the known-broken artifact is a blind test. Composes §11.4.1 / §11.4.2 / §11.4.5 / §11.4.69 / §11.4.107 / §11.4.4 / §11.4.7 / §11.4.43 / §11.4.50 / §11.4.108 / §11.4.114. Propagation gate CM-

COVENANT-114-115-PROPAGATION (literal 11.4.115) + recommended gate CM-RED-POLARITY-SWITCH-PRESENT + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.115. Non-compliance is a release blocker. No escape hatch — no --green-only-test, --skip-red-reproduction, --separate-happy-path-suffices, --synthetic-red-OK flag.

§11.4.116 — Real-time conductor↔autonomous-test-framework sync channel mandate (1.1.8-dev remediation, 2026-06-03)

Any autonomous, long-running test/QA/validation framework an external orchestrator (conductor agent / operator) depends on for real-time decisions MUST expose a real-time sync channel: (1) a structured append-only event stream (JSONL or equivalent, one event per line, never rewritten) emitting at minimum session-start / phase-transition / per-test-or-challenge-start / captured-evidence-path / external-call (LLM / vision / sink-probe) / error / per-item-verdict events; (2) an atomically-rewritten status snapshot (single small file written write-temp-then-rename so a reader never sees a torn write) carrying current session/phase/item + counters + last verdict. Verdicts use the closed vocabulary PASS / FAIL / SKIP / OPERATOR-BLOCKED (§11.4.45). The conductor tails it live so it stays in real-time sync, can §11.4.4-interrupt on a fresh defect, and never idles blindly (§11.4.94 / §11.4.97). Anti-bluff: a verdict event MUST carry the evidence path that backs it (§11.4.69) — a PASS event with no evidence path is a channel-layer PASS-bluff; a snapshot reporting PASS while the stream shows no evidence event for that item is a contradiction → treat as FAIL; an item with no start-event cannot have a verdict-event. When the framework is an owned project-agnostic submodule (§11.4.28), the channel stays project-neutral — the consumer registers its data (endpoints, package ids, sink hosts) at runtime via the public API, never hardcoded. Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.27 / §11.4.28 / §11.4.45 / §11.4.52 / §11.4.89 / §11.4.94 / §11.4.97. Propagation gate CM-COVENANT-114-116-PROPAGATION (literal 11.4.116) + recommended gate CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.116. Non-compliance is a release blocker. No escape hatch — no --no-sync-channel, --end-of-session-report-suffices, --verdict-without-evidence-path, --torn-status-write-OK flag.

§11.4.117 — Computer-vision / OCR pixel-oracle fallback for non-introspectable UIs mandate (1.1.8-dev remediation, 2026-06-03)

Any test that needs to drive a UI control OR assert on-screen content MUST NOT assume the accessibility/semantic/DOM hierarchy is the source of truth for what the user sees. When the hierarchy is blank/partial/known-unreliable for the app under test (TV-Compose, leanback, canvas/SurfaceView/GL, games, custom-rendered UIs), the test MUST fall back to a PIXEL ORACLE: (1) DRIVE input by computer-vision template-match (locate a control by its rendered appearance → tap its screen coordinates), not by a hierarchy node id; (2) ASSERT content by ROI OCR (read the rendered text the user reads — caption strip, title, error overlay) with a per-word confidence floor + region-of-interest per §11.4.107(12), not a hierarchy text attribute that may be absent/stale. The tool MUST both drive input AND read pixels — a hierarchy-only tool is NOT a content oracle. This makes §11.4.52's "near-empty hierarchy → INFEASIBLE" constructive: pixel-drive, not only SKIP. Anti-bluff (§11.4.107(10)): the CV/OCR analyzer is self-validated — golden-good fixture PASSES, golden-bad fixture FAILs, wired into meta-test; thresholds calibrated on the project's own frames, not hardcoded (§11.4.6). Honest boundary: when BOTH hierarchy is blank AND pixel oracle is infeasible (secure surface black-captures per §11.4.112, geo-unreachable), SKIP-with-reason per §11.4.3 + tracked operator-attended migration item per §11.4.52 — never fake PASS. Composes §11.4.5 / §11.4.6 / §11.4.48 / §11.4.49 / §11.4.52 / §11.4.107 / §11.4.112. Propagation gate CM-COVENANT-114-117-PROPAGATION (literal 11.4.117) + recommended gate CM-CV-OCR-PIXEL-ORACLE-FALLBACK + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.117. Non-compliance is a release blocker. No escape hatch — no --hierarchy-is-content-oracle, --skip-pixel-fallback, --unvalidated-ocr-OK, --hardcoded-ocr-threshold-OK flag.

§11.4.118 — Discovery-pressure to confirm known-issue-set completeness mandate (1.1.8-dev remediation, 2026-06-03)

A remediation/release cycle MUST NOT treat "every reported defect is fixed" as "the build is good" — the reported set is biased by what the operator happened to test ("we only see what we test"). After/alongside fixing the reported set, the cycle MUST run a discovery + stress pass across ALL target devices/environments that deliberately exercises subsystems, journeys, and edge cases BEYOND the reported defects — to CONFIRM the reported set is the COMPLETE critical set and surface unreported defects before the end user does. The pass MUST produce PROVABLE coverage:

an enumerated list of the subsystems/user-journeys/stress scenarios actually exercised, each with its outcome (no-new-issue, or a new tracker entry per §11.4.15 / §11.4.16). "We found no other issues" is a §11.4 bluff unless accompanied by "here is the enumerated set we exercised" — absence of evidence of looking is not evidence of absence. New findings trigger §11.4.4 interrupt + the §11.4.114/§11.4.115 isolation→RED→fix loop. Honest boundary (§11.4.6): discovery reduces the unknown-unknown surface but does not prove zero remaining defects — the earned claim is "reported set + enumerated discovery set addressed," with un-exercised subsystems stated as honest coverage gaps (§11.4.3), never silently implied clean. Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.25 / §11.4.40 / §11.4.42 / §11.4.52 / §11.4.85 / §11.4.114 / §11.4.115 / §11.4.119. Propagation gate CM-COVENANT-114-118-PROPAGATION (literal 11.4.118) + recommended gate CM-DISCOVERY-COVERAGE-ENUMERATED + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.118. Non-compliance is a release blocker. No escape hatch — no --reported-set-suffices, --skip-discovery-pass, --no-issues-without-coverage-list, --assume-complete flag.

§11.4.119 — Single-resource-owner partitioning for parallel hardware testing mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.58 / §11.4.103 for hardware contention. When multiple parallel work/test/discovery streams exercise SHARED hardware or any exclusive-access resource (a media-playback path, a single HDMI/audio sink, an exclusive device handle, a serial/JTAG line, a GPU under exclusive capture), exactly ONE stream MUST own each such resource at a time. The exclusive owner drives it (playback, input injection, capture); every other concurrent stream targeting the same resource MUST be READ-ONLY (passive probes — dumphys / /proc / /sys reads / sink-side network probes / log tails). Parallelism is partitioned by resource: distinct devices/sinks/handles run fully concurrent (stream-per-device), but the same device's exclusive resource is single-owner. Ownership MUST be enforced by an advisory lock/token (§11.4.58 L3 + the hardware analogue of §11.4.84 quiescence), event-driven (claim when the resource frees, release the moment work completes so the next queued stream claims it per §11.4.103 auto-backfill). Why: concurrent drivers of one exclusive resource produce CROSS-CONTAMINATED evidence (sink reports the wrong stream's audio, foreground belongs to whichever am start landed last, input events interleave) — a PASS under contention is a §11.4 evidence-integrity bluff. Honest boundary (§11.4.6): a "read-only" probe stream MUST issue NO state-changing command against a device it does not own; when a resource genuinely cannot be partitioned, streams serialize on it

(single-owner over time), not run concurrently and hope.
Composes §11.4.5 / §11.4.69 / §11.4.13 / §11.4.50 / §11.4.58 / §11.4.82 / §11.4.84 / §11.4.103 / §11.4.118. Propagation gate CM-COVENANT-114-119-PROPAGATION (literal 11.4.119) + recommended gate CM-SINGLE-RESOURCE-OWNER-PARTITION + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.119. Non-compliance is a release blocker. No escape hatch — no --allow-concurrent-resource-drivers, --skip-ownership-lock, --read-only-may-mutate, --contended-evidence-OK flag.

§11.4.120 — Fix-breaks-its-own-gate reconciliation mandate (1.1.8-dev remediation, 2026-06-03)

When a correct fix causes a pre-existing gate/test to FAIL because that gate asserted the OLD (now-removed-or-changed) behaviour, the gate FAIL is the CORRECT signal the fix landed — it MUST NOT be suppressed by either forbidden response: (1) FAKE-PASSING the gate (editing it to always pass, weakening its assertion to a tautology, or deleting it — a §11.4 gate-layer bluff + a §11.4.84 mutation-residue risk); (2) REVERTING the correct fix to satisfy the stale gate (re-introducing the defect). The required response is RECONCILIATION: rewrite the gate to assert the NEW mechanism the fix introduced, backed by captured evidence of the new correct behaviour, AND update its paired §1.1 mutation so the mutation breaks the NEW invariant. Discriminator vs bluffing: after reconciliation the gate + mutation still form a valid §1.1 pair (mutate the new invariant → gate FAILs); a bluffed gate's mutation no longer makes it FAIL (the assertion became a tautology). The reconciliation MUST be a visible, evidence-cited change, never a silent assertion-weakening. Honest boundary (§11.4.6): a post-fix gate FAIL is NOT automatically "stale gate, reconcile" — investigate per §11.4.102 first; the FAIL may be the gate correctly catching a REGRESSION the fix introduced, in which case the FIX is wrong, not the gate. Reconcile ONLY when investigation PROVES the gate asserted old-correct-now-removed behaviour AND the new behaviour is the intended, evidence-confirmed mechanism. Composes §11.4.1 / §11.4.4 / §11.4.6 / §11.4.84 / §11.4.102 / §11.4.108 / §1.1. Propagation gate CM-COVENANT-114-120-PROPAGATION (literal 11.4.120) + recommended gate CM-GATE-RECONCILED-NOT-FAKE-PASSED + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#) §11.4.120. Non-compliance is a release blocker. No escape hatch — no --fake-pass-stale-gate, --revert-fix-for-gate, --weaken-assertion-to-pass, --delete-failing-gate flag.

§11.4.121 — No-commit-while-build-writes-tracked-artifacts mandate (1.1.8-dev remediation, 2026-06-03)

A commit (especially `git add -A` / any broad stage) MUST NOT run while a build/packaging/generation step is actively writing artifacts INTO tracked (version-controlled) directories — doing so races the writer and stages a PARTIAL or stale artifact (a §11.4.108 SOURCE→ARTIFACT integrity failure landed in version control). The commit MUST be deferred until the build step that writes tracked artifacts has COMPLETED, so the tree is quiescent at the artifact layer AND the committed artifacts are the FRESH, whole outputs (no stale pre-rebuild artifact, no half-written file). Before committing tracked build outputs, verify the writing step finished (process exit / completion marker / per-artifact mtime ≥ build-start) — a build still in flight writing tracked dirs is a HOLD on the commit, not a race to win. Build-output analogue of §11.4.84 (no commit while a mutation gate is in flight): both close the "commit captures transient non-final tree state" class at two write-sources. Where build outputs land OUTSIDE version control (`gitignored out/ / dist/`), the race does not apply. Honest boundary (§11.4.6): "the build probably finished" is not "the build finished" — verify with a completion signal; committing source changes that DON'T touch the build's tracked-artifact directories is fine mid-build. The PROJECT-SPECIFIC instance (which tracked directory + which build steps write it) is recorded in the consuming project's own governance per §11.4.35. Composes §11.4.6 / §11.4.30 / §11.4.58 / §11.4.84 / §11.4.88 / §11.4.96 / §11.4.103 / §11.4.108. Propagation gate CM-COVENANT-114-121-PROPAGATION (literal 11.4.121) + recommended gate CM-NO-COMMIT-DURING-ARTIFACT-WRITE + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule [Constitution.md](#)
§11.4.121. Non-compliance is a release blocker. No escape hatch — no `--commit-during-build`, `--stage-partial-artifact-OK`, `--assume-build-finished`, `--skip-build-completion-check` flag.